FURTHER PROPERTIES OF THREE-DIMENSIONAL SU(2) GAUGE THEORY WITH MASSIVE FERMIONS

Abstract. We study further properties of the massive fundamental SU(2) theory in three dimensions. A bound for the mixed dependence of the vacuum energy on fermion sources gives a uniform variance estimate for conditional loops at coinciding supports. It also gives quantitative comparisons between gauge laws at different masses and mutual absolute continuity of their simultaneous limits on a fixed physical torus. We enlarge the scalar meson family to include diquarks, construct its coordinate-reflection Hilbert space, and prove lower energy bounds in the charged sectors. We also obtain decay of parity-odd meson correlations and the short-distance two-point law for conserved currents. On a fixed physical torus, subsequences give action-density and meson correlations holomorphic in coupling and mass away from the total diagonal. At fixed real coupling, the separated correlations continue meromorphically in the mass, with a quantitative zero-free neighbourhood of a longer positive real interval. The construction paper supplies the chamber model and its renormalization-group estimates; the additional source estimates and limit arguments are proved here.

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1. Introduction

We study the massive fundamental SU(2) theory constructed in [QCDM1]. Its regularization uses the chamber Dirac operator with Wilson parameter $r = 1/2$. Fix $N_f \geq 1$ and a physical mass window $0 < m_- \leq m_+ < 1/2$. At spacing $a = L^{-K}$ the bare parameters are $\beta = \mathsf{b}/a$ and $m_0 = a\mathsf{m}$. The gauge measure on a torus is

$$\nu_K(dU) = Z_K^{-1} e^{-\beta A(U)} |\det(D_+[U] + m_0)|^{2N_f}\, dU, \qquad A(U) = \sum_p (1 - \tfrac{1}{2}\operatorname{tr} U_p).$$

Here $D_+[U]$ is one internal block of the massless chamber operator for one flavour.

Theorem 0. For sufficiently weak gauge coupling and masses in a fixed compact positive interval, conditional fermion loops have cutoff-independent variance bounds, including at coinciding supports. On a fixed physical torus, gauge laws at two masses that converge through the same observables and cutoff sequence have mutually absolutely continuous limits.

The scalar-bilinear theory extends to diquarks and has lower energy bounds in its charged sectors. The conserved currents have a definite short-distance two-point law. On a fixed torus, subsequences give correlations holomorphic in coupling and mass off the total diagonal. At fixed real coupling, the separated correlations continue meromorphically in mass and are holomorphic on a quantitative neighbourhood of a longer positive real interval. ■□

The precise statements are Thm. A and Prop. 3.4, the results of sections 5, 6 and 8, and Thms. H and I.

At fixed K, an infinite-lattice state means any weak limit of these measures as the torus side tends to infinity through the sides permitted by [QCDM1, Prop. 10.20]. Such states exist by [QCDM1, Prop. 2.5]. For a bounded compactly supported function u put

$$\mathsf{G}_+ = a(D_+[U] + a\mathsf{m})^{-1}, \qquad \mathsf{L}(u) = \mathrm{Tr}(\mathsf{G}_+ \hat{u}).$$

The hat denotes multiplication by u at the site of each chamber. The trace is over chambers, spin and colour. It is finite at each cutoff, even on the infinite lattice.

Theorem A (Fluctuations and changes of mass)**.** Choose the odd integer $L \geq 13$ and the auxiliary constants as in [QCDM1, Prop. 10.20]. There is $\mathsf{b}_1 < \infty$ such that, for every fixed $\mathsf{b} \geq \mathsf{b}_1$ and every $R < \infty$, there is a constant $C_R < \infty$ with

$$\mathsf{E}_\nu \big|\mathsf{L}(u) - \mathsf{E}_\nu \mathsf{L}(u)\big|^2 \leq C_R \|u\|_\infty^2.$$

This holds for every $K \geq 1$, every $\mathsf{m} \in [m_-, m_+]$, every infinite-lattice torus-limit state ν, and every bounded real u whose support lies in a ball of radius R. The constant is independent of the location of the ball and the chosen state. On a fixed allowed physical torus the same assertion holds with a constant depending on its volume.

On a fixed physical torus, fix $0 < g \leq \gamma$, $\ell \in \mathfrak{L}(g)$ and an allowed coupling label b, as in section 2. Let f_K be Borel maps from the cutoff configuration spaces to a fixed Polish space. If their laws at two masses $\mathsf{m}, \mathsf{m}' \in [m_-, m_+]$ converge along a common cutoff sequence to P, Q, then P and Q are mutually absolutely continuous and

$$\mathrm{TV}(P, Q) \leq C_{g,\ell} |\mathsf{m}' - \mathsf{m}|.$$

The constant is independent of the cutoff, the two masses and b. The cutoff measure at each mass uses that mass in the determinant defining ν_K. ■□

The trace need not have a finite limit without subtraction. The theorem controls its fluctuation at coinciding supports. Its proof uses a determinant with a complex source in one internal block and the conjugate source in the other. The parts of the vacuum energy depending on just one source are holomorphic. Removing them leaves a mixed energy bounded uniformly in the cutoff. The first coefficient of the resulting holomorphic determinant gives the variance bound by Parseval's identity. Thm. B gives the explicit constants and proves the variance assertion of Thm. A. For a constant source the determinant changes the mass. The same circle bound controls the density for a short change of mass. Chaining these bounds across the mass interval gives the comparison of limiting gauge laws in Prop. 3.4.

This estimate also enters the analysis of parity-odd scalar bilinears. We treat those bilinears together with mesons, diquarks and antidiquarks. Their separated correlations have a common continuum subsequence and a compact symplectic flavour action by $G = \mathrm{USp}(2N_f)^2$. Coordinate reflection positivity gives an enlarged Hilbert space and a time-translation semigroup. Bounds obtained by counting fermion propagators give a lower bound $H \geq |q|\mathsf{m}$ on quark number q, and $H \geq 2\mathsf{m}$ on the complement of the G-invariant vectors. Both sectors $q = 2$ and $q = -2$ are nonzero. These claims and the distinction between coordinate reflection positivity and full rotation covariance are stated in sections 5 and 6.

We also prove a short-distance two-point law for the conserved currents. The current is a point-split lattice bilinear. Its comparison with a site bilinear requires a source that acts on spin, as well as on internal and flavour indices. The source construction extends to those matrices; an energy estimate for the propagator then controls the point splitting. The result concerns current two-point functions and lattice reflection-positive current states. It does not assume a joint reconstruction of all the currents with the scalar fields of [QCDM1]; see section 8.

Finally, for the original action-density and meson family on a fixed physical torus, we allow supports with empty total intersection. Individual pairs of supports may overlap. A polymer contributing to a mixed derivative is either large or far from at least one insertion; in both cases the local-energy estimate supplies decay. A common subsequence then gives correlations off the total diagonal, holomorphic in the coupling and mass near the permitted real parameter window. The neighbourhood depends on the physical volume. This is a separate fixed-volume conclusion from the volume-first construction used for the charged sectors.

At fixed real coupling we continue the separated correlations meromorphically in the mass. The same subsequence and discrete exceptional set work for all their orders and test functions. A quantitative lower bound for the normalized partition function gives a zero-free neighbourhood of a longer real mass interval, with width at least $c/[\ell^3(1+\log \ell)]$. The real-mass cumulant estimates remain uniform in the volume. These statements, including the continued short-distance laws, are proved in section 10.

All statements concern the massive chamber model of [QCDM1]. We use its construction and the estimates explicitly recalled below, and give the additional determinant, source and continuation arguments in the same notation.

2. Model, sources and parameter ranges

We recall the definitions needed for the observables and the estimates. The lattice construction is in [QCDM1, §2]. Spatial coordinates in operator formulas are in lattice units; supports and distances are in physical units unless stated otherwise.

2.1. **The chamber operator.** Let W_3 be the group of signed permutation matrices. A place is $v = (x, q)$ with $x \in \mathbb{Z}^3$ and $q \in W_3$. Put

$$(N_0, N_1, N_2, N_3) = (e_1, e_2 - e_1, e_3 - e_2, -e_3), \qquad t_s = 3|N_s|,$$

and let w_s be reflection in $N_s^{\perp}$. The adjacent places and the corresponding unit vectors are

$$v \cdot s = (x + \delta_{s0} q e_1, q w_s), \qquad u_s(v) = q N_s/|N_s|.$$

Set $U(v, v\cdot s) = 1$ for $s = 1, 2, 3$ and $U(v, v\cdot 0) = U_{\langle x, x+qe_1\rangle}$. The spinor fibre is

$$\mathbb{C}^2_{\mathrm{s}} \otimes \mathbb{C}^2_{\tau} \otimes \mathbb{C}^2_{\mathrm{c}} \otimes \mathbb{C}^{N_f}.$$

The factors denote spin, internal index, colour and flavour. With the Pauli matrices σ_μ and τ_μ write

$$\gamma_\mu = \sigma_\mu \otimes \tau_3, \qquad \gamma_4 = 1 \otimes \tau_1, \qquad e_\pm = (1 \pm \tau_3)/2.$$

The massless operator is

$$(D[U]f)(v) = \frac{1}{2}\sum_{s=0}^{3} t_s f(v) - \sum_{s=0}^{3} t_s(\tfrac{1}{2} - \gamma(u_s(v)))U(v, v\cdot s) f(v\cdot s). \tag{1}$$

The two internal blocks satisfy $D_- = D_+^*$ and $\mathrm{Re}(D_\pm + m_0) \geq m_0$. In particular $\|\mathsf{G}_\pm\| \leq \mathsf{m}^{-1}$, where $\mathsf{G}_\pm = a(D_\pm + m_0)^{-1}$.

On $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$ the links are periodic and spinors are antiperiodic in the three coordinate directions. The finite-volume functional is

$$\Omega_K(F) = \int dU \int d\bar\psi\, d\psi\, e^{-\beta A(U) - \bar\psi(D[U]+m_0)\psi} F, \qquad \langle F\rangle_K = \Omega_K(F)/\Omega_K(1).$$

Haar measures have total mass one. The Berezin integral has the convention giving $\det(D[U]+m_0)$ at $F=1$. Conditional expectations at fixed U are the Gaussian Berezin functional with covariance $(D[U]+m_0)^{-1}$. On the infinite lattice we integrate this functional against an infinite-lattice torus-limit gauge state. All polynomial insertions have finite support at fixed a.

2.2. **Physical sources.** Choose Hermitian flavour matrices t_j with $t_1 = N_f^{-1/2}1$ and $\operatorname{tr}(t_j t_l) = \delta_{jl}$. Set

$$(c_0,c_1,c_2,c_3) = (1,i,i,1), \quad \Gamma_{a,j} = 1_{\mathrm{s}} \otimes c_a\tau_a \otimes 1_{\mathrm{c}} \otimes t_j, \quad \mathfrak{B} = \{\Gamma_{a,j} : 0 \le a \le 3,\ 1 \le j \le N_f^2\}.$$

These matrices have operator norm at most one. The physical action and meson insertions are

$$O_K(w) = a^{-1}\sum_p w(x(p))(1 - \tfrac{1}{2}\operatorname{tr} U_p), \tag{2}$$

$$\Phi_K(J) = a\sum_x \sum_{v:x_v=x} \bar\psi(v) J(ax)\psi(v), \tag{3}$$

where J takes values in the complex span of $\mathfrak{B}$. Here $x(p)$ is the physical plaquette centre, and each positive coordinate orientation occurs once. The source partition function is

$$Z_K[w,J] = \Omega_K(e^{O_K(w)+\Phi_K(J)}) = \int e^{-\beta A(U)+O_K(w)} \det(D[U]+m_0-a\hat J)\, dU.$$

For $\alpha = 0$ set $\Phi_{\alpha,K}(f) = O_K(f)$, and for $\alpha = \Gamma \in \mathfrak{B}$ set $\Phi_{\alpha,K}(f) = \Phi_K(f\Gamma)$. We use

$$\|f\|_0 = \|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}, \qquad \|f\|_\Gamma = \|f\|_\infty.$$

The constants M, ϱ, ϱ_J and r_M belong to the source estimates of [QCDM1, §10].

Joint cumulants are derivatives of the logarithm of the normalized source partition function at zero. Their centered moments are the sums over set partitions with no singleton blocks. We use the same definition for the enlarged bilinear family introduced below.

2.3. **Permitted parameters and limits.** Fix the mass window and N_f. Choose the constants of [QCDM1, Prop. 10.20], and decrease its upper bound γ on g as specified in section 4. These choices depend on the fixed mass window and N_f, and not on K or the volume. Put

$$I_g = [g^{-2}+\bar B, L(g^{-2}+B_\sharp) - \bar B], \qquad 0 < g \le \gamma, \quad \mathsf{b} \in I_g, \quad \mathsf{m} \in [m_-, m_+].$$

The allowed sides form the cofinal set $\mathfrak{L}(g) = \{2L^m : m \ge m_g\}$ of that proposition. More explicitly, with $M = L^{m'}$,

$$X_g = L(g^{-2}+B_\sharp), \qquad R(x) = \min\{L^q : q \ge 0,\ L^q \ge (\log x)^{r_c}\},$$
$$m_g = 1 + m' + \max\{\lceil \log_L R(X_g+\bar B)\rceil, \lceil \log_L C_{\mathrm{s}}\rceil\}.$$

Here $r_c \ge 2$ and $C_{\mathrm{s}} \ge 2$ are the fixed local-chart constants of the construction. This condition supplies the room required by the local estimates. The intervals I_g cover all sufficiently large b: for example take $g^{-2} = \mathsf{b} - \bar B$ and increase the lower threshold on b so that the upper endpoint also exceeds b. This explains the parameter formulation in Thm. A.

For infinite-volume conclusions we first take torus limits at fixed K, then let $K \to \infty$. Constants may depend on g unless a stronger uniformity is stated. For the analytic results in section 9, the physical torus is instead kept fixed throughout the cutoff limit.

3. Determinant bounds at coinciding supports

The estimates for disjoint supports do not bound the variance of a loop against itself. We obtain this bound from a different use of the fermion determinant. A source in one species gives a holomorphic determinant; its conjugate source in the other species gives the squared absolute value. After removing the two terms of the vacuum energy which depend on just one species, the remaining energy is bounded uniformly in the cutoff. Cauchy's formula then bounds the coefficients of the determinant, including the loop.

We retain the setting of section 2, with the further choices in section 4. Write ν_K for the gauge measure, and let $\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$ be the infinite-lattice states obtained as limits of these torus measures. All the constants below are uniform in the admitted data except where dependence on g, a torus side, or a support radius is stated.

The two physical resolvents and the simplest loop are

$$\mathsf{G}_\pm = a(D_\pm[U] + m_0)^{-1}, \qquad \mathsf{G}_- = \mathsf{G}_+^*, \qquad \mathsf{L}(f) = \mathrm{Tr}(\mathsf{G}_+ \hat{f}).$$

Here $\hat{f}$ denotes multiplication by f, and the trace includes the position, spin and colour variables but no flavour variable. When a source acts on flavour, we extend $\mathsf{G}_\pm$ by the identity on that space. The resolvent bound is $\|\mathsf{G}_\pm\| \le \mathsf{m}^{-1}$. On the infinite lattice all functions in a determinant or trace have finite support; consequently the relevant operator has finite rank. Set

$$\Xi^f_\nu(z) = \mathsf{E}_\nu |\det(1 + z\mathsf{G}_+ \hat{f})|^{2N_f}. \tag{4}$$

We say that the circle bound holds at (f, ρ, C) if there is a function h, holomorphic near $|z| \le \rho$, such that $h(0) = 0$ and

$$\Xi^f_\nu(z) \le C \exp(2N_f \operatorname{Re} h(z)), \qquad |z| = \rho. \tag{5}$$

The function h absorbs the mean of the loop. Its size need not be uniform in K.

Theorem B (Circle bound)**.** There are constants C_{mix} and $C'_{\mathscr{C}}$, independent of the admitted data, for which the following holds. Put

$$\rho_* = 2\varrho_J \le \tfrac{1}{2} m_-, \qquad \rho_* \le \frac{r_M}{2L}, \qquad c_V(g) = C + 2 + C_P(1 + \log(L(g^{-2} + B_\sharp))).$$

The constants on the right are those of the partition bounds in [QCDM1, Props. 10.14 and 10.17].

(1) On a torus of side ℓ, every bounded real f satisfies the circle bound with

$$\rho = \rho_*/\|f\|_\infty, \qquad C = \exp((c_V(g) + C_{\mathrm{mix}})\ell^3).$$

The assertion for $f = 0$ is understood without the quotient. One can take

$$N_f h_K(z) = E_K(0, -e_+ z f 1) - E_K(0, 0), \tag{6}$$

which is holomorphic on $|z|\|f\|_\infty < r_M/L$.

(2) For $R > 0$, let $i(R)$ be the least integer $i \ge 1$ with $L^i \ge 2R + 3$, and put

$$C(R) = \exp\left((c_V(g) + C_{\mathrm{mix}})L^{3i(R)} + 2C'_{\mathscr{C}}(1 + R^3)\right). \tag{7}$$

If f is real, $\|f\|_\infty \le 1$, and its support lies in a ball of radius R, the circle bound holds at $(f, \rho_*, C(R))$ on every admitted torus of side $\ell \ge 2L^{i(R)}$ and for every $\nu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$.

(3) For any such measure and any real u supported in that ball,

$$\mathrm{Var}_\nu \mathsf{L}(u) \le \frac{\|u\|^2_\infty}{\rho_*^2} \min\left\{ C(R)^{1/N_f}, \frac{C(R) - 1}{N_f^2} \right\}. \tag{8}$$

The same assertion on a fixed torus holds with $C(R)$ replaced by $\exp((c_V(g) + C_{\mathrm{mix}})\ell^3)$.

■□

The estimate concerns random variables of the gauge field. It does not identify the Grassmann bilinear with a random distribution. The relation between bilinear correlations and conditional loops is supplied by the conditional Gaussian identities.

3.1. **Coefficients of a determinant.**

Proposition 3.1. If the circle bound holds at (f, ρ, C), then

$$\mathsf{E}_\nu |\mathsf{L}(f) - h'(0)|^{2N_f} \le \rho^{-2N_f} C, \tag{9}$$

$$\mathsf{E}_\nu |\mathsf{L}(f) - h'(0)|^{2} \le \frac{C-1}{N_f^2 \rho^2}. \tag{10}$$

In particular,

$$\operatorname{Var}_\nu \mathsf{L}(f) \le \rho^{-2} \min\{C^{1/N_f}, (C-1)/N_f^2\}.$$

More generally, if Z_U is a random holomorphic polynomial with $Z_U(0) = 1$, and $\mathcal{F}$ is holomorphic near the closed disc with $\mathcal{F}(0) = 0$, write $e^{-\mathcal{F}(z)} Z_U(z) = \sum_{n\ge 0} d_n(U) z^n$. Then

$$\sum_{n\ge 0} \rho^{2n} \mathsf{E}_\nu |d_n|^2 = \frac{1}{2\pi} \int_0^{2\pi} e^{-2\operatorname{Re}\mathcal{F}(\rho e^{i\theta})} \mathsf{E}_\nu |Z_U(\rho e^{i\theta})|^2 \, d\theta. \tag{11}$$

■

Proof. Parseval's identity holds at each U; Tonelli's theorem permits integration of its nonnegative terms. Apply it first to $Z_U(z) = \det(1 + z\mathsf{G}_+ \hat{f})^{N_f}$ and $\mathcal{F} = N_f h$. The first two coefficients are $d_0 = 1$ and $d_1 = N_f(\mathsf{L}(f) - h'(0))$. This gives (10).

For the higher moment, use the first coefficient of $e^{-h(z)} \det(1 + z\mathsf{G}_+ \hat{f})$ instead. Cauchy's formula and Jensen's inequality give

$$|\mathsf{L}(f) - h'(0)|^{2N_f} \le \rho^{-2N_f} \frac{1}{2\pi} \int_0^{2\pi} |e^{-h(z)} \det(1 + z\mathsf{G}_+ \hat{f})|^{2N_f} \, d\theta, \qquad z = \rho e^{i\theta}.$$

Integrate over ν. Finally the mean minimizes the squared distance from a constant, and the L^{2N_f} bound implies the L^2 bound with constant C^{1/N_f}. □

For the passage to an infinite-lattice state it is preferable to avoid extracting a convergent family of the functions h. The next observation expresses their existence in terms of the determinant alone.

Lemma 3.2. Let $\rho \|f\|_\infty < \mathsf{m}$. The function $\log \Xi_\nu^f$ is subharmonic on $|z|\|f\|_\infty < \mathsf{m}$, and the circle bound holds at (f, ρ, C) if and only if

$$\frac{1}{2\pi} \int_0^{2\pi} \log \Xi_\nu^f(\rho e^{i\theta}) \, d\theta \le \log C. \tag{12}$$

■

Proof. Put $Z_U(z) = \det(1 + z\mathsf{G}_+ \hat{f})^{N_f}$. The resolvent bound makes this determinant nonzero in the stated disc. Tilting ν by $|Z_U(z)|^2 / \Xi_\nu^f(z)$ gives a probability measure ν_z, and differentiation gives

$$\partial_z \partial_{\bar z} \log \Xi_\nu^f(z) = \mathsf{E}_{\nu_z} |\partial_z \log Z_U(z)|^2 - |\mathsf{E}_{\nu_z} \partial_z \log Z_U(z)|^2 \ge 0.$$

The circle bound implies (12) by the mean value property for $\operatorname{Re} h$. Conversely the boundary function $\log \Xi_\nu^f$ is real analytic. Let H be holomorphic near the closed disc, with real part equal to this boundary function and $H(0)$ equal to its real circle mean M. Such an H is obtained by its Fourier series. Then $h = (H - M)/(2N_f)$ gives the circle bound with $C = e^M$. □

3.2. **The two parts of the vacuum energy.** Write $e_\pm = (1 \pm \tau_3)/2$. A split source is a pair $J_1 = e_+ J_1 e_+$, $J_2 = e_- J_2 e_-$. For a function F of the source set

$$D^{\mathrm{mix}} F(J_1, J_2) = F(J_1 + J_2) - F(J_1) - F(J_2) + F(0). \tag{13}$$

The vacuum energy $E_K(0, J)$ is the energy defined by the source construction in [QCDM1, Prop. 10.6]; put $E_K^{\mathrm{mix}} = D^{\mathrm{mix}} E_K(0, \cdot)$.

Proposition 3.3 (Mixed vacuum energy)**.** For split sources with $\|J_b\|_\infty \le \rho_*$,

$$|E_K^{\mathrm{mix}}(J_1, J_2)| \le C_{\mathrm{mix}} \ell^3. \tag{14}$$

More precisely, set

$$\lambda_j = L^{j-K}, \qquad n_j = \ell^3 \lambda_j^{-3}, \qquad \mathfrak{l}_k = \mathsf{t}_k^{-1/2} (\log \mathsf{t}_k)^{-p_0}, \qquad \mathsf{t}_k = L^{K-k} \mathsf{b},$$

and retain $\phi_k = \phi(x_k)$ from the source construction. The energy contribution at level j satisfies, for $\lambda_j \|J_b\|_\infty < r_M$,

$$\left| \sum_X D^{\mathrm{mix}} e_j(X; 0, \cdot)(J_1, J_2) \right| \le C_{\mathrm{gm3}} M^{-3} n_j \left(\phi_{j-1}^3 + \phi_{j-1}^2 e^{-c_{\mathrm{t}} \ell / (2\lambda_j)} + \mathfrak{l}_{j-1}^2 \right) \frac{\lambda_j^2 \|J_1\|_\infty \|J_2\|_\infty}{r_M^2}. \tag{15}$$

The bound is on the sum of the energy contributions at a level. ∎

Its proof occupies section 4. The three terms have different origins. Independent colour averages remove the product of two linear gauge vertices. The remaining terms with fermion legs have three powers of ϕ, apart from a term caused by polymers crossing a torus seam. Two scalar vertices need not cancel; their contribution is controlled by $\mathfrak{l}^2$.

Proof of Thm. B. For $J = e_+ A + e_- B$, block diagonalization gives

$$\frac{\det(D + m_0 + a\hat{J})}{\det(D + m_0)} = \det(1 + \mathsf{G}_+ \hat{A}) \det(1 + \mathsf{G}_- \hat{B}). \tag{16}$$

At $A = zf1$, $B = \bar{z} f1$, integration gives

$$\Xi_{\nu_K}^f(z) = \frac{Z[0, -e_+ z f1 - e_- \bar{z} f1]}{Z[0, 0]}.$$

The partition bounds in [QCDM1, Prop. 10.14] bound this quotient by the exponential of

$$\mathrm{Re}\big(E_K(0, -e_+ z f1 - e_- \bar{z} f1) - E_K(0, 0)\big) + c_V(g) \ell^3.$$

The two pure-source energies are complex conjugates. Their sum is $2N_f \,\mathrm{Re}\, h_K(z)$, with h_K as in (6); the mixed part is bounded by Prop. 3.3. This proves the fixed-torus assertion.

For the local assertion place the support in a cube of side $L^{i(R)}$, at distance at least one from its boundary. The chessboard estimate in [QCDM1, proof of Prop. 10.17] reduces the source quotient to the Nth root of the quotient for the reflected source on the whole board, where $\ell^3 = N L^{3i(R)}$. Reflection alternates the source z with its conjugate. Thus the upper species source has the form $z f_{\mathrm{even}} + \bar{z} f_{\mathrm{odd}}$, and the lower source is its conjugate. Its mixed energy is bounded by $C_{\mathrm{mix}} \ell^3$. The comparison of the energy of separated source copies with the sum of their energies, by [QCDM1, Prop. 10.16 and equation (46)] at separation one, costs at most $2NC'_{\mathscr{C}}(1 + R^3)$. The pure-source terms, after conjugating the odd copies, give twice the real part of a holomorphic function which vanishes at zero. Taking the Nth root gives (7).

At fixed K and fixed support the determinants are polynomials of a common finite degree. Torus convergence of their coefficients follows from the convergence of the local Gaussian functionals in [QCDM1, Prop. 2.5]; their uniform resolvent bounds also give uniform convergence on the circle. Since $\rho_* \le m_-/2$, the determinants are nonzero there and their squared absolute values have positive expectations. The logarithms of these expectations therefore converge uniformly. Pass to the limit

in (12) and apply Lem. 3.2. The variance assertion now follows from Prop. 3.1, after dividing u by its supremum norm. □

3.3. **Loops with two or more propagators.** Some longer loops require less information about the energy. A flavour cycle forces the first mixed source monomial to have degree at least four. The ordinary energy estimate is already summable at this degree.

Theorem C. Suppose $2 \le k \le N_f$, and let $f_1, \dots, f_k$ be real bounded functions on the torus. Set

$$J_+ = \sum_{i=1}^{k} f_i E_{i,i+1}, \qquad E_{k,k+1} = E_{k,1}, \qquad \lambda_{\mathsf{q}} = \operatorname{Tr}(\mathsf{G}_+ \hat f_1 \cdots \mathsf{G}_+ \hat f_k).$$

If $\rho \max_i \|f_i\|_\infty \le \rho_*$, there is a holomorphic $\mathcal{F}$, with $\mathcal{F}(0) = 0$, such that

$$\mathsf{E}_{\nu_K} |\det(1 + z\mathsf{G}_+ \hat J_+)|^2 \le e^{(c_V(g)+C_4)\ell^3} |e^{\mathcal{F}(z)}|^2, \qquad |z| \le \rho.$$

Consequently

$$\operatorname{Var}_{\nu_K} \lambda_{\mathsf{q}} \le \rho^{-2k} e^{(c_V(g)+C_4)\ell^3}.$$

If $N_f \ge 2$, f, h are real and bounded, and $\rho \|f\|_\infty \|h\|_\infty \le \rho_*^2$, the same conclusion holds for

$$\Upsilon(\zeta) = \mathsf{E}_{\nu_K} |\det(1 - \zeta \mathsf{G}_+ \hat f \mathsf{G}_- \hat h)|^2, \quad |\zeta| \le \rho,$$

with a holomorphic factor, and

$$\operatorname{Var}_{\nu_K} \operatorname{Tr}(\mathsf{G}_+ \hat f \mathsf{G}_- \hat h) \le \rho^{-2} e^{(c_V(g)+C_4)\ell^3}.$$

One can take

$$C_4 = C_E C_{\mathfrak{C}} M^{-3} (2L)^{-4} \frac{L}{L-1} (1 - (2L)^{-2})^{-2}.$$

■□

Proof. Give the k upper-species edges separate parameters z_i, and the lower-species edges separate parameters w_i. Diagonal flavour rotations leave the determinants unchanged and multiply the edge parameters by phases u_i with $\prod_i u_i = 1$. Average the energy bound over this phase group in each species. The surviving monomials are powers of $\zeta = \prod_i z_i$ and $\xi = \prod_i w_i$. The terms depending only on ζ or only on ξ form the holomorphic factor. Every remaining monomial has a positive power of each and hence at least $2k \ge 4$ source factors.

The energy coefficients in [QCDM1, Prop. 10.6] are holomorphic for $\lambda_j \|J\| < r_M$, with polymer bound $C_{\mathfrak{C}} e^{-\kappa d_j(X)}$, with $\sum_X e^{-\kappa d_j(X)} \le C_E n_j M^{-3}$. Cauchy's estimate, at source size $\rho_* \le r_M/(2L)$, bounds the mixed sum at level j by

$$C_E C_{\mathfrak{C}} M^{-3} n_j (2L)^{-4} \lambda_j^4 (1 - (2L)^{-2})^{-2}.$$

Since $\sum_{j=1}^{K} n_j \lambda_j^4 \le \ell^3 L/(L-1)$, its sum is at most $C_4 \ell^3$. The geometric mean of the source bounds gives the asserted determinant bound. The flavour trace vanishes until the cycle closes, so

$$\det(1 + z\mathsf{G}_+ \hat J_+) = 1 + (-1)^{k-1} z^k \lambda_{\mathsf{q}} + O(z^{2k}).$$

Use the coefficient of z^k in (11); centering can only decrease its squared norm.

For the second assertion use the two-flavour source

$$(zf\tau_+ + z' h \tau_-) \otimes E_{11} + (wf\tau_+ + w' h \tau_-) \otimes E_{22}.$$

The Schur determinant identity gives factors $\det(1 - zz' \mathsf{G}_+ \hat f \mathsf{G}_- \hat h)$ and its conjugate when $(w, w') = (\bar z, \bar z')$. Phase averaging leaves only powers of zz' and ww'; a mixed term again has degree at least four. For $\zeta = zz'$ choose $|z| \|f\|_\infty = |z'| \|h\|_\infty = \sqrt{|\zeta| \|f\|_\infty \|h\|_\infty}$, which stays in the source domain. The same energy sum applies. The coefficient of ζ is the negative of the stated trace, so (11) gives its variance bound. □

3.4. **Comparing two masses.** A constant source changes the mass at every site. The circle bound therefore compares the gauge measures at two masses. We first control the derivative of the determinant and the density for a short change of mass. Iterating this comparison gives bounds throughout the permitted mass interval, uniformly in the cutoff. If the same observables converge at both masses along one cutoff sequence, their limiting laws are mutually absolutely continuous; see Prop. 3.4.

Keep the chamber parameter $r = 1/2$ and the constants of sections 2 and 4. Fix $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$ and $\mathsf{b} \in I_g$. The physical torus T_ℓ remains fixed. At cutoff K, write $a = L^{-K}$ and, for $s > 0$, put

$$F_s(U) = \det(D_+[U] + as), \qquad Z_K(s) = \int e^{-(\mathsf{b}/a)A(U)} |F_s(U)|^{2N_f}\, dU, \tag{17}$$

$$\nu_{K,s}(dU) = Z_K(s)^{-1} e^{-(\mathsf{b}/a)A(U)} |F_s(U)|^{2N_f}\, dU, \tag{18}$$

$$\mathfrak{l}_s(U) = \partial_s \log |F_s(U)|^2 = 2a \operatorname{Re} \operatorname{Tr}(D_+[U] + as)^{-1}. \tag{19}$$

The determinant and trace include chambers, spin and colour, but no flavour variable. We suppress K in F_s and $\mathfrak{l}_s$, and write E_s for expectation under $\nu_{K,s}$. The resolvent bound makes F_s nonzero. At fixed K these densities are strictly positive and continuous on the compact configuration space, and depend smoothly on $s > 0$.

With ρ_* and $c_V(g)$ from Thm. B, set

$$\begin{aligned} M_g &= \exp\bigl((c_V(g) + C_{\mathrm{mix}})\ell^3\bigr), \\ \mathfrak{X}_\star &= \frac{64 M_g}{e^2 N_f^2 \rho_*^2}, \qquad \mathfrak{X}_n^\sharp = 4^n (1 + M_g)^3 \mathfrak{X}_\star \quad (n \ge 1). \end{aligned} \tag{20}$$

These constants are independent of K, of $\mathsf{b} \in I_g$ and of the masses in $[m_-, m_+]$. Their dependence on the physical volume enters through M_g.

Theorem D (Moments and changes of mass)**.** For every cutoff $K \ge 1$ the following assertions hold.

(i) Let $t \in [m_-, m_+]$, $s \in \mathbb{R}$, $|s - t| \le r < \rho_*$ and $p \ge 1/2$. There is a real constant c, independent of U and p, such that

$$\mathsf{E}_t |\mathfrak{l}_s - c|^{2p} \le \left(\frac{2}{N_f}\right)^{2p} \left(\frac{4\rho_*}{(\rho_* - r)^2}\right)^{2p} \left(\frac{p}{e}\right)^{2p} M_g. \tag{21}$$

In particular,

$$\operatorname{Var}_{\nu_{K,t}} \mathfrak{l}_t \le \mathfrak{X}_\star. \tag{22}$$

(ii) If $t \in [m_-, m_+]$, $s \in \mathbb{R}$ and $|s - t| \le \rho_*/9$, then the density $\rho_{s|t} = d\nu_{K,s}/d\nu_{K,t}$ satisfies

$$\mathsf{E}_t \rho_{s|t}^2 \le (1 + M_g)^3, \qquad \mathsf{E}_t \rho_{s|t}^{-2} \le (1 + M_g)^3. \tag{23}$$

(iii) Fix $m_- \le \mathsf{m} < \mathsf{m}' \le m_+$ and put

$$n_{\mathrm{ch}} = \left\lceil \frac{9(\mathsf{m}' - \mathsf{m})}{\rho_*} \right\rceil. \tag{24}$$

For every $s, t \in [\mathsf{m}, \mathsf{m}']$,

$$\operatorname{Var}_{\nu_{K,t}} \mathfrak{l}_s \le \mathfrak{X}_{n_{\mathrm{ch}}}^\sharp, \tag{25}$$

$$|\operatorname{Cov}_{\nu_{K,t}}(\mathfrak{l}_s, \mathfrak{l}_t)| \le \mathfrak{X}_{n_{\mathrm{ch}}}^\sharp. \tag{26}$$

For every Borel set E of configurations,

$$\nu_{K,t}(E) \le (1 + M_g)^3 \nu_{K,s}(E)^{2^{-n_{\mathrm{ch}}}}. \tag{27}$$

■□

Proof. Fix K and $t \in [m_-, m_+]$. Apply Thm. B on T_ℓ with $f = 1$, and denote its holomorphic function by h_t. Since $\rho_* \le t/2$ and $\rho_* \le r_M/(2L)$,

$$\Omega_U(z) = N_f \left[\operatorname{Tr} \log(1 + za(D_+[U] + at)^{-1}) - h_t(z) \right]$$

is holomorphic near $|z| \le \rho_*$ and vanishes at zero. The matrix logarithm is the power series at the identity. The exponential of its trace is F_{t+z}/F_t. The circle bound says

$$\mathsf{E}_t e^{2 \operatorname{Re} \Omega_U(z)} \le M_g, \qquad |z| \le \rho_*. \tag{28}$$

For interior points this follows from the maximum principle applied to $\log \Xi^1_{\nu_{K,t}} - 2N_f \operatorname{Re} h_t$, which is subharmonic by Lem. 3.2. Differentiation at real z gives

$$\mathfrak{l}_{t+z} = \frac{2}{N_f} \operatorname{Re} \Omega_U'(z) + 2 \operatorname{Re} h_t'(z). \tag{29}$$

Write

$$Y_U^+ = \frac{1}{2\pi} \int_0^{2\pi} \left(\operatorname{Re} \Omega_U(\rho_* e^{i\theta}) \right)^+ d\theta.$$

The real part of Ω_U has mean zero on the circle, so the mean of its absolute value is $2Y_U^+$. Schwarz's integral formula, with $w = \rho_* e^{i\theta}$, therefore gives

$$\Omega_U(z) = \frac{1}{2\pi} \int_0^{2\pi} \operatorname{Re} \Omega_U(w) \frac{2z}{w - z} \, d\theta,$$
$$\Omega_U'(z) = \frac{1}{2\pi} \int_0^{2\pi} \operatorname{Re} \Omega_U(w) \frac{2w}{(w - z)^2} \, d\theta.$$

Consequently, for $|z| \le r < \rho_*$,

$$|\Omega_U(z)| \le \frac{4r}{\rho_* - r} Y_U^+, \qquad |\Omega_U'(z)| \le \frac{4\rho_*}{(\rho_* - r)^2} Y_U^+. \tag{30}$$

For $p \ge 1/2$, Jensen's inequality and $(y^+)^{2p} \le (p/e)^{2p} e^{2y}$ give

$$(Y_U^+)^{2p} \le \left(\frac{p}{e} \right)^{2p} \frac{1}{2\pi} \int_0^{2\pi} e^{2 \operatorname{Re} \Omega_U(\rho_* e^{i\theta})} \, d\theta.$$

Similarly $e^{2y^+} \le 1 + e^{2y}$ and Jensen's inequality imply

$$e^{2Y_U^+} \le 1 + \frac{1}{2\pi} \int_0^{2\pi} e^{2 \operatorname{Re} \Omega_U(\rho_* e^{i\theta})} \, d\theta.$$

Integrating these nonnegative bounds and using (28), we obtain

$$\mathsf{E}_t (Y_U^+)^{2p} \le (p/e)^{2p} M_g, \qquad \mathsf{E}_t e^{2Y_U^+} \le 1 + M_g. \tag{31}$$

Take $c = 2 \operatorname{Re} h_t'(s - t)$ in (29). Equations (30) and (31) prove (21). At $s = t$, take $r = 0$ and $p = 1$ and use the minimizing property of the mean to get (22).

For (ii), put $z = s - t$. The factor $e^{2N_f \operatorname{Re} h_t(z)}$ is independent of U and cancels from the normalized density:

$$\rho_{s|t}(U) = \frac{|F_s(U)/F_t(U)|^{2N_f}}{\mathsf{E}_t |F_s/F_t|^{2N_f}} = \frac{e^{2 \operatorname{Re} \Omega_U(z)}}{\mathsf{E}_t e^{2 \operatorname{Re} \Omega_U(z)}}. \tag{32}$$

If $|z| \le \rho_*/9$, then $4|z|/(\rho_* - |z|) \le 1/2$, and hence $4|\Omega_U(z)| \le 2Y_U^+$. In particular,

$$\mathsf{E}_t e^{4|\Omega_U(z)|} \le 1 + M_g, \qquad \mathsf{E}_t e^{2|\Omega_U(z)|} \le 1 + M_g.$$

Cauchy–Schwarz, applied to $e^{|\Omega_U(z)|}$ and its reciprocal, gives

$$(1 + M_g)^{-1} \le \mathsf{E}_t e^{-2|\Omega_U(z)|} \le \mathsf{E}_t e^{2 \operatorname{Re} \Omega_U(z)} \le M_g.$$

Substitution in (32) yields

$$\mathsf{E}_t\rho_{s|t}^2 \le (1+M_g)^3, \qquad \mathsf{E}_t\rho_{s|t}^{-2} \le (1+M_g)M_g^2 \le (1+M_g)^3.$$

For (iii), first let $s \neq t$, set $n' = \lceil 9|t-s|/\rho_* \rceil \le n_{\mathrm{ch}}$ and divide the segment from s to t into n' equal parts, with endpoints $u_i = s + i(t-s)/n'$. All these masses lie in the admitted interval. For any nonnegative Borel function Y, (23) gives

$$\mathsf{E}_{u_{i+1}} Y = \mathsf{E}_{u_i}(Y\rho_{u_{i+1}|u_i}) \le (1+M_g)^{3/2}(\mathsf{E}_{u_i}Y^2)^{1/2}.$$

Iterating, the exponents $1/2, 1/4, \dots, 2^{-n'}$ sum to $1-2^{-n'}$, so

$$\mathsf{E}_t Y \le (1+M_g)^{3(1-2^{-n'})}(\mathsf{E}_s Y^{2^{n'}})^{2^{-n'}}. \tag{33}$$

With $Y = 1_E$ this proves (27). For the variance take $Y = (\mathfrak{l}_s - c_s)^2$, where c_s is the constant in (i) at the centre s, and use (i) with $r=0$ and $p = 2^{n'}$. Then

$$\begin{aligned}\mathrm{Var}_{\nu_{K,t}}\mathfrak{l}_s &\le \mathsf{E}_t Y \\ &\le (1+M_g)^3 \frac{64\,4^{n'}}{e^2 N_f^2 \rho_*^2} M_g^{2^{-n'}} \le \mathfrak{X}^\sharp_{n_{\mathrm{ch}}}.\end{aligned}$$

The last inequality uses $M_g \ge 1$. Cauchy–Schwarz and (22) give (26). If $s=t$, the variance estimate is already (i), and the event bound follows from $\nu_{K,s}(E) \le \nu_{K,s}(E)^{2^{-n_{\mathrm{ch}}}}$. □

We now pass from the cutoff measures to any pair of limits formed with the same observables. For probability measures P, Q on one measurable space, write

$$\mathrm{TV}(P,Q) = \sup_E |P(E)-Q(E)|, \qquad \mathrm{Aff}(P,Q) = \int \sqrt{pq}\, dR,$$

where R dominates P, Q and $p = dP/dR$, $q = dQ/dR$. The affinity is independent of R. We use

$$\mathrm{Ent}(P \mid Q) = \int \log \frac{dP}{dQ}\, dP$$

when $P \ll Q$, and assign the value $+\infty$ otherwise.

Proposition 3.4 (Limit laws at two masses)**.** Retain the fixed g, ℓ, b above, and choose $m_- \le \mathsf{m} < \mathsf{m}' \le m_+$. Let Y be a Polish space and let f_K be Borel maps from the cutoff configuration spaces to Y. Suppose that, along one sequence $K_j \to \infty$,

$$(f_{K_j})_*\nu_{K_j,\mathsf{m}} \implies P, \qquad (f_{K_j})_*\nu_{K_j,\mathsf{m}'} \implies Q.$$

Put $\Delta = \mathsf{m}' - \mathsf{m}$, let n_{ch} be (24), and set

$$x = \frac{N_f}{2}\sqrt{\mathfrak{X}_\star}\,\Delta = \frac{4\sqrt{M_g}}{e\rho_*}\,\Delta, \qquad H = \frac{N_f^2}{2}\mathfrak{X}^\sharp_{n_{\mathrm{ch}}}\Delta^2.$$

Then P and Q are mutually absolutely continuous, and for every Borel set $E \subset Y$,

$$Q(E) \le (1+M_g)^3 P(E)^{2^{-n_{\mathrm{ch}}}}, \qquad P(E) \le (1+M_g)^3 Q(E)^{2^{-n_{\mathrm{ch}}}}. \tag{34}$$

Moreover,

$$\mathrm{TV}(P,Q) \le x, \tag{35}$$

$$\mathrm{Ent}(P \mid Q),\ \mathrm{Ent}(Q \mid P) \le H, \tag{36}$$

$$\mathrm{Aff}(P,Q) \ge e^{-H/2}, \tag{37}$$

and $\mathrm{Aff}(P,Q) \ge \cos x$ when $x \le \pi/2$. ■

Proof. We first prove the estimates before taking the limit. At fixed K, let $p_t = d\nu_{K,t}/dU$. Differentiation of (18) gives its logarithmic derivative,

$$\partial_t \log p_t = N_f(\mathfrak{l}_t - \mathsf{E}_t\mathfrak{l}_t), \qquad \partial_t \mathsf{E}_t F = N_f \operatorname{Cov}_{\nu_{K,t}}(F, \mathfrak{l}_t) \tag{38}$$

for a bounded F independent of t. Every differentiation here is justified on a compact positive mass interval by the resolvent bound and compactness of the configuration space.

The functions $p_t^{1/2}$ form a continuously differentiable curve on the unit sphere of $L^2(dU)$, with speed

$$\|\partial_t p_t^{1/2}\|_2 = \frac{N_f}{2}\sqrt{\operatorname{Var}_{\nu_{K,t}} \mathfrak{l}_t}.$$

The angle between its endpoints is at most its length. Their inner product is the affinity, so

$$\arccos \operatorname{Aff}(\nu_{K,\mathsf{m}}, \nu_{K,\mathsf{m}'}) \le \frac{N_f}{2}\int_{\mathsf{m}}^{\mathsf{m}'} \sqrt{\operatorname{Var}_{\nu_{K,t}} \mathfrak{l}_t}\, dt \le x. \tag{39}$$

For any two probability densities, Cauchy–Schwarz gives

$$\operatorname{TV}(P,Q) = \frac{1}{2}\int |\sqrt{p} - \sqrt{q}|(\sqrt{p} + \sqrt{q})\, dR \le \sqrt{1 - \operatorname{Aff}(P,Q)^2} \le \arccos \operatorname{Aff}(P,Q).$$

Thus the total variation at the cutoff is at most x as well.

For entropy, integration of the first identity in (38) gives

$$\log p_{\mathsf{m}} - \log p_{\mathsf{m}'} = N_f \int_{\mathsf{m}}^{\mathsf{m}'} (\mathsf{E}_s\mathfrak{l}_s - \mathfrak{l}_s)\, ds.$$

Taking its expectation at m and applying the second identity in (38), with s held fixed, yields

$$\operatorname{Ent}(\nu_{K,\mathsf{m}} \mid \nu_{K,\mathsf{m}'}) = N_f^2 \int_{\mathsf{m}}^{\mathsf{m}'} ds \int_{\mathsf{m}}^{s} dt\ \operatorname{Cov}_{\nu_{K,t}}(\mathfrak{l}_s, \mathfrak{l}_t), \tag{40}$$

$$\operatorname{Ent}(\nu_{K,\mathsf{m}'} \mid \nu_{K,\mathsf{m}}) = N_f^2 \int_{\mathsf{m}}^{\mathsf{m}'} ds \int_{s}^{\mathsf{m}'} dt\ \operatorname{Cov}_{\nu_{K,t}}(\mathfrak{l}_s, \mathfrak{l}_t). \tag{41}$$

Each triangle has area $\Delta^2/2$. Equation (26) therefore bounds both entropies by H.

We recall the measure-theoretic facts needed to preserve these estimates. Pushforward by a Borel map decreases total variation and entropy and increases affinity. For total variation the supremum is over fewer sets. For affinity, take $R = (P+Q)/2$ and condition the densities p, q on the map: conditional Cauchy–Schwarz gives

$$\mathsf{E}_R(\sqrt{pq} \mid f) \le \sqrt{\mathsf{E}_R(p \mid f)\mathsf{E}_R(q \mid f)}.$$

For entropy use the variational formula

$$\operatorname{Ent}(P \mid Q) = \sup_v \left(\int v\, dP - \log \int e^v\, dQ \right). \tag{42}$$

On a Polish space the supremum may be taken either over bounded Borel functions or over bounded continuous functions; see [DM22, equation (1.1) and the following paragraph]. Using $v \circ f$ in the Borel version proves the pushforward inequality. The continuous version makes entropy lower semicontinuous under simultaneous weak convergence. The same is true for total variation, since

$$\operatorname{TV}(P,Q) = \frac{1}{2} \sup_{\substack{v \in C_b(Y) \\ \|v\|_\infty \le 1}} \left| \int v\, dP - \int v\, dQ \right|.$$

Affinity is upper semicontinuous under that convergence. Indeed,

$$\operatorname{Aff}(P,Q) = \inf_{\substack{v \in C_b(Y) \\ \inf v > 0}} \left(\int v\, dP \right)^{1/2} \left(\int v^{-1}\, dQ \right)^{1/2}. \tag{43}$$

Cauchy–Schwarz proves one inequality. For the reverse, truncate $\sqrt{q/p}$ to $[\epsilon,\epsilon^{-1}]$, taking the value ϵ^{-1} where $p=0$. If the result is v_ϵ, then

$$\int v_\epsilon\,dP \le \mathrm{Aff}(P,Q)+2\epsilon, \qquad \int v_\epsilon^{-1}\,dQ \le \mathrm{Aff}(P,Q)+2\epsilon.$$

Approximate v_ϵ in $L^1(P+Q)$ by bounded continuous functions with values in the same interval, using Lusin's theorem and extension from compact sets. Their reciprocals converge in $L^1(P+Q)$ too. Letting ϵ decrease to zero proves (43). Its right-hand side is an infimum of continuous functions of (P,Q), as required.

Apply these pushforward and semicontinuity statements to

$$P_j=(f_{K_j})_*\nu_{K_j,\mathsf{m}}, \qquad Q_j=(f_{K_j})_*\nu_{K_j,\mathsf{m}'}.$$

The cutoff estimates give (35), (36), and the cosine lower bound. Finite entropy in both orders gives mutual absolute continuity. Jensen's inequality then gives

$$-\log\mathrm{Aff}(P,Q)=-\log\int\left(\frac{dQ}{dP}\right)^{1/2}dP\le\frac12\,\mathrm{Ent}(P\mid Q)\le H/2,$$

which proves (37).

It remains to retain the event comparison. By (27), with preimages under f_{K_j}, every Borel $E\subset Y$ satisfies

$$Q_j(E)\le(1+M_g)^3P_j(E)^\alpha, \qquad \alpha=2^{-n_{\mathrm{ch}}}.$$

For a nonempty closed set $F\subset Y$, choose a compatible metric and put $G_\delta=\{y:\mathrm{dist}(y,F)<\delta\}$ and $F_\delta=\{y:\mathrm{dist}(y,F)\le\delta\}$. The weak-convergence inequalities for open and closed sets give

$$\begin{aligned}Q(F)&\le Q(G_\delta)\le\liminf_j Q_j(G_\delta)\\&\le(1+M_g)^3\limsup_j P_j(F_\delta)^\alpha\le(1+M_g)^3P(F_\delta)^\alpha.\end{aligned}$$

As $\delta\downarrow0$, $P(F_\delta)\downarrow P(F)$. Inner regularity extends the inequality from closed sets to all Borel sets. Interchanging the masses proves the other inequality in (34). □

4. The mixed source energy

The ordinary source estimate gives one small factor for each source. Applied to the mixed energy, these two factors leave a nonsummable power of the scale. We need additional smallness in the coefficient of the two sources. We prove it in two stages. First, at zero gauge source, the activities with fermion legs are quadratic in ϕ_k. Second, at the trivial gauge configuration, independent colour averages in the two species remove the mixed terms of weighted degree two which have fermion legs. The scalar term left at this degree is bounded by $\mathfrak{l}_k^2$.

We recall the scale quantities of [QCDM1, §4]:

$$\lambda_j=L^{j-K}, \qquad \mathsf{t}_j=\lambda_j^{-1}\mathsf{b}, \qquad x_0=L^K\mathsf{b}, \quad x_{j+1}=L^{-1}x_j-b_{j+1}, \quad |b_{j+1}|\le B_\sharp,$$
$$\phi(t)=t^{-1/2}(1+\log t)^{p_0}, \qquad \phi_j=\phi(x_j), \qquad \delta_j=A_1\mathsf{t}_j^{-1/2}(\log\mathsf{t}_j)^{p_0}.$$

The fixed exponent is $p_0\ge6r_c+1$. The running and calibration coefficients satisfy $|x_j-\mathsf{t}_j|\le\bar B$, and $\phi_i/\phi_j\le(2L^{-1/2})^{j-i}$ for $i\le j$. A level-j polymer is a connected union of blocks of physical side $M\lambda_j$; $d_j(X)$ is the least number of edges in a block tree meeting every block of X. In the formulas below δ_k is the fluctuation radius, h_F is the fixed Grassmann norm parameter, and c_{cl} is the radius on which the cluster expansion converges.

We use the norms, local extraction and source coordinates of [QCDM1, Prop. 10.1 and its proof]. The level-j norm of a polymer family is the supremum, over a level-j cube, of the sum of the Grassmann norms of all its summands containing that cube, weighted by $e^{\kappa d_j(X)}$. Summands include the local quadratic forms introduced by extraction. A pointed family is obtained by retaining a

specified fermion leg or specified scalar bond in the localization; its norm uses the same weights. The constant K_μ^{s} bounds the increase of this norm under local mass extraction.

A regular pair $(\mathbf{U}, \mathbf{J})$ consists of the complex connection and gauge-current coordinate in the analytic core of [QCDM1, §4]. This domain includes the bounds for the local gauge potentials and the reduced block towers, as well as those for curvature and current. We use radius α_0 at every level. The gauge-current coordinate $\mathbf{J}$ is distinct from the physical fermion source J below; the trivial pair is $(1, 0)$. There is no gauge source. The doubled coordinate domain is

$$2\mathscr{D}_k^\circ = \{\mathfrak{c} = (0, \mathsf{n}, (\sigma_i)_{i\le k}) : L^{k+1}\|\mathsf{n}\| < 2r_M, \|\sigma_i\| < 2B_\sigma\phi_i\}. \tag{44}$$

Its inner half is $\mathscr{D}_k^\circ$. The bare coordinate is $\mathsf{n} = -aJ$. The matrices σ_i record the changes in the extracted local mass at the preceding levels. At zero gauge source the coordinate distance appearing below is

$$\|\mathfrak{c} - \mathfrak{c}'\|_{k,y} = \sup_x e^{-\kappa_\sigma|x-y|/L^{k+1}} \max\left\{\frac{L^{k+1}|\mathsf{n} - \mathsf{n}'|(x)}{r_M}, \max_{i\le k}\frac{|\sigma_i - \sigma_i'|(x)}{B_\sigma\phi_i}\right\},$$

where x, y are in fine lattice units. The matrices σ_i remain free in the first-order domain; the stronger estimate proved below is a conclusion, not an assumption on these coordinates.

The role of these coordinates can be seen in one step of [QCDM1, Prop. 10.1]. At level k, the remaining fields have a quadratic fermion form and local interaction activities. Completing the quadratic form after insertion of the block-average kernel gives, on the variables being integrated,

$$\Psi_k = H_k\Psi_{k+1} + W, \qquad \bar\Psi_k = \bar\Psi_{k+1}\bar H_k + \bar W.$$

Here H_k extends the prescribed coarse field to its fermion background, $\bar H_k$ is its row twin, and $W, \bar W$ are integrated with covariance Γ_k^{-1}. The row twin need not be the adjoint. On a history, this notation suppresses the stored variables, which remain unchanged. Substituting these expressions into the old activities expresses their fermion legs in terms of the new background and the fluctuation. The gauge fluctuation is denoted by B. The background pair $(1, 0)$ fixes the background connection and current; setting $B = 0$ is a further evaluation, not part of the definition of that pair.

The connected cluster expansion for the joined gauge and Berezin integration produces activities $\widetilde{E}^{(k+1)}(X; \mathfrak{c})$ on level-$(k+1)$ polymers. Each is an even polynomial in the remaining fermion fields, with coefficients depending on the background pair and the coordinates $\mathfrak{c}$. Its terms of degree zero contribute to the scalar energy. More precisely, let $\mathfrak{E}_{k+1}(X; \mathfrak{c})$ be their value at the trivial pair, including the explicit normalizer and Gaussian terms. The source-dependent constant removed from the activity is

$$e_{k+1}(X; \mathfrak{c}) = \mathfrak{E}_{k+1}(X; \mathfrak{c}) - \mathfrak{E}_{k+1}(X; 0).$$

Its sum contributes to the source energy. Terms of degree zero may still depend on the gauge background after this constant has been removed.

The quadratic part has a matrix kernel. Its symmetric zeroth moment determines the local mass increment $\nu_{k+1}(\mathfrak{c})$, with $\mu_{k+1} = \nu_{k+1}(0)$ the zero-source increment. The extraction rewrites this local quadratic contribution as a mass term and retains the remaining kernel and its localized correction forms in the activities. This is why the activity norms below also include those forms. For a physical source J, the mass change relative to zero source at level k is $-\lambda_k J + \sigma_k(J)$. The corrections are determined recursively by

$$\sigma_0(J) = 0, \qquad \sigma_{k+1}(J) = L\sigma_k(J) + \nu_{k+1}(\mathfrak{c}_k(J)) - \mu_{k+1}, \qquad \mathfrak{c}_k(J) = (0, -aJ, (\sigma_i(J))_{i\le k}).$$

The factor L rescales the preceding mass correction. The two extractions therefore give the outputs needed here: a scalar source energy and a new local mass coordinate, together with the activities carried to the next step.

We first estimate the step with n and the σ_i independent in (44). We then substitute $\mathfrak{c}_k(J)$, as in [QCDM1, Lem. 10.4]. For split sources, restrict an expression by zeroing one species' entries while holding the other entries fixed; for activities with fermion legs, zero that species' generators as well.

The mixed part is the full expression minus its two one-species restrictions, plus its restriction with both species removed. Turning off one physical source instead recomputes the entire chain: the other species' corrections σ_i can change as well. The later comparison of source chains estimates this additional change. Thus a bound for the mixed coordinate expression is an input to the bound for the mixed physical-source energy.

4.1. **Quadratic bounds at zero gauge source.**

Proposition 4.1. There are constants C_2^{s}, $B_\nu^{(2)}$ and $B_\sigma^{(2)}$, chosen independently of the admitted data, such that the following holds after decreasing γ. For $1 \le j \le K$, the new activity with fermion legs, including its extracted summands and every pointed family, has norm at most $C_2^{\mathrm{s}}\phi_{j-1}^2$ throughout $2\mathscr{D}_{j-1}^\circ$. The increment satisfies

$$|\nu_{k+1}(y;\mathfrak{c}) - \nu_{k+1}(y;\mathfrak{c}')| \le B_\nu^{(2)}\phi_k^2 \|\mathfrak{c} - \mathfrak{c}'\|_{k,y}, \tag{45}$$

for $\mathfrak{c} \in \mathscr{D}_k^\circ$ and $\mathfrak{c}' \in 2\mathscr{D}_k^\circ$. The norm on the right is the weighted coordinate norm of [QCDM1, Prop. 10.1]. Along the source chain,

$$\|\sigma_j(0,J)\| \le B_\sigma^{(2)}\phi_{j-1}^2 \frac{\lambda_j\|J\|}{r_M}, \qquad B_\sigma^{(2)} = \frac{B_\nu^{(2)}}{1-4/L}. \tag{46}$$

■

Proof. We give the additional estimates needed in the proof of [QCDM1, Prop. 10.1]. Write the potential as the sum of the scalar term V_0, the gauge–fermion vertex V_F, the old activity V_{old} with fermion legs, and the quadratic source term V_s. On the field disc,

$$\|V_F\| \le C_V h_F^2 \mathsf{C}_{\mathcal{A}}\delta_k, \quad \|V_0\| \le K_0^{\mathrm{s}}\delta_k, \quad \|V_s\| \le c'_\psi c_{\mathsf{S}} h_F^2(\bar{B}_M + 2B_\sigma)\phi_k, \quad \delta_k \le \sqrt{2}A_1\phi_k.$$

Here one can take

$$K_0^{\mathrm{s}} = 7K_P + c_{\mathsf{s}}K_{\mathrm{old}}^{\mathrm{s}} + \frac{13}{10}\mathsf{C}_{\mathcal{A}}K_\Lambda^{\mathrm{s}}K_M c_{\mathrm{cl}}/\alpha_0,$$

using the scalar localization constants at radius α_0. The old-activity argument, including its cell sums, gives

$$\|V_{\mathrm{old}}\| \le c'_\psi\Sigma_k, \qquad \Sigma_k = (1+c_{\mathrm{cs}})\sum_{i\le k}\omega_{k+1-i}\epsilon_i + C_{0,\sharp}^{\mathrm{cs}}h_F^2\phi_k^2,$$

where ϵ_i includes the extraction factor K_μ^{s} and the zero-source bound $2C_2\phi_{i-1}^2$. The additional term in $C_{0,\sharp}^{\mathrm{cs}} = C_0^{\mathrm{cs}} + 3L^{1/2}C_{\mathrm{pc}}S_{1/4}c_{\mathsf{S}}\bar{B}_M$ uses $|\mu_i| \le 3\bar{B}_M\phi_i^2$ in the cell calculation. Here μ_i is the zero-source mass increment. Its local mass forms cost $c_{\mathsf{S}}h_F^2|\mu_i|$ before they are read at a later level. Their sum is bounded by

$$\sum_{i\le k}\omega_{k+1-i}c_{\mathsf{S}}h_F^2|\mu_i| \le 3c_{\mathsf{S}}\bar{B}_M h_F^2\phi_k\sum_{i\le k}\omega_{k+1-i}\phi_i \le 3L^{1/2}C_{\mathrm{pc}}S_{1/4}c_{\mathsf{S}}\bar{B}_M h_F^2\phi_k^2.$$

The change of a cell moment under the bare matrix mass has an additional factor $L^i\|\mathsf{n}\|/r_M$. Reading that cell at level k costs L^{k-i}, and

$$L^{k-i}\frac{L^i\|\mathsf{n}\|}{r_M} < \frac{2}{L} \quad \text{on } 2\mathscr{D}_k^\circ.$$

Thus its sum is controlled by $h_F^2\sum_{i\le k}\phi_i^2$, which is at most $h_F^2\phi_k^2/(1-2L^{-1/2})$. This is the term included in C_0^{cs}. The other cell terms are linear in the old activity norms ϵ_i and give the factor $1+c_{\mathrm{cs}}$. These are the reasons the old-activity estimate has the quadratic remainder $C_{0,\sharp}^{\mathrm{cs}}h_F^2\phi_k^2$.

Introduce the four-variable potential

$$V(z) = z_3V_0(B) + V_F(z_1B) + z_2V_{\mathrm{old}}(z_1B) + z_4V_s(z_1B).$$

Take the radii

$$r_1 = \frac{\min\{c_{\mathrm{cl}}/(8C_V h_F^2), \beta_0/2\}}{\mathsf{C}_{\mathcal{A}}\delta_k},$$
$$r_2 = \frac{c_{\mathrm{cl}}}{8c'_\psi \Sigma_k}, \qquad r_3 = \frac{c_{\mathrm{cl}}}{8K_0^{\mathrm{s}}\delta_k}, \qquad r_4 = \frac{c_{\mathrm{cl}}}{8c'_\psi c_{\mathsf{S}} h_F^2(\bar{B}_M + 2B_\sigma)\phi_k}.$$

Each of the four terms has norm at most $c_{\mathrm{cl}}/8$ on this polydisc. In particular, scaling B by z_1 keeps the fine field in its analytic domain; the extra bound $\beta_0/2$ in r_1 also keeps the changes of the background legs within their Cauchy radius. The cluster expansion of [QCDM1, Lem. 7.5] therefore gives a holomorphic family

$$\widetilde{E}_{\geq 2}(X; z) = \sum_{n\in\mathbb{N}^4} a_n(X) z^n, \qquad \|a_n(X)\| \leq K'_M c_{\mathrm{cl}} \prod_{i=1}^{4} r_i^{-n_i} e^{-\kappa_+ d_{k+1}(X)}.$$

The coefficients take values in a finite Grassmann algebra, so Cauchy's estimate applies in its norm. After summing polymers, K'_M is replaced by K_M.

Assign weights $1, 2, 1, 1$ to z_1, z_2, z_3, z_4. We examine the coefficients before estimating their sum. When $z_1 = z_2 = z_4 = 0$ the potential is scalar, so every coefficient involving only z_3 has Grassmann degree zero. When $z_2 = z_3 = z_4 = 0$, the substitution $B \mapsto -B$ replaces z_1 by $-z_1$. The zero-source gauge Gaussian, its tests and its local quadratic term are even in B; the decoupled fermion covariance is independent of B. The cluster construction respects this symmetry polymer by polymer. Its coefficient of z_1 is therefore zero.

For the coefficient linear in z_4, put $z_1 = z_2 = 0$. Let $\psi^{\mathrm{bg}} = H_{0,k+1}\Psi_{k+1}$ and $\mathsf{N} = s_k + \sigma_k$, where s_k is the zero-source mass coordinate. At $B = 0$ the source vertex is

$$V_s(0) = -L^{-k}\left[\mathsf{S}_1(\mathsf{N}(\psi^{\mathrm{bg}} + H_{0,k}W)) - \mathsf{S}_1(\mathsf{N}\psi^{\mathrm{bg}})\right].$$

Here $\mathsf{S}_1(\mathsf{N}\psi)$ denotes the quadratic mass form with coefficient N. Expanding this difference gives two terms linear in the fluctuation generators and one term quadratic in those generators. The linear terms have zero Berezin mean; the quadratic term is contracted to a scalar. At any power of z_3, a coefficient linear in z_4 has exactly one such vertex, independent of B, beside scalar vertices. Its refined Gaussian integral thus has degree zero, and the subsequent cluster operations preserve that degree. This proves that every coefficient of weighted degree at most one has no part with fermion legs.

The coefficient of z_2 has weight two and must be kept separately: it is the first variation of the old family at $B = 0$. For an even potential V with legs, the cluster activity at vV is holomorphic for $|v| < c_{\mathrm{cl}}/\|V\|$ and has bound $(17/16)K'_M c_{\mathrm{cl}} e^{-\kappa_+ d_{k+1}(X)}$. Cauchy's estimate at $v = 0$ bounds its first variation by $(17/16)K'_M\|V\|e^{-\kappa_+ d_{k+1}(X)}$. At $V = V_{\mathrm{old}}(0)$ this yields $(17/16)K_M c'_\psi \Sigma_k$ in the polymer norm.

Choose C_r^{s} so that $r_1^{-1}, r_3^{-1}, r_4^{-1} \leq x = C_r^{\mathrm{s}}\phi_k$, and put $y = r_2^{-1} = 8c'_\psi\Sigma_k/c_{\mathrm{cl}}$. For $n_2 = 0$, all remaining coefficients have $n_1 + n_3 + n_4 \geq 2$. Their Cauchy factors sum to

$$(1-x)^{-3} - 1 - 3x \leq 22x^2, \qquad 0 \leq x \leq \tfrac{1}{2}.$$

The quotient by x^2 has nonnegative coefficients and takes the value 22 at $x = 1/2$. For $n_2 \geq 1$, remove the first-variation coefficient already treated. The remaining sum is at most

$$y((1-x)^{-3}(1-y)^{-1} - 1) \leq y(28x + 2y), \qquad 0 \leq x, y \leq \tfrac{1}{2},$$

using $(1-x)^{-3} - 1 \leq 14x$ and $(1-y)^{-1} \leq 2$. Together these give

$$\|\widetilde{E}_{\geq 2}^{(k+1)}\|_{k+1} \leq \tfrac{17}{16}K_M c'_\psi \Sigma_k + K_M c_{\mathrm{cl}}(22x^2 + 28xy + 2y^2). \tag{47}$$

For a pointed family, the insertion $(V_\psi)_z$ multiplies each monomial by a number in $[0, 1]$ and therefore preserves its degree and does not increase its norm. Replace the part with legs by $tV_\psi + \zeta(V_\psi)_z$, for

$|t| < 9/8$ and $|\zeta| < 1$. The pointed activity is

$$\int_0^1 \partial_\zeta \widetilde{E}_{\geq 2}[z_3 V_0, tV_\psi + \zeta (V_\psi)_z]\Big|_{\zeta=0}\, dt.$$

The scalar part costs $c_{\mathrm{cl}}/8$, and the part with legs costs at most $(9/8+1)3c_{\mathrm{cl}}/8$; their sum is $59c_{\mathrm{cl}}/64$. Cauchy's estimate in ζ gives the same Taylor coefficient bound as above. The three vanishing-coefficient arguments still hold because pointing retains the monomials and commutes with scaling B. Thus (47) holds at every pointer, with the same constants.

For completeness, the constants can be chosen in the following order. Put

$$R_a = 8c'_\psi(1+c_{\mathrm{cs}})C_{\mathrm{pc}}S_{1/4}/c_{\mathrm{cl}}, \qquad R_b = 8c'_\psi C^{\mathrm{cs}}_{0,\sharp}h_F^2/c_{\mathrm{cl}},$$

$$C^{\mathrm{s}}_{(2)} = \theta^{\mathrm{s}}_E C^{\mathrm{cs}}_{0,\sharp}h_F^2 + K^{\mathrm{s}}_\mu K_M c_{\mathrm{cl}}(22(C^{\mathrm{s}}_r)^2 + 15R_b), \qquad C^{\mathrm{s}}_2 = \max\{2C^{\mathrm{s}}_{(2)}, 2C_2\}.$$

The source norm parameter h_F is fixed with $4\theta^{\mathrm{s}}_E C_{\mathrm{pc}}S_{1/4}(1+c_{\mathrm{cs}}) \leq 1/2$, as permitted in [QCDM1, Prop. 10.20]. Decrease γ so that, with $\phi_* = \phi(\gamma^{-2}/2)$,

$$C^{\mathrm{s}}_r\phi_* \leq \tfrac{1}{2}, \qquad (R_aC^{\mathrm{s}}_2 + R_b)\phi_* \leq C^{\mathrm{s}}_r, \qquad 240K^{\mathrm{s}}_\mu K_M c_{\mathrm{cl}}C^{\mathrm{s}}_r R_a\phi_* \leq 1.$$

To close the induction, suppose $\epsilon_i \leq C^{\mathrm{s}}_2\phi_{i-1}^2$ for $i \leq k$. The old-family sum then satisfies

$$\sum_{i\leq k}\omega_{k+1-i}\epsilon_i \leq C^{\mathrm{s}}_2\phi_k \sum_{i\leq k}\omega_{k+1-i}\phi_{i-1} \leq C_{\mathrm{pc}}S_{1/4}C^{\mathrm{s}}_2\phi_k^2.$$

Consequently $y \leq (R_aC^{\mathrm{s}}_2 + R_b)\phi_k^2 \leq x \leq 1/2$. After extraction the linear old term is bounded by

$$\tfrac{17}{16}K^{\mathrm{s}}_\mu K_M c'_\psi \Sigma_k \leq \tfrac{1}{8}C^{\mathrm{s}}_2\phi_k^2 + \theta^{\mathrm{s}}_E C^{\mathrm{cs}}_{0,\sharp}h_F^2\phi_k^2.$$

For the terms involving y, use $28x + 2y \leq 30C^{\mathrm{s}}_r\phi_k$. The third smallness condition and $C^{\mathrm{s}}_r\phi_k \leq 1/2$ give

$$K^{\mathrm{s}}_\mu K_M c_{\mathrm{cl}} y(28x+2y) \leq \tfrac{1}{8}C^{\mathrm{s}}_2\phi_k^2 + 15K^{\mathrm{s}}_\mu K_M c_{\mathrm{cl}}R_b\phi_k^2.$$

Adding the $22x^2$ term leaves $(C^{\mathrm{s}}_2/4 + C^{\mathrm{s}}_{(2)})\phi_k^2 \leq 3C^{\mathrm{s}}_2\phi_k^2/4$. The other term in the definition of ϵ_{k+1} is $2C_2\phi_k^2 \leq C^{\mathrm{s}}_2\phi_k^2$. This closes the induction for the activity and every pointed family at once.

The increment is the local zeroth moment of the quadratic kernel. We describe how its coordinate estimate uses the improved norm. A term f in the moment at y comes from a polymer X and a walk ω. It is holomorphic on $2\mathscr{D}_k^\circ$ and bounded by M_0, the product of its polymer bound and the row-sum bound of the walk. Restrict the bare-coordinate difference to $X \cup \omega$ and each σ_i difference to X; call the resulting coordinate difference Δ. Let T be the maximum of the normalized absolute differences on these sets. The value of f depends only on these entries, so $f(\mathfrak{c}') = f(\mathfrak{c}+\Delta)$. Since $\mathfrak{c}$ is in the inner domain, the line $\mathfrak{c} + \zeta\Delta/T$, $|\zeta| < 1$, lies in the doubled domain. Schwarz's lemma gives

$$|f(\mathfrak{c}') - f(\mathfrak{c})| \leq 2M_0T;$$

for $T > 1$ this follows directly from the two bounds by M_0. Summing these inequalities is exactly the kernel and walk sum in [QCDM1, Prop. 10.1]: the reserved factor $e^{-\kappa d_{k+1}(X)/4}$ absorbs the polynomial $1+d_{k+1}(X)$, and $e^{-|\omega|/4}$ absorbs the distance in the weighted coordinate norm. Replacing the polymer size there by $C^{\mathrm{s}}_2\phi_k^2$ proves (45) with $B^{(2)}_\nu = 2c_{\mathrm{inc}}h_F^{-2}C^{\mathrm{s}}_2$.

To evaluate this estimate on the source chain, we first check its domain by induction using the first-order increment estimate of [QCDM1, Prop. 10.1 and Lem. 10.4]. The inductive bound is

$$\|\sigma_i(0,J)\| \leq B_\nu S_\uparrow \phi_i \frac{\lambda_i\|J\|}{r_M}, \qquad B_\nu S_\uparrow \leq LB_\sigma,$$

where $S_\uparrow$ bounds $\sum_{i\leq j}\phi_i/\phi_j$. It holds at level zero, where $\sigma_0 = 0$. On $\lambda_j\|J\| < 2r_M$, every earlier entry has $\lambda_i\|J\| < 2r_M/L$ and is therefore inside the doubled coordinate domain. Moreover, the

normalized coordinate distance at step j is at most $\lambda_j\|J\|/r_M$: this is its bare-coordinate entry, and for $i<j$,

$$\frac{\|\sigma_i(0,J)\|}{B_\sigma\phi_i}\le L\frac{\lambda_i\|J\|}{r_M}\le\frac{\lambda_j\|J\|}{r_M}.$$

The ordinary increment estimate is applicable with one point $0\in\mathscr{D}^\circ_{j-1}$ and the other $\mathfrak{c}_{j-1}(0,J)\in 2\mathscr{D}^\circ_{j-1}$. It gives

$$\|\nu_j(\mathfrak{c}_{j-1}(0,J))-\mu_j\|\le B_\nu\phi_j\frac{\lambda_j\|J\|}{r_M}.$$

Summing this in the recurrence preserves the inductive bound because $L^{j-i}\lambda_i=\lambda_j$ and $\sum_{i\le j}\phi_i\le S_\uparrow\phi_j$. The next increment is therefore defined, and holomorphy follows by composition. The stronger estimate is applied to a chain whose existence and domain have already been established. Since $\nu_j(0)=\mu_j$, it gives

$$\|\nu_j(\mathfrak{c}_{j-1}(0,J))-\mu_j\|\le B^{(2)}_\nu\phi^2_{j-1}\frac{\lambda_j\|J\|}{r_M}.$$

Finally, iterate $\sigma_j=L\sigma_{j-1}+\nu_j-\mu_j$ from $\sigma_0=0$. For $1\le i\le j$,

$$L^{j-i}\lambda_i\phi^2_{i-1}=\lambda_j\phi^2_{i-1}\le\lambda_j\phi^2_{j-1}(4/L)^{j-i}.$$

It follows that

$$\|\sigma_j(0,J)\|\le B^{(2)}_\nu\frac{\lambda_j\|J\|}{r_M}\sum_{i=1}^{j}\phi^2_{i-1}\le\frac{B^{(2)}_\nu}{1-4/L}\phi^2_{j-1}\frac{\lambda_j\|J\|}{r_M},$$

which is (46). □

4.2. **Separation of the species.** Restrict to split coordinates in (44). Let R_b set both the coordinates and the Grassmann generators of species b to zero. Define

$$R^{(0)}=R_1R_2,\quad R^{(1)}=(1-R_1)R_2,\quad R^{(2)}=R_1(1-R_2),\quad R^{\mathrm{mix}}=(1-R_1)(1-R_2).$$

Let P_{col} average independent constant SU(2) colour rotations in the two species at the trivial pair. These operations are contractions except for the explicit factors 2 in $R^{(b)}$ and 4 in R^{mix}.

Lemma 4.2. At split coordinates the covariances, background maps and their walk terms are block diagonal, with each block depending only on its own coordinates. The normalizers and Gaussian logarithms are sums of two separate block contributions. The source chain stays split. The new activities are neutral under the independent phase rotations of the two species. At the trivial pair P_{col} commutes with reading at later levels and is the identity on the quadratic part. ■

Proof. Each assertion about operators follows by taking the two blocks in their defining inverse, average or walk formula. A logarithmic trace also separates into the two blocks. The extraction map respects independent phase rotations; hence its quadratic kernel and its source increment are block diagonal. This proves the assertion about the chain by induction. At the trivial pair the background maps are scalar in colour. A quadratic monomial which is neutral in each species lies wholly in one block; its colour-invariant kernel is a multiple of the identity in colour, and independent colour averaging fixes it. □

Lemma 4.3. Throughout the regular pair domain, the mixed scalar fluctuation family and its pointed families satisfy

$$\|R^{\mathrm{mix}}F^{(k+1)}_0\|_{k+1}\le C_{\mathrm{gm}}\phi^2_k.$$

The activities with fermion legs have mixed norm at most $4C^{\mathrm{s}}_2\phi^2_k$. ■

Proof. The latter assertion follows from Prop. 4.1. Write $V = R^{(0)}V + R^{(1)}V + R^{(2)}V + R^{\mathrm{mix}}V$. The preceding bounds give $\|R^{(b)}V\| \le V_1\phi_k$ and

$$\|R^{\mathrm{mix}}V\| \le 4c'_\psi\bar\Sigma\phi_k^2 + C_\Lambda C_{\mathrm{gm}}\phi_k^3, \qquad \bar\Sigma = (1+c_{\mathrm{cs}})C_{\mathrm{pc}}S_{1/4}C_2^{\mathrm{s}} + C_{0,\sharp}^{\mathrm{cs}}h_F^2.$$

Here $C_\Lambda = (13/12)K_\Lambda^{\mathrm{s}}\mathsf{C}_{\mathcal{A}}\sqrt{2}A_1/\alpha_0$ is the scalar localization constant in the proof of [QCDM1, Prop. 10.1], and

$$V_1 = \sqrt{2}A_1(C_V h_F^2\mathsf{C}_{\mathcal{A}} + 2K_0^{\mathrm{s}}) + c'_\psi c_{\mathsf{S}} h_F^2(\bar B_M + 2B_\sigma) + 2c'_\psi\bar\Sigma.$$

Use parameters t_1, t_2, u multiplying the last three parts, of radii $c_{\mathrm{cl}}/(8V_1\phi_k)$ and $c_{\mathrm{cl}}/(8\|R^{\mathrm{mix}}V\|)$, respectively. At $u = 0$ the mixed activity vanishes if either $t_b = 0$: the remaining Berezin covariance and all vertices then read only the other species. Schwarz's lemma in t_1, t_2 costs $O(\phi_k^2)$; the difference between $u = 1$ and $u = 0$ costs $O(\|R^{\mathrm{mix}}V\|)$.

Scalar subtraction compares the actual pair with the trivial pair and costs a factor two. The same argument applies to pointed terms by the auxiliary differentiation used in the previous proof, and to scalar bond pointers by the localization bound [QCDM1, Lem. 7.4]. In its notation one may take

$$C_{\mathrm{gm}} = 17K_M^{\bullet}(64V_1^2/c_{\mathrm{cl}} + 64c'_\psi\bar\Sigma).$$

Decreasing γ makes the last, cubic term at most half the proposed quadratic bound and all parameter radii at least two. The induction starts with no old mixed family. □

Lemma 4.4. At the trivial pair put $S_b = R^{(b)}V_0$. Then

$$|S_b(Y;B)| \le v_{\mathsf{S}}\mathfrak{l}_k\left(M^{-3}\sum_{b'\subset Y}\mathfrak{t}_k|B(b')|^2\right)e^{-\lambda_V d_k(Y)}, \qquad v_{\mathsf{S}} = \frac{2K_0^{\mathrm{s}}M^3}{\mathsf{C}_{\mathcal{A}}^2A_1}. \tag{48}$$

■

Proof. At $k = 0$ there is no old scalar part, so $S_b = 0$. For $k \ge 1$, scalar localization gives $S_b(Y;0) = 0$ and the bound $2K_0^{\mathrm{s}}\delta_k e^{-\lambda_V d_k(Y)}$ on $|B(b')| < \mathsf{C}_{\mathcal{A}}\delta_k$. Its derivative at zero is invariant under $B \mapsto \mathrm{Ad}(u)B$. The Haar average of the nontrivial adjoint representation of SU(2) is zero, so this derivative vanishes. Apply Schwarz's lemma of order two to $S_b(Y;\zeta B)$, using $\max_{b'\subset Y}|B(b')|^2 \le \sum_{b'\subset Y}|B(b')|^2$ and $\delta_k\mathfrak{t}_k = A_1\mathfrak{t}_k^{1/2}(\log\mathfrak{t}_k)^{p_0}$. □

4.3. **A representative of the total mixed activity.** We must use cancellations in the sum of the activities at one level. They need not hold in each polymer of its chosen localization. We therefore propagate a representative of that sum. The representative is given only at the trivial pair; no holomorphic extension in the pair is claimed or needed. To express a level-j family at level k, we substitute the composed fermion background maps in its legs. At the next step the substitution is $\Psi_k = H_k\Psi_{k+1} + W$, where H_k is the fermion background extension and W is the Gaussian Berezin fluctuation. Its fluctuation difference is obtained by subtracting the same expression with $W = 0$. The sequence in [QCDM1, Lem. 7.9] is $\omega_n = C_\omega L^{n/4}$.

The bound below separates the quadratic terms that contain a difference of background fields from those that contain only their values at a cell centre. The former have the gain supplied by the regularity of the background maps. The latter retain the cell's zeroth moment and must be estimated separately. Equality of two total activities does not provide this separation: the polymer norm takes absolute values before summing over polymers. We therefore specify both the local kernels of a representative and the grouped cell moments which will enter its bound.

Lemma 4.5. Let $1 \le j \le k < K$ and let n satisfy $L^k\|\mathsf{n}\| < 2r_M/L$. Let F be an even family at the trivial pair, with no scalar part and with legs of levels at most j. Write $F(X) = \sum_\sigma F_\sigma(X)$, assigning each summand to a cell of X, and put

$$\epsilon_F = \sup_{\square}\sum_{X\supset\square}e^{\kappa d_j(X)}\sum_\sigma\|F_\sigma(X)\|_{h^{(j)}}.$$

Express its quadratic part in the level-j field Φ as

$$\sum_{u}\sum_{y,y'}\bar{\Phi}(y)\beta_u(y,y')\Phi(y').$$

Each kernel β_u is assigned to a cell $c(u)$, with centre $z_{c(u)}$. Assume, with a fixed constant C, that

$$\sum_{u:c(u)=c}\sum_{y,y'}\|\beta_u(y,y')\|(1+|y-z_c|_j+|y'-z_c|_j)\le Ch_F^{-2}\epsilon_F \tag{49}$$

for every cell c. Here $|\cdot|_j$ and dist_j use level-j lattice units. The Grassmann weights $h^{(j)}$ and the background-map constants A_κ, C_H, α_D are those of [QCDM1, Lem. 7.9]. The units are obtained by moving diagonal terms $\bar{\Phi}(y)a_X(y)\Phi(y)$ from the cell assigned to X to the cell containing y. Assume

$$\|a_X(y)\|\le Ch_F^{-2}(\|F(X)\|+\|F_1(X)\|+\|G(X)\|)e^{-\alpha_D\,\mathrm{dist}_j(y,X)/2}, \tag{50}$$

where $F=F_1+F_2$ and $\|F_1\|_j,\|G\|_j\le\epsilon_F$. For sums of summands, replace each norm in this inequality by the sum of the corresponding norms. Each $a_X(y)$ is also required to be a fixed linear combination, with sum of absolute coefficients at most 2, of row and column sums at y of the quadratic kernels of $F_1(X)$ and $G(X)$. Their legs are expressed in Φ by the background maps. The families F_1 and G have the same even local structure as F; G may use a second mass matrix, with the same background-map bounds. All these matrices are in the admitted mass domain of [QCDM1, Lem. 7.9].

Let m_c^{tor} be the sum of the zeroth moments of these units at c. On the torus the columns are transported to z_c along the lifted paths, including their antiperiodic signs. For the resulting decomposition $m_c^{\mathrm{tor}}=\sum_p m_{c,p}$ into terms supported on polymers Y_p, use

$$\|m_c^{\mathrm{tor}}\|_{\mathrm{nat}}:=\sum_p\|m_{c,p}\|e^{\kappa L^{-(k-j)}d_j(Y_p)}(1+d_j(Y_p))^3.$$

The summands here are the grouped coefficients obtained after the cell cancellations. Let n_{lag} be the least nonnegative integer such that $2(2A_\kappa C_H)^2L^{-n_{\mathrm{lag}}}\le 1$. When F is expressed at level k by the composed background maps, at the trivial pair and with no shift, its fluctuation difference satisfies

$$\|V_F\|\le c_\psi\left(\omega_{k+1-j}\epsilon_F+C_{\mathrm{hl}}L^{k-j}h_F^2\sup_c\|m_c^{\mathrm{tor}}\|_{\mathrm{nat}}\right) \tag{51}$$

when $k-j\ge n_{\mathrm{lag}}$. For $k-j<n_{\mathrm{lag}}$ the first term alone suffices. The constants depend on the fixed RG parameters and the constant C in the two hypotheses. The total of V_F is $P(H_k\Psi_{k+1}+W)-P(H_k\Psi_{k+1})$, where P is the total of the read family. The first variation G_F of the cluster activities at potential vV_F has norm at most $(17/16)K_M\|V_F\|$, and its total is the Berezin expectation of the total of V_F. ∎

Proof. Put $n=k-j$. We use the leg and walk estimates in the proof of [QCDM1, Lem. 7.9], at the trivial pair and with no shift. The comparison there between the coefficient at a variable pair and its value at the trivial pair is identically zero here. The remaining estimates concern the Grassmann coefficients, the background maps and the cell units in the hypotheses.

Suppose first that $n\ge n_{\mathrm{lag}}$. Express each old leg in the level-j field Φ, and then substitute the composed background map from level k into Φ. Each leg contributes L^{-n} relative to its old weight. Its two expansions into local and walk terms cost at most $(2A_\kappa C_H)^2$. The definition of n_{lag} makes their product at most $1/2$. For Grassmann degree $2p\ge 4$, the factors beyond the fourth leg are therefore bounded by $2^{-(2p-4)}$; this also pays the polynomials in the number of legs in the walk count. The four remaining legs give L^{-4n}, while a level-k cube contains L^{3n} level-j cells. Thus the part of degree at least four is bounded by $C\epsilon_F L^{-n}$ per level-k cube, with the same localization estimates as in that lemma.

For the quadratic part, fix a unit u assigned to c and put $z=z_c$. Carry its fields and kernel to z along the prescribed lifted paths. In the next display the transported quantities are written

without additional notation; the transports include the antiperiodic signs. Adding and subtracting the reference fields gives the exact identity

$$\sum_{y,y'} \bar\Phi(y)\beta_u(y,y')\Phi(y') = \bar\Phi(z)m_u\Phi(z) + \sum_{y,y'}[\bar\Phi(y)-\bar\Phi(z)]\beta_u(y,y')\Phi(y') + \sum_{y,y'}\bar\Phi(z)\beta_u(y,y')[\Phi(y')-\Phi(z)], \qquad m_u = \sum_{y,y'}\beta_u(y,y').$$

After summing the units assigned to c, the first term is $\bar\Phi(z_c)m_c^{\mathrm{tor}}\Phi(z_c)$. The other two terms each contain a difference. This is the point at which the grouped zeroth moment enters: the norm of this zeroth-moment term is estimated using m_c^{tor} after the cancellations in the cell have been made.

The background estimate with Hölder exponent $3/4$ bounds a transported difference along a path of length s in level-j units by $Ch_F sL^{-7n/4}$. The other leg costs $Ch_F L^{-n}$. For a moved unit, the path lies in its assigned cell and its length is paid by (49). For a polymer unit, the paths run through its polymer and the walk polymers of its two legs. Their lengths are bounded by a fixed multiple of $M(1+d_j(X))$ plus the walk lengths, where X is the original polymer. The unused polymer decay and the walk weights pay these factors, as in the proof of [QCDM1, Lem. 7.9]. We obtain

$$\text{sum of norms of the difference terms at one cell} \le C\epsilon_F L^{-11n/4}.$$

Summing over the L^{3n} old cells gives $C\epsilon_F L^{n/4}$. The estimate uses the actual lifted paths; no wrapping path is replaced by a shorter distance on the torus.

We also keep track of the diagonal terms that were moved between cells. A term $\bar\Phi(y)a_X(y)\Phi(y)$ is subtracted from the unit assigned to X and added to the unit assigned to the cell of y. These two copies cancel in the total. For the local estimate, split $a_X(y)=\sum_p a_{X,p}(y)$ according to the local or walk term in each of the two legs of the kernels from which it was formed. Let Z_p be their supporting polymer. The row-and-column hypothesis ensures that this is the same decomposition used for the kernels of F_1 and G, with uniformly bounded coefficients, and that $a_{X,p}(y)=0$ unless $y\in Z_p$. The localized row-sum estimate gives

$$\sum_{p,y}\|a_{X,p}(y)\|e^{\kappa(d_j(Z_p)-d_j(X))} \le CA_\kappa^2 h_F^{-2}(\|F_1(X)\|+\|G(X)\|).$$

For families with several summands, the right-hand side is the sum of their norms. The same estimate holds for G at its second mass matrix, since its background maps have the same constants. This is the estimate for moved pieces in [QCDM1, Prop. 10.1]. It controls their actual supports as well as their coefficients.

Changing the cell assignment changes a level-k cube sum only through terms whose two assignments lie on opposite sides of its boundary. The count in [QCDM1, Lem. 7.9] bounds these terms by $C\epsilon_F$: the boundary contains $O(L^{2n})$ old cells, the two legs give L^{-2n}, and the exponential in (50) pays the distance of a moved site from its polymer. The polymer and walk weights pay the remaining diameter factors. This contribution is included in $\omega_{k+1-j}\epsilon_F$.

To localize the difference terms, take the level-k hull of each supporting polymer and enlarge it by the walk polymers of its two final legs. The hull estimate

$$d_k(Z^{\mathrm{up}}) \le L^{-n}d_j(Z)$$

and the weighted walk sums give the local polymer norm. The spare rate $\kappa(1-L^{-n})d_j(Z)$ pays the polynomials left by the path and cell counts. Together with the preceding degree and boundary estimates, these are the terms bounded by $\omega_{k+1-j}\epsilon_F$.

It remains to localize the zeroth-moment terms. Use the decomposition $m_c^{\mathrm{tor}} = \sum_p m_{c,p}$ from the statement, after its cell cancellations. Assign

$$\bar{\Phi}(z_c) m_{c,p} \Phi(z_c)$$

to the level-k hull of Y_p, enlarged by the walk polymers of the two legs. Those legs give $C h_F^2 L^{-2n}$, and the count of old cells gives L^{3n}. The hull estimate accounts for the exponential $e^{\kappa L^{-n} d_j(Y_p)}$ in the norm of the moment; the factor $(1 + d_j(Y_p))^3$ pays the number of cells from which its hull can reach a fixed output cube. The hull and walk count of [QCDM1, Prop. 10.1] therefore gives

$$C_{\mathrm{hl}} L^n h_F^2 \sup_c \|m_c^{\mathrm{tor}}\|_{\mathrm{nat}}.$$

This is the second term of (51). It retains the torus moment itself, including any contribution of wrapping pieces.

For $n < n_{\mathrm{lag}}$, use the original summands without expanding their legs about cell centres. The count is at most $L^{3n_{\mathrm{lag}}}$, a fixed factor already included in C_ω. This proves the asserted bound without a separate moment term at these lags.

Finally substitute $\Psi_k = H_k \Psi_{k+1} + W$ and subtract the value at $W = 0$. Every surviving monomial contains a fluctuation leg. The factor c_ψ follows from the fluctuation weights, rather than from counting these monomials individually. Let h be the original leg weight, h_a a bound for its background image, and h_b the weight of its fluctuation image. Choose a positive allowance h_b' with $h_a + h_b' \le h$ and $h_b \le h_b'$. For a homogeneous product of degree q,

$$(h_a + h_b)^q - h_a^q \le \frac{h_b}{h_b'} ((h_a + h_b')^q - h_a^q).$$

The background image uses at most half the leg weight; take the other half for h_b'. The localized W bound then gives $h_b / h_b' \le c_\psi$, as in the old-potential estimate of [QCDM1, §7]. Applying this inequality to the localized terms above proves (51), including its short-lag assertion.

All decompositions used here preserve the sum: the moved diagonal terms were added and subtracted, the cell expansion is an identity, and the walk series converge absolutely. Thus the total of V_F is $P(H_k \Psi_{k+1} + W) - P(H_k \Psi_{k+1})$. For the first variation, the cluster activity at potential $v V_F$ is holomorphic on $|v| < c_{\mathrm{cl}} / \|V_F\|$ and has bound

$$\frac{17}{16} K_M' c_{\mathrm{cl}} e^{-\kappa_+ d_{k+1}(X)}.$$

Cauchy's estimate, followed by the polymer sum, yields $\|G_F\|_{k+1} \le (17/16) K_M \|V_F\|$. Since V_F is independent of B, the joined integral factors into its gauge integral and $\mu_{\Gamma_k^{-1}}(e^{v \sum_X V_F(X)})$. Differentiating its logarithm at $v = 0$, and using $\mu_{\Gamma_k^{-1}}(1) = 1$, identifies the total of G_F with $\mu_{\Gamma_k^{-1}}(\sum_X V_F(X))$. □

Put

$$q_k = \phi_k^3 + \phi_k^2 e^{-c_{\mathrm{t}} \ell / (2\lambda_{k+1})}, \qquad \bar{q}_k = q_k + \mathfrak{l}_k^2.$$

The following estimate contains the cancellation needed for Prop. 3.3.

Lemma 4.6. There are constants $E_3, N_3, C_{\mathfrak{d}}$, independent of the admitted data, with these properties. At level j, the total of $P_{\mathrm{col}} R^{\mathrm{mix}} E_{\ge 2}^{(j)}$ at the trivial pair has an even representative F^j, neutral in each species and invariant under P_{col}, with no scalar part and

$$\epsilon_j^{(3)} \le E_3 q_{j-1}. \tag{52}$$

Its cell sums satisfy

$$\|m_c^{\mathrm{tor}}\|_{\mathrm{nat}} \le C_{\mathrm{pw}} h_F^{-2} \epsilon_j^{(3)} \left[\eta_{\min}^{-2} \left(C_{\mathrm{bg}}^{\mathrm{w}} \frac{L^j \|\mathfrak{n}\|}{r_M} + C_A e^{-4 c_{\mathrm{t}} \ell / \lambda_j} \right) + e^{-c_{\mathrm{t}} \ell / \lambda_j} \right] + C_{\mathfrak{d}} h_F^{-2} \phi_{j-1}^2 e^{-c_{\mathrm{t}} \ell / \lambda_j}. \tag{53}$$

If $f_j(\mathfrak{c})$ is the total scalar output of step j at the trivial pair, including its normalizers and Gaussian terms, then

$$|R^{\mathrm{mix}} f_j(\mathfrak{c})| \le E_3 M^{-3} n_j \bar{q}_{j-1}. \tag{54}$$

For split $\mathfrak{c} \in 2\mathscr{D}^{\circ}_{j-1}$, write $\nu_{j,1} = e_+ \nu_j e_+$ and $\nu_{j,2} = e_- \nu_j e_-$. If $b' \neq b$, then

$$\sup_y \|\nu_{j,b}(\mathfrak{c})(y) - \nu_{j,b}(R_{b'}\mathfrak{c})(y)\| \le N_3 q_{j-1}. \tag{55}$$

Thus the coordinates of species b are held fixed in this comparison. No source factor is included in this bound. ∎

Proof. The induction starts with no old mixed activity. Fix a split coordinate $\mathfrak{c} \in 2\mathscr{D}^{\circ}_k$ and suppose the three assertions hold through level k. All quantities in this proof are evaluated at the trivial pair. We first identify the terms whose totals must be cancelled, then construct a local family representing the remaining total.

The deformed potential. Let $V_g = R^{(0)}V_0$ be the zero-source scalar potential. For species b, let $V_{F,b}$ and $V_{s,b}$ be the gauge–fermion vertex and mass vertex restricted to its generators; let $V_{\mathrm{old},b} = R^{(b)}V_{\mathrm{old},\ge 2}$; and let $S_b = R^{(b)}V_0$ as in Lem. 4.4. The old mixed family is split into its value at $B = 0$ and its change from that value. With ten parameters $p = (z_1, z_2, x_1, x_2, y_1, y_2, d_1, d_2, u, u')$, put

$$V_g + \sum_{b=1}^{2} \big(V_{F,b}(z_b B) + x_b V_{s,b}(z_b B) + y_b V_{\mathrm{old},b}(z_b B) + d_b S_b(B)\big) + u V_{\mathrm{mix},0} + u' V_{\mathrm{mix},1}(B).$$

Here the last two terms mean

$$V_{\mathrm{mix},0} = R^{\mathrm{mix}} V_{\mathrm{old},\ge 2}(0), \qquad V_{\mathrm{mix},1}(B) = R^{\mathrm{mix}}\big(V_{\mathrm{old},\ge 2}(B) - V_{\mathrm{old},\ge 2}(0)\big) + R^{\mathrm{mix}} V_0(B).$$

In a term evaluated at $z_b B$, all its background maps are evaluated at the fine connection reconstructed from $z_b B$. At $p = (1, \dots, 1)$ this is the original potential. The parameter u isolates the part which can be propagated by the old representative; u' groups terms which already have three small factors. Indeed, the bound $\|R^{\mathrm{mix}} V_{\mathrm{old},\ge 2}\| \le 4c'_\psi \bar{\Sigma} \phi_k^2$, followed by Schwarz's lemma in the scaling of B, gives

$$\|R^{\mathrm{mix}}(V_{\mathrm{old},\ge 2}(B) - V_{\mathrm{old},\ge 2}(0))\| \le 8c'_\psi \bar{\Sigma} \phi_k^2 r_z^{-1}.$$

The scalar localization estimate in Lem. 4.3 supplies the other term. Thus

$$\|V_{\mathrm{mix},1}\| \le \left(8c'_\psi \bar{\Sigma} C'_r + C_\Lambda C_{\mathrm{gm}}\right) \phi_k^3.$$

Here $C'_r = \sqrt{2} A_1 \mathsf{C}_{\mathcal{A}} \max\{16 C_V h_F^2 / c_{\mathrm{cl}}, 2/\beta_0\}$. The first term has acquired an extra factor ϕ_k from $r_z^{-1} \le C'_r \phi_k$; the second has that factor from scalar localization.

Choose parameter radii

$$r_z = \frac{\min\{c_{\mathrm{cl}}/(16 C_V h_F^2), \beta_0/2\}}{\mathsf{C}_{\mathcal{A}} \delta_k}, \qquad r_x = \frac{c_{\mathrm{cl}}}{16 c'_\psi c_{\mathsf{S}} h_F^2 (\bar{B}_M + 2B_\sigma)\phi_k},$$

$$r_y = \frac{c_{\mathrm{cl}}}{32 c'_\psi \bar{\Sigma} \phi_k^2}, \qquad r_d = \frac{c_{\mathrm{cl}}}{16 v_{\mathsf{S}} \mathfrak{l}_k},$$

$$r_u = \frac{c_{\mathrm{cl}}}{64 c'_\psi \bar{\Sigma} \phi_k^2}, \qquad r_{u'} = \frac{c_{\mathrm{cl}}}{16(8c'_\psi \bar{\Sigma} C'_r + C_\Lambda C_{\mathrm{gm}})\phi_k^3}.$$

The eleven displayed parts of the potential have total norm at most $11 c_{\mathrm{cl}}/16$; the cluster tail costs at most $c_{\mathrm{cl}}/16$. Assign weight one to z_b, x_b, d_b, weight two to y_b, u, and weight three to u'. Since $\mathfrak{l}_k \le \sqrt{2}\phi_k$, there is a fixed $C_r^{\mathrm{b}} \ge 2$ such that the inverse radii are bounded by $(C_r^{\mathrm{b}} \phi_k)$ to their

respective weights. Decrease γ until $C_r^{\mathrm{b}}\phi_* \le 1/2$. For a multi-index $v \in \mathbb{N}^{10}$ denote its weighted degree by $|v|_w$. Cauchy's estimate gives

$$\widetilde{E}(X;p) = \sum_v a_v(X)p^v, \qquad \|a_v(X)\| \le K'_M c_{\mathrm{cl}} r^{-v} e^{-\kappa_+ d_{k+1}(X)}, \qquad r^{-v} \le (C_r^{\mathrm{b}}\phi_k)^{|v|_w}. \tag{56}$$

Every radius is at least two, so these series converge at $p = (1,\dots,1)$.

To identify a coefficient of the total, let μ_B denote the normalized zero-source gauge integral of this step, including its scalar potential V_g, and let μ_1, μ_2 be the decoupled Berezin means of the two species. The remaining background generators are not integrated. The total of the cluster activities is the logarithm of the joined integral. Hence the coefficient of two distinct linear parameters is their connected expectation under

$$\langle A\rangle_* := \mu_B\mu_1\mu_2(A), \qquad \kappa_*(A,B) = \langle AB\rangle_* - \langle A\rangle_*\langle B\rangle_*.$$

This identity holds in the even algebra of the retained background generators. It describes the sum over X of a coefficient in (56), not its separate polymer terms.

The coefficients of weight two. Call an index mixed if it contains a parameter of each species, or contains u or u'. Every other coefficient is killed by R^{mix} polymer by polymer. For example, with $u = u' = 0$ and all parameters of species two zero, the potential has no generator of that species and does not depend on its coordinates. Its Berezin integrals use only μ_1, the unused normalized mean contributing one. Thus its activities also have no dependence on species two.

There are precisely ten mixed indices of weight at most two: the index u, and the nine products formed by one of z_1, x_1, d_1 and one of z_2, x_2, d_2. The vertices in these products are

$$l_b(B) = \partial_{z_b} V_{F,b}(z_b B)\big|_{z_b=0}, \qquad M_b = V_{s,b}(0), \qquad S_b(B).$$

The first is linear in B and quadratic in the fields of species b; M_b is independent of B and has only that species' generators; S_b is scalar.

The five products containing a mass vertex have zero total. For example, the independence of μ_1 and μ_2 gives

$$\langle M_1 M_2\rangle_* = \mu_1(M_1)\mu_2(M_2) = \langle M_1\rangle_*\langle M_2\rangle_*.$$

Since M_2 is independent of B, the same factorization holds for $\langle l_1(B)M_2\rangle_*$ and $\langle S_1(B)M_2\rangle_*$. Exchanging the species proves the cancellations for $x_1x_2, z_1x_2, x_1z_2, x_1d_2, d_1x_2$.

For the other three products containing a gauge vertex, integrate the fermion fluctuations first. Put $L_b(B) = \mu_b(l_b(B))$. The corresponding totals are

$$\kappa_B(L_1,L_2), \qquad \kappa_B(L_1,S_2), \qquad \kappa_B(S_1,L_2),$$

where $\kappa_B(A,B) = \mu_B(AB) - \mu_B(A)\mu_B(B)$. These need not vanish before colour averaging. Let $\alpha_g^{(b)}$ rotate the background generators of species b by the constant colour matrix g. At the trivial pair the vertex and covariance satisfy

$$L_b(\mathrm{Ad}(g)B) \circ \alpha_g^{(b)} = L_b(B).$$

The covariance is scalar in colour, which permits commuting the rotation through μ_b. Since L_b is linear in B, its average in the background colour variables is

$$P^{(b)}L_b(B) = L_b\left(\int_{\mathrm{SU}(2)} \mathrm{Ad}(g^{-1})B\, dg\right) = 0.$$

The adjoint representation has no invariant vector. The averages of the two species are independent, so averaging the first species kills $\kappa_B(L_1,L_2)$ and $\kappa_B(L_1,S_2)$, and averaging the second kills $\kappa_B(S_1,L_2)$. In Grassmann degree zero the colour average is the identity: the scalar part of L_b is itself an invariant linear form in B and vanishes. Thus these cancellations hold in every degree after $P_{\mathrm{col}}R^{\mathrm{mix}}$.

The ninth product, d_1d_2, is different. Its total is the gauge covariance $\kappa_B(S_1,S_2)$, so it has degree zero. Both vertices vanish when their own species coordinate is zero; therefore R^{mix} fixes this coefficient. There is no cancellation asserted for it. Its Cauchy estimate is

$$\sum_X |a_{d_1d_2}(X)| \le K_M c_{\mathrm{cl}} r_d^{-2} n_{k+1} M^{-3} = \frac{256 K_M v_{\mathfrak{S}}^2}{c_{\mathrm{cl}}} \mathfrak{l}_k^2 n_{k+1} M^{-3}. \tag{57}$$

Propagation of the old mixed total. The remaining coefficient of weight two is u. Because $V_{\mathrm{mix},0}$ is independent of B, its total is $\mu_1\mu_2$ of the total of that vertex. If P_i is the total of the level-i family after it has been read at level k, the fluctuation vertex has total

$$R^{\mathrm{mix}} \sum_{i\le k} \big(P_i(H_k\Psi_{k+1}+W) - P_i(H_k\Psi_{k+1})\big).$$

The projection P_{col} commutes with these background maps, with R^{mix} and with $\mu_1\mu_2$. Thus the induction hypothesis permits replacing each total by that of F^i. Apply Lem. 4.5 to each F^i and add the resulting families. If V_F denotes this sum, the coefficient of u, after colour averaging, has the total of its first-variation family G_F. This replacement uses equality of totals, not equality of the chosen localizations.

In the cell bound, terms containing $L^i\|\mathfrak{n}\|/r_M$ acquire the factor L^{k-i} from reading at level k; their product is less than $2/L$. The exponential terms have uniformly bounded products with L^{k-i}. These factors are included in c_{cs}, exactly as in the ordinary old-family count. The last term of (53) must instead be summed separately. We obtain

$$\|V_F\| \le c_\psi \Big[(1+c_{\mathrm{cs}}) \sum_{i\le k} \omega_{k+1-i}\epsilon_i^{(3)} + C_{\mathrm{hl}} C_{\mathfrak{d}} \sum_{i\le k} L^{k-i}\phi_{i-1}^2 e^{-c_{\mathrm{t}}\ell/\lambda_i} \Big].$$

Both sums are bounded by a multiple of q_k. For the first, monotonicity of ϕ_i and of the torus exponential gives

$$q_{i-1} \le \frac{\phi_{i-1}}{\phi_k} q_k, \qquad \sum_{i\le k} \omega_{k+1-i}\phi_{i-1} \le C_{\mathrm{pc}} S_{1/4} \phi_k.$$

For the second, write $v=k-i$ and divide by the torus term $\phi_k^2 e^{-c_{\mathrm{t}}\ell/(2\lambda_{k+1})}$ in q_k. The quotient is at most

$$\sum_{v\ge 0} L^v \exp\Big[-\frac{c_{\mathrm{t}}\ell}{\lambda_{k+1}}(L^{v+1}-\tfrac12)\Big] \le \sum_{v\ge 0} L^v e^{-2c_{\mathrm{t}}(L^{v+1}-1/2)},$$

because $\ell/\lambda_{k+1}\ge 2$. Consequently

$$\|V_F\| \le c_\psi \Big((1+c_{\mathrm{cs}}) C_{\mathrm{pc}} S_{1/4} E_3 + C_{\mathrm{hl}} C_{\mathfrak{d}} S_{\mathrm{tor}} \Big) q_k, \qquad S_{\mathrm{tor}} = \sum_{v\ge 0} L^v e^{-2c_{\mathrm{t}}(L^{v+1}-1/2)} < \infty. \tag{58}$$

The factor from the first variation is $(17/16)K_M$.

The remaining coefficients. For a multi-index of weighted degree at least three the Cauchy product has at least three powers of ϕ_k. Summing these bounds uses

$$N_{10} = \sum_{m\ge 3} \binom{m+10}{10} 2^{-(m-3)} < \infty.$$

It bounds their total norm by $4K_M c_{\mathrm{cl}} N_{10} (C_r^{\mathrm{b}})^3 \phi_k^3$ after $P_{\mathrm{col}} R^{\mathrm{mix}}$. Adding their degree-at-least-two parts to the first variation gives a pre-extraction representative R with

$$\|R\|_{k+1} \le \Big[\tfrac{17}{4} K_M c_\psi \big((1+c_{\mathrm{cs}}) C_{\mathrm{pc}} S_{1/4} E_3 + C_{\mathrm{hl}} C_{\mathfrak{d}} S_{\mathrm{tor}} \big) + 4 K_M c_{\mathrm{cl}} N_{10} (C_r^{\mathrm{b}})^3 \Big] q_k.$$

For clarity, the pre-extraction representative in this display is the explicitly defined family

$$R(X) = \big(R^{\mathrm{mix}} G_F(X)\big)_{\geq 2} + \sum_{\substack{v \text{ mixed} \\ |v|_w \geq 3}} P_{\mathrm{col}} R^{\mathrm{mix}}(a_v(X))_{\geq 2}.$$

It has the total of the mixed activity with legs, because all omitted low-weight totals were just shown to vanish or to be scalar. Each retained term is neutral in both species. The first-variation family is formed from colour-invariant representatives, so it is also invariant under P_{col}.

The scalar estimate can already be closed. The normalizers and Gaussian logarithms have no mixed part. In degree zero the colour projection is the identity, and the coefficient bound for R^{mix} costs at most four. The scalar output is therefore bounded by the first-variation contribution, the sum of coefficients of weight at least three, and (57). Their sum is

$$M^{-3} n_{k+1} \left[(E_3/2 + A_3) q_k + \frac{256 K_M v_{\mathfrak{S}}^2}{c_{\mathrm{cl}}} \mathfrak{l}_k^2 \right] \leq E_3 M^{-3} n_{k+1} \bar{q}_k,$$

with the constants chosen below. This proves (54); it also explains why the scalar term $\mathfrak{l}_k^2$ need not enter the size of the representative with fermion legs.

Extraction and cell sums. In the following estimates put $j = k+1$. Write $K_X^R = \sum_p K_{X,p}^R$ for the quadratic kernel on X, decomposed by expanding the two background legs into their local and walk terms, and let Z_p be their supporting polymers. For such a kernel define

$$M_y[K] = \frac{1}{2} \sum_{y'} \big(K(y, y') + K(y', y) \big),$$

with columns transported to y inside the fundamental cell centered at y. This is a linear function of K. The moved coefficients are $a_{X,p}(y) = M_y[K_{X,p}^R]$. The row-sum bound from the proof of [QCDM1, Prop. 10.1] is

$$\sum_{p,y} \|a_{X,p}(y)\| e^{\kappa [d_j(Z_p) - d_j(X)]} \leq C A_\kappa^2 h_F^{-2} \|R(X)\|.$$

After summing over X, it gives, for every cell c, every cube $\square$, and $n \geq 1$,

$$\sum_{y \in c, X, p} \|a_{X,p}(y)\| e^{\kappa L^{-n} d_j(Z_p)} (1 + d_j(Z_p))^3 \leq \tfrac{1}{2} C_{\mathrm{pw}} h_F^{-2} \|R\|_j, \tag{59}$$

$$\sum_{Z_p \supset \square, y} \|a_{X,p}(y)\| e^{\kappa d_j(Z_p)} \leq \tfrac{1}{2} C_{\mathrm{pw}} h_F^{-2} \|R\|_j. \tag{60}$$

The factor $1/2$ appears because C_{pw} in that proof bounds the two families at the source and at zero together; here there is one family. At lag zero use (60). Equality of the totals identifies the sum of these moments with the moment of the actual total. This does not identify the moments polymer by polymer. The source increment uses only polymers of diameter less than one quarter of the torus side and reads their moments along lifted paths. Denote the actual mixed quadratic kernels by $K_{X,p}$. By [QCDM1, Prop. 10.1, equation (35)],

$$-\mathfrak{s}_j R^{\mathrm{mix}} \nu_j(y) = \sum_{\mathrm{diam}\, X < \ell/4, p} \mathrm{ord}_0[K_{X,p}](y),$$

where $\mathfrak{s}_j \geq \eta_{\min}^2$ is the zero-source normalization of the extraction map. On a nonwrapping Z_p, the moment ord_0 read along the lifted paths agrees with M_y. Equality of the total kernels therefore gives the exact identity

$$-\mathfrak{s}_j R^{\mathrm{mix}} \nu_j(y) = \sum_{X,p} M_y[K_{X,p}^R] - d(y), \tag{61}$$

where

$$d(y) = \sum_{X,p}{}' M_y[K_{X,p}] - \sum_{X,p}{}'' \operatorname{ord}_0[K_{X,p}](y).$$

The first sum contains actual pieces for which either $\operatorname{diam} X \geq \ell/4$ or Z_p wraps around the torus. The second contains pieces with $\operatorname{diam} X < \ell/4$ whose Z_p wraps around the torus. This formula accounts for both the omitted large polymers and the difference between the two moment conventions on wrapping paths. Every supporting polymer in this difference satisfies

$$d_{k+1}(Z_p) \geq \frac{\ell}{4c_{\mathcal{D}} M \lambda_{k+1}} - 1.$$

Here $c_{\mathcal{D}}$ is the fixed diameter constant, $\operatorname{diam} Y \leq c_{\mathcal{D}} M \lambda_j (1 + d_j(Y))$. Indeed, either the original polymer has diameter at least one quarter of the torus side, or Z_p contains a path wrapping around the torus. The tail of (59) therefore gains $e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}$, by the choice of c_{t} in [QCDM1, Prop. 10.1]. Apply this estimate to the actual mixed quadratic activity, whose norm is at most $4C_2^{\mathrm{s}}\phi_k^2$ by Prop. 4.1. Consequently

$$\sum_{y \in c} |d(y)| \leq 4C_{\mathrm{pw}} h_F^{-2} C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}. \tag{62}$$

We now perform the extraction on the representative. With $\psi^{\mathrm{bg}} = H_{0,j}\Psi$, define the quadratic forms

$$P^R_{z;X,p} = -\mathfrak{s}_j^{-1} L^{-j} \mathsf{S}_1\big(M_z[K^R_{X,p}] 1_z \psi^{\mathrm{bg}}\big),$$
$$P^d_z = \mathfrak{s}_j^{-1} L^{-j} \mathsf{S}_1\big(d(z) 1_z \psi^{\mathrm{bg}}\big).$$

Here 1_z restricts the coefficient to z and S_1 is the mass bilinear used in the source extraction. Assign both forms to the cell containing z; their supporting polymers are Z_p and the cube of z, respectively. By (61),

$$\sum_{z;X,p} P^R_{z;X,p} + \sum_z P^d_z = L^{-j} \mathsf{S}_1\big(R^{\mathrm{mix}} \nu_j \psi^{\mathrm{bg}}\big).$$

This is precisely the local form that the actual extraction adds to the mixed activity. Thus

$$F^{(j)}_{\mathfrak{c}} = R + \sum_{z;X,p} P^R_{z;X,p} + \sum_z P^d_z$$

represents the required extracted total. Its forms are neutral in both species and invariant under P_{col}, since their coefficients are split and the reconstruction maps at the trivial pair act trivially on colour. The first forms add at most $\eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}} \|R\|_{k+1}$ to the polymer norm by (60). The correction forms add at most

$$4\eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}} C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}.$$

Thus a common bound for the norm and the cell-unit hypotheses is

$$\epsilon^{(3)}_{k+1} = K^{\mathrm{s}}_{\mu} \|R\|_{k+1} + 4(1 + \eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}}) C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}. \tag{63}$$

For the unit hypotheses, split the quadratic kernel into $K^R_X - \delta_{yy'} M_y[K^R_X]$, assigned to the cell of X, and the moved diagonal pieces, assigned to the cell of y. Their variation and first moments obey (49) by the kernel bound and (59). The coefficients of the two types of extraction forms, summed over a cell, are bounded respectively by

$$\tfrac{1}{2}\eta_{\min}^{-2} C_{\mathrm{pw}} h_F^{-2} \|R\|_{k+1}, \qquad 4\eta_{\min}^{-2} C_{\mathrm{pw}} h_F^{-2} C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}.$$

Each is at most $\eta_{\min}^{-2} C_{\mathrm{pw}} h_F^{-2} \epsilon^{(3)}_{k+1}$. The moved coefficient is exactly one half of a row sum plus one half of a column sum of $K^R_{X,p}$, as required in Lem. 4.5; its second family G is zero.

For a nonwrapping Z_p, the zeroth moment of $K^R_{X,p} - \delta_{yy'} M_y[K^R_{X,p}]$ is zero. The two moment conventions agree on this support, so the subtracted diagonal has exactly the same total moment

as the original kernel. A wrapping piece contributes at most twice the double sum of its kernel. Assigning it to the hull of Z_p and using the tail of (59) bounds these contributions in the cell norm by

$$C_{\mathrm{pw}} h_F^{-2} \|R\|_j e^{-c_{\mathrm{t}}\ell/\lambda_j}.$$

A moved piece and its extraction form have joint coefficient

$$\left(1 - \mathfrak{s}_{k+1}^{-1} \sum_y A_{k+1}^{\mathfrak{n}}(y,z)\right) M_z[K_{X,p}^R],$$

where $A_j^{\mathfrak{n}}$ is the zeroth-moment map of the background mass form. Its source variation and its torus comparison from [QCDM1, proof of Prop. 10.1] give the bound

$$\eta_{\min}^{-2} \left(C_{\mathrm{bg}}^{\mathrm{w}} \frac{L^j \|\mathfrak{n}\|}{r_M} + C_A e^{-4c_{\mathrm{t}}\ell/\lambda_j} \right) \frac{1}{2} C_{\mathrm{pw}} h_F^{-2} \|R\|_j$$

for the grouped moved pieces and their forms. Keeping them grouped is what produces the small coefficient $1 - \mathfrak{s}_j^{-1} \sum_y A_j^{\mathfrak{n}}(y,z)$.

The correction form P_z^d contributes $\mathfrak{s}_j^{-1} \sum_y A_j^{\mathfrak{n}}(y,z) d(z)$. At zero source,

$$\left| \sum_y A_j^0(y,z) - \mathfrak{s}_j \right| \le C_A e^{-4c_{\mathrm{t}}\ell/\lambda_j}.$$

The change with $\mathfrak{n}$, expanded in the local and walk terms of the two background legs, is at most $C_{\mathrm{bg}}^{\mathrm{w}} L^j \|\mathfrak{n}\|/r_M$ in the same weighted cell norm. Since $L^j\|\mathfrak{n}\|/r_M < 2$, (62) bounds these correction forms by

$$C_{\mathfrak{d}} h_F^{-2} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_j}, \qquad C_{\mathfrak{d}} = 4(1 + \eta_{\min}^{-2}(C_A + 2C_{\mathrm{bg}}^{\mathrm{w}})) C_{\mathrm{pw}} C_2^{\mathrm{s}}.$$

Combining the three types of terms proves (53).

Choose

$$\begin{aligned} C_{\mathfrak{d}} &= 4(1 + \eta_{\min}^{-2}(C_A + 2C_{\mathrm{bg}}^{\mathrm{w}})) C_{\mathrm{pw}} C_2^{\mathrm{s}}, \\ A_3 &= K_\mu^{\mathrm{s}} \left(\tfrac{17}{4} K_M c_\psi C_{\mathrm{hl}} C_{\mathfrak{d}} S_{\mathrm{tor}} + 4 K_M c_{\mathrm{cl}} N_{10} (C_r^{\mathrm{b}})^3 \right) \\ &\quad + 4(1 + \eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}}) C_2^{\mathrm{s}}, \\ E_3 &= 2A_3 + 256 K_M v_{\mathsf{S}}^2 / c_{\mathrm{cl}}. \end{aligned}$$

After extraction the coefficient of $E_3 q_k$ is at most $4\theta_E^{\mathrm{s}}(1 + c_{\mathrm{cs}}) C_{\mathrm{pc}} S_{1/4} \le 1/2$. Thus the norm is at most $(E_3/2 + A_3) q_k$. These choices close both (52) and (54). Finally, (61), the first moment bound and (62) give

$$\begin{aligned} |R^{\mathrm{mix}} \nu_j(y)| &\le \eta_{\min}^{-2} C_{\mathrm{pw}} h_F^{-2} \left(\tfrac{1}{2} \|R\|_j + 4 C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_j} \right) \\ &\le N_3 q_k, \qquad N_3 = \eta_{\min}^{-2} C_{\mathrm{pw}} h_F^{-2} (2E_3 + 4C_2^{\mathrm{s}}). \end{aligned}$$

To read this as an estimate for one species, apply R^{mix} to its local quadratic term. For $b' \neq b$ the result is

$$R^{\mathrm{mix}}(\bar\psi_b \nu_{j,b}(\mathfrak{c}) \psi_b) = \bar\psi_b (\nu_{j,b}(\mathfrak{c}) - \nu_{j,b}(R_{b'}\mathfrak{c})) \psi_b.$$

The preceding kernel estimate is therefore (55). □

4.4. **The source chain and the sum over scales.** The preceding lemma concerns free split coordinates. The vacuum energy is evaluated along the actual source chain. We next control the change in one chain when the source in the other species is added.

Lemma 4.7. For split sources and $\lambda_i\|J_b\| < r_M$,

$$\|\sigma_{i,1}(J_1,J_2) - \sigma_{i,1}(J_1,0)\| \le C_{\mathrm{s}3} q_{i-1} \frac{\lambda_i\|J_2\|}{r_M}, \qquad C_{\mathrm{s}3} = \frac{3N_3}{1-4/L}, \tag{64}$$

and the analogous assertion holds for species two. If two inner coordinates have the same n, and

$$T = \max_{i\le j-1} \frac{\|\sigma_i - \sigma_i'\|}{B_\sigma \phi_i} \le 1,$$

then

$$|f_j(\mathfrak{c}) - f_j(\mathfrak{c}')| \le 8K_M M^{-3} n_j C_{\mathrm{Lip}} \phi_{j-1} T, \tag{65}$$

where $C_{\mathrm{Lip}} = c'_\psi c_{\mathsf{S}} h_F^2 B_\sigma + 2c'_\psi \bar{\Sigma} + 2\sqrt{2} A_1 K_0^{\mathrm{s}}$. ∎

Proof. The chains exist and are holomorphic on $\lambda_j\|J_b\| < 2r_M$ by Prop. 4.1. On the smaller domain in the statement their coordinates are split and lie in the inner coordinate domain. Write

$$\mathfrak{c}_i^+ = \mathfrak{c}_i(J_1,J_2), \qquad \mathfrak{c}_i^0 = \mathfrak{c}_i(J_1,0), \qquad \Delta_l = \sigma_{l,1}(J_1,J_2) - \sigma_{l,1}(J_1,0).$$

The bare first-species coordinate is $-aJ_1$ in both chains. It cancels from their difference, leaving

$$\Delta_l = \sum_{i=1}^{l} L^{l-i} D_i, \qquad D_i = \nu_{i,1}(\mathfrak{c}_{i-1}^+) - \nu_{i,1}(\mathfrak{c}_{i-1}^0). \tag{66}$$

Insert the coordinates with their second block set to zero into each increment:

$$\begin{aligned} D_i = {} & [\nu_{i,1}(\mathfrak{c}_{i-1}^+) - \nu_{i,1}(R_2\mathfrak{c}_{i-1}^+)] \\ & + [\nu_{i,1}(R_2\mathfrak{c}_{i-1}^+) - \nu_{i,1}(R_2\mathfrak{c}_{i-1}^0)] \\ & + [\nu_{i,1}(R_2\mathfrak{c}_{i-1}^0) - \nu_{i,1}(\mathfrak{c}_{i-1}^0)]. \end{aligned}$$

The first and third brackets are each bounded by $N_3 q_{i-1}$, by (55). The third bracket must be retained: even when the bare source J_2 is zero, the chain can have nonzero second-species mass corrections. The two coordinates in the middle bracket have the same bare coordinate, and their earlier first-species entries differ by Δ_l, $l < i$. Hence (45) gives

$$\|D_i\| \le 2N_3 q_{i-1} + B_\nu^{(2)} \phi_{i-1}^2 \max_{l<i} \frac{\|\Delta_l\|}{B_\sigma \phi_l}. \tag{67}$$

The maximum is zero at $i = 1$.

Let d_i be the supremum of $\|D_i\|$ on $\lambda_i\|J_2\| < r_M$, with the admissible J_1 fixed. The increment difference is holomorphic in J_2 and vanishes at $J_2 = 0$. Schwarz's lemma on each complex line through the origin, followed by (66), gives

$$\|D_i\| \le d_i \frac{\lambda_i\|J_2\|}{r_M}, \qquad \|\Delta_l\| \le \frac{\lambda_l\|J_2\|}{r_M} \sum_{i=1}^{l} d_i. \tag{68}$$

Here the identity $L^{l-i}\lambda_i = \lambda_l$ accounts for the scale factor in the second inequality.

We prove $d_i \le 3N_3 q_{i-1}$ by induction. The scale comparison from [QCDM1, Prop. 10.1] gives $\phi_{i'-1} \le q_\phi^{l-i'} \phi_{l-1}$ with $q_\phi^2 \le 4/L$. The exponential in $q_{i'-1}$ is at most the corresponding exponential in q_{l-1}. It follows that

$$\sum_{i'=1}^{l} q_{i'-1} \le \frac{q_{l-1}}{1-4/L}. \tag{69}$$

When $\lambda_i\|J_2\| < r_M$ and $l < i$, the earlier scale satisfies $\lambda_l\|J_2\| < r_M/L$. The induction hypothesis and (68) therefore imply

$$\|\Delta_l\| \le \frac{3N_3}{L(1-4/L)} q_{l-1}.$$

Moreover,

$$\frac{q_{l-1}}{\phi_l} \le \phi_{i-1}\left(\phi_{i-1} + e^{-c_{\mathrm{t}}\ell/(2\lambda_i)}\right), \qquad l < i.$$

Thus the last term of (67) is at most

$$\frac{3B_\nu^{(2)}\phi_{i-1}N_3}{L(1-4/L)B_\sigma}\, q_{i-1} \le N_3 q_{i-1}.$$

This proves the induction. Substitution into (68) and (69) proves (64); the other species is treated by interchanging the two blocks. The smallness conditions used here and below are

$$3B_\nu^{(2)}\phi_* \le L(1-4/L)B_\sigma, \qquad 2C_{\mathrm{s}3}\phi_* \le B_\sigma.$$

For the scalar estimate put $\Delta V = V(\mathfrak{c}') - V(\mathfrak{c})$. The two coordinates have the same n, so their fermion covariances, reconstruction maps, Gaussian terms and normalizing factors coincide. The potential can change only through the mass member and the old activities. The mass difference is bounded by

$$c'_\psi c_{\mathsf{S}} h_F^2 \|\sigma'_{j-1} - \sigma_{j-1}\| \le c'_\psi c_{\mathsf{S}} h_F^2 B_\sigma \phi_{j-1} T.$$

If $T = 0$ there is nothing to prove. Otherwise, the line

$$\mathfrak{c} + \zeta(\mathfrak{c}' - \mathfrak{c})/T, \qquad |\zeta| < 1,$$

lies in the doubled coordinate domain. On this line the old members with fermion variables have norm at most $c'_\psi \bar{\Sigma}\phi_{j-1}^2$, while their scalar members have norm at most $K_0^{\mathrm{s}}\delta_{j-1}$. Schwarz's lemma applied to each difference from its value at $\zeta = 0$ supplies the factor $2T$. Consequently

$$\begin{aligned}\|\Delta V\| &\le \left(c'_\psi c_{\mathsf{S}} h_F^2 B_\sigma \phi_{j-1} + 2c'_\psi \bar{\Sigma}\phi_{j-1}^2 + 2K_0^{\mathrm{s}}\delta_{j-1}\right) T \\ &\le C_{\mathrm{Lip}}\phi_{j-1} T,\end{aligned}$$

where we used $\phi_{j-1} \le 1$ and $\delta_{j-1} \le \sqrt{2}A_1\phi_{j-1}$.

At the trivial pair the potential $V(\mathfrak{c})$ has norm at most $11c_{\mathrm{cl}}/16$. On the disc

$$|v| < \frac{c_{\mathrm{cl}}}{4\|\Delta V\|},$$

the potential $V(\mathfrak{c}) + v\Delta V$ has norm at most $15c_{\mathrm{cl}}/16$. The tail has norm at most $c_{\mathrm{cl}}/16$, so the cluster theorem applies throughout this disc. The scalar total is holomorphic there and bounded by $K_M c_{\mathrm{cl}} n_j M^{-3}$: the local cluster bound is summed over the $n_j M^{-3}$ cubes. Schwarz's lemma for its difference from the value at $v = 0$ gives

$$|f_j(\mathfrak{c}') - f_j(\mathfrak{c})| \le 2K_M c_{\mathrm{cl}} n_j M^{-3} \frac{4\|\Delta V\|}{c_{\mathrm{cl}}}.$$

If the displayed radius is at most one, the same estimate follows from the individual bounds at the two admissible endpoints. Substituting the bound for $\|\Delta V\|$ proves (65). □

Proof of Prop. 3.3. Let $\mathfrak{c}_{j-1}(J_1, J_2)$ be the source-chain coordinate, and write

$$G_j(J_1, J_2) = D^{\mathrm{mix}}[f_j \circ \mathfrak{c}_{j-1}](J_1, J_2).$$

This is the sum over polymers of the mixed step energy. The bound for $R^{\mathrm{mix}} f_j$ does not apply to G_j directly: a single-source part of f_j can still depend on the other bare source through the mass corrections in its own species. We compare with a coordinate in which the two species depend on separate sources.

Define $\mathfrak{c}^\flat(J_1, J_2)$ by taking every first-species entry from $\mathfrak{c}_{j-1}(J_1, 0)$ and every second-species entry from $\mathfrak{c}_{j-1}(0, J_2)$. It is in the inner domain, and its bare coordinate agrees with that of $\mathfrak{c}_{j-1}(J_1, J_2)$. In the decomposition

$$f_j = f_j(0) + R^{(1)} f_j + R^{(2)} f_j + R^{\mathrm{mix}} f_j,$$

the composition of $R^{(b)} f_j$ with $\mathfrak{c}^\flat$ depends only on J_b. Further, $R^{\mathrm{mix}} f_j(\mathfrak{c}^\flat)$ vanishes on each source axis, because the corresponding block of $\mathfrak{c}^\flat$ is taken from the zero-source chain. Consequently

$$D^{\mathrm{mix}}[f_j(\mathfrak{c}^\flat)] = R^{\mathrm{mix}} f_j(\mathfrak{c}^\flat).$$

At each corner of the mixed difference the original and separated coordinates have the same bare coordinate. At (J_1, J_2) their two mass corrections differ by

$$\sigma_{i,1}(J_1, J_2) - \sigma_{i,1}(J_1, 0), \qquad \sigma_{i,2}(J_1, J_2) - \sigma_{i,2}(0, J_2).$$

At $(J_1, 0)$ only the second block differs, by $\sigma_{i,2}(J_1, 0)$; at $(0, J_2)$ only the first block differs, by $\sigma_{i,1}(0, J_2)$. At $(0, 0)$ there is no difference. The axis differences are also bounded by (64), with the other source set to zero. Thus at each of the three nonzero corners the normalized coordinate difference satisfies

$$T \le \frac{C_{\mathrm{s3}}}{B_\sigma} \max_{i \le j-1} \frac{q_{i-1}}{\phi_i}, \qquad \phi_{j-1} T \le \frac{C_{\mathrm{s3}}}{B_\sigma} q_{j-1}.$$

For the second inequality, use

$$\frac{q_{i-1}}{\phi_i} \le \phi_{i-1} \left(\phi_{i-1} + e^{-c_{\mathrm{t}} \ell/(2\lambda_i)} \right) \le \phi_{j-1} \left(\phi_{j-1} + e^{-c_{\mathrm{t}} \ell/(2\lambda_j)} \right).$$

The condition $2C_{\mathrm{s3}}\phi_* \le B_\sigma$ gives $T \le 1$. Apply (65) at the three corners and (54) to $\mathfrak{c}^\flat$. Since $q_{j-1} \le \bar{q}_{j-1}$, we obtain

$$\begin{aligned} |G_j(J_1, J_2)| &\le E_3 M^{-3} n_j \bar{q}_{j-1} + 3 \cdot 8 K_M M^{-3} n_j C_{\mathrm{Lip}} \frac{C_{\mathrm{s3}}}{B_\sigma} q_{j-1} \\ &\le C_{\mathrm{gm3}} M^{-3} n_j \bar{q}_{j-1}, \end{aligned}$$

where

$$C_{\mathrm{gm3}} = E_3 + 24 K_M C_{\mathrm{Lip}} C_{\mathrm{s3}} / B_\sigma.$$

This estimate holds throughout $\lambda_j \|J_b\| < r_M$. For nonzero J_1, J_2, set

$$G(\zeta_1, \zeta_2) = G_j(\zeta_1 J_1, \zeta_2 J_2), \qquad R_b = \frac{r_M}{\lambda_j \|J_b\|}.$$

The function G is holomorphic on $|\zeta_b| < R_b$ and zero on both coordinate axes. First apply Schwarz's lemma in ζ_1. The quotient G/ζ_1 has a holomorphic extension through $\zeta_1 = 0$, is bounded by $\sup |G|/R_1$, and is zero when $\zeta_2 = 0$. A second application in ζ_2 gives

$$|G_j(J_1, J_2)| \le C_{\mathrm{gm3}} M^{-3} n_j \bar{q}_{j-1} \frac{\lambda_j \|J_1\|}{r_M} \frac{\lambda_j \|J_2\|}{r_M}.$$

If either source is zero, this inequality is immediate. Since the energy contribution is $f_j(\mathfrak{c}_{j-1}(J)) - f_j(0)$, the inequality is (15).

For the sum over scales, the two source factors supply λ_j^2, whereas the number of sites contributes $n_j = \ell^3 \lambda_j^{-3}$. Their product leaves one inverse length:

$$n_j \lambda_j^2 = \ell^3 \lambda_j^{-1} = \ell^3 L^{K-j}.$$

Since $\rho_*/r_M \le (2L)^{-1}$, it follows that

$$|E_K^{\mathrm{mix}}(J_1, J_2)| \le C_{\mathrm{gm3}} M^{-3} (2L)^{-2} \ell^3 \sum_{j=1}^{K} L^{K-j} \bar{q}_{j-1}. \tag{70}$$

The three terms in $\bar{q}_{j-1}$ are summed separately. Put $n = K - j$. Then $\mathsf{t}_{j-1} = L^{n+1}\mathsf{b}$ and

$$x_{j-1} \geq \mathsf{t}_{j-1} - \bar{B} \geq \tfrac{1}{2}L^{n+1}\mathsf{b},$$

because $\mathsf{b} > \gamma^{-2} \geq L\bar{B}$. The function $\phi(t) = t^{-1/2}(1+\log t)^{p_0}$ is decreasing on $[e^{2p_0}, \infty)$, which contains $[\frac{1}{2}\gamma^{-2}, \infty)$ by the choice of γ. Consequently

$$\begin{aligned}\phi_{j-1} &\leq \phi(\tfrac{1}{2}L^{n+1}\mathsf{b}) \\ &\leq \sqrt{2}L^{-(n+1)/2}\phi(\mathsf{b})(1+(n+1)\log L)^{p_0}.\end{aligned}$$

For the second inequality use $1 + a + b \leq (1+a)(1+b)$ for nonnegative a, b on the logarithmic factor. Since $\phi(\mathsf{b}) \leq \phi_* \leq 1$,

$$L^n\phi_{j-1}^3 \leq 2^{3/2}L^{-3/2}L^{-n/2}(1+(n+1)\log L)^{3p_0}.$$

The geometric factor $L^{-n/2}$ makes this series summable despite its logarithmic polynomial.

For the scalar covariance left at weight two, the definition of $\mathfrak{l}_{j-1}$ gives the exact identity

$$L^n\mathfrak{l}_{j-1}^2 = \frac{1}{L\mathsf{b}}(\log\mathsf{b} + (n+1)\log L)^{-2p_0} \leq L^{-1}((n+1)\log L)^{-2p_0}.$$

The scale powers cancel. Since $2p_0 > 1$, the upper bounds sum to

$$\frac{\zeta(2p_0)}{L(\log L)^{2p_0}}.$$

Finally, $\ell \geq 2$ and $\phi_{j-1} \leq 1$ imply

$$L^n\phi_{j-1}^2 e^{-c_{\mathrm{t}}\ell/(2\lambda_j)} = L^n\phi_{j-1}^2 e^{-c_{\mathrm{t}}\ell L^n/2} \leq L^n e^{-c_{\mathrm{t}}L^n}.$$

This series is summable because its exponential contains L^n. Thus all three contributions in (70) are bounded uniformly in K and ℓ. One may take

$$\begin{aligned}S_{\mathrm{mix}} &= 2^{3/2}L^{-3/2}\sum_{n\geq 0} L^{-n/2}(1+(n+1)\log L)^{3p_0} + \frac{\zeta(2p_0)}{L(\log L)^{2p_0}} + \sum_{n\geq 0} L^n e^{-c_{\mathrm{t}}L^n}, \\ C_{\mathrm{mix}} &= C_{\mathrm{gm3}}M^{-3}(2L)^{-2}S_{\mathrm{mix}}.\end{aligned}$$

This proves (14). □

Order of the constants. The mass window remains fixed. The new constants are chosen after the mass-independent source norm and localization constants of [QCDM1, Prop. 10.20], and before the final choice of γ. First choose $K_0^{\mathrm{s}}, C_r^{\mathrm{s}}, C_2^{\mathrm{s}}, B_\nu^{(2)}$; then $\bar{\Sigma}, V_1, C_{\mathrm{gm}}, v_{\mathsf{S}}, C_r^{\mathrm{b}}$; then $C_{\mathfrak{d}}, A_3, E_3, N_3, C_{\mathrm{s}3}, C_{\mathrm{Lip}}$ and C_{mix}. All are functions of constants already chosen. In addition to the three smallness conditions in the proof of Prop. 4.1, impose

$$C_r^{\mathrm{b}}\phi_* \leq \tfrac{1}{2}, \qquad \tfrac{17}{2}\,16K_M^{\bullet}C_\Lambda\phi_* \leq \tfrac{1}{2}, \qquad 3B_\nu^{(2)}\phi_* \leq L(1-4/L)B_\sigma, \qquad 2C_{\mathrm{s}3}\phi_* \leq B_\sigma.$$

They hold after decreasing γ, since $\phi_* \to 0$. They do not depend on the cutoff, volume, source or scale.

5. Scalar mesons and diquarks

We retain the setting of section 2 and write $D_U = D[U] + m_0$ for the massive operator. Constants may depend on the fixed coupling interval and mass window, but are independent of the cutoff, volume and infinite-volume state. In this section and the next two, sites are labelled by $x \in a\mathbb{Z}^3$. Thus (x, q) corresponds to the place $(a^{-1}x, q)$ in (1), and x_v denotes the physical site of v.

The mesons $\bar{\psi}X\psi$ do not form an invariant family under all internal symmetries of the model. Pseudoreality of the colour representation permits a symmetry to replace a quark by a charge-conjugate antiquark. We first include the resulting diquarks among the fields, and then prove estimates for arbitrary joint correlations of this larger family.

5.1. **The labels and the exact symmetries.** Put $\epsilon = i\sigma_2$ on colour and $C_0 = \sigma_2 \otimes \epsilon$ on spin and colour. The identities

$$C_0^{\mathsf{T}} = C_0, \qquad C_0^{-1} = -C_0, \qquad C_0^{\dagger} = -C_0, \qquad \sigma_\mu^{\mathsf{T}} C_0^{-1} = -C_0^{-1}\sigma_\mu$$

will be used below. With $\chi = C_0\bar{\psi}^{\mathsf{T}}$, write $\widehat{\Psi} = (\psi, \chi)$. The internal label space of this column is

$$W = \mathbb{C}^2_\tau \otimes \mathbb{C}^2_{\mathrm{d}} \otimes \mathbb{C}^{N_f}.$$

The second factor distinguishes ψ from χ. Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on this factor, let $\mathcal{J} = 1 \otimes J \otimes 1$, and let $P_\pm = (1 \pm \tau_3)/2$. The complex label space and its reflection-real form are

$$\mathscr{S} = \{M \in \operatorname{End} W : \mathcal{J} M^{\mathsf{T}} \mathcal{J}^{-1} = M\}, \qquad M^* = \tau_3 M^\dagger \tau_3, \qquad \mathscr{S}_{\mathbb{R}} = \{M \in \mathscr{S} : M^* = M\}. \quad (71)$$

Thus $\mathcal{J}M$ is antisymmetric and $\dim_{\mathbb{C}} \mathscr{S} = 2N_f(4N_f - 1)$. For X on $\mathbb{C}^2_\tau \otimes \mathbb{C}^{N_f}$, let $\widehat{X}$ act as X on the ψ components and as X^{T} on the χ components. It belongs to $\mathscr{S}$. When this notation is applied to a meson label Γ, we suppress its identity factors on spin and colour.

For a chamber v over the physical site x_v, define

$$\phi[M](v) = \frac{1}{2}\widehat{\Psi}(v)^{\mathsf{T}}(\mathcal{J}M \otimes C_0^{-1})\widehat{\Psi}(v), \qquad \Phi[M](f) = a\sum_x f(x) \sum_{v : x_v = x} \phi[M](v).$$

In particular $\phi[\widehat{X}] = \bar{\psi} X \psi$. The blocks of $\mathcal{J}M$ with two ψ indices give the diquarks; those with two χ indices give the antidiquarks. The label 0 will continue to denote the action-density field $O(f)$ and is not the zero matrix in $\mathscr{S}$.

Let

$$G_{\mathbb{C}} = \operatorname{Sp}(2N_f, \mathbb{C})^2, \qquad G = \operatorname{USp}(2N_f)^2. \quad (72)$$

Each factor acts on $P_\pm W$ with its symplectic form $J \otimes 1$. We regard (S_+, S_-) as the block diagonal matrix $S = P_+S_+ + P_-S_-$ on W, and define the algebra automorphism α_S by $\alpha_S\widehat{\Psi}(v) = S^{-1}\widehat{\Psi}(v)$. It acts trivially on functions of the links.

Proposition 5.1 (The doubled field and its symmetries)**.** At every cutoff the following identities hold on a torus and for every infinite-volume state used in [QCDM1]. For $S \in G_{\mathbb{C}}$,

$$\alpha_S \phi[M] = \phi[SMS^{-1}], \qquad \omega \circ \alpha_S = \omega.$$

At a lattice reflection with mirror P,

$$\Theta_P \Phi[M](f) = \Phi[M^*](\overline{f} \circ \theta_P).$$

Every α_S with $S \in G$ commutes with Θ_P. All these fields are gauge invariant and even.

On a finite torus, let $\widehat{D}_U = \sum_{h=+,-} P_h \otimes D_{U,h}$ be the massive operator on the doubled space and let M be a function of the chambers with values in $\mathscr{S}$. Its Berezin integral

$$Z_U[M] = \int d\bar{\psi}\, d\psi \, \exp\left(-\bar{\psi} D_U \psi - \sum_v \phi[M(v)](v)\right)$$

is a polynomial satisfying

$$Z_U[M]^2 = \det(\widehat{D}_U + M), \qquad Z_U[0] = \det D_U.$$

These identities also hold for complex link variables whenever the displayed expressions are defined. ∎

Proof. The relation $\epsilon u^{\mathsf{T}} \epsilon^{-1} = u^{-1}$ for $u \in \mathrm{SL}(2, \mathbb{C})$ and the Pauli identities give $C_0 D_{U,h}^{\mathsf{T}} C_0^{-1} = D_{U,h}$, with the transpose also interchanging the chamber arguments. Consequently $K_h = C_0^{-1} D_{U,h}$ is symmetric. Since odd columns satisfy $b^{\mathsf{T}} A a = -a^{\mathsf{T}} A^{\mathsf{T}} b$,

$$\bar{\psi}_h D_{U,h} \psi_h = \frac{1}{2}\widehat{\Psi}_h^{\mathsf{T}}(J \otimes K_h)\widehat{\Psi}_h.$$

The full quadratic coefficient is therefore $(\mathcal{J}\otimes C_0^{-1})(\widehat{D}_U+M)$, which is antisymmetric. Its Berezin integral is a fixed sign times its Pfaffian. Squaring gives the determinant identity; the normalization at $M=0$ fixes the square of the sign. For $\widehat{X}$, the two off-diagonal blocks of $\mathcal{J}\widehat{X}$ are $-X^{\mathsf{T}}$ and X. Reversing the odd variables in the first term gives $\phi[\widehat{X}]=\chi^{\mathsf{T}}C_0^{-1}X\psi=\bar{\psi}X\psi$.

The identity $S^{\mathsf{T}}\mathcal{J}S=\mathcal{J}$ gives $S^{-\mathsf{T}}\mathcal{J}=\mathcal{J}S$. Hence substitution of $S^{-1}\widehat{\Psi}$ fixes the quadratic action, because S commutes with $P_\pm$, and carries the source coefficient $\mathcal{J}M$ into $\mathcal{J}SMS^{-1}$. Its Berezin Jacobian is one. This proves invariance on a torus. On the infinite lattice the doubled Gaussian functional has covariance

$$\sum_h(-P_h\mathcal{J})\otimes D_{U,h}^{-1}C_0.$$

The same symplectic identity preserves this covariance and therefore every Wick contraction. Averaging in U proves the assertion for the state.

If $\hat{n}$ is the unit normal to P, direct substitution in the reflection of the generators gives

$$\Theta_P\widehat{\Psi}(v)=(\mathcal{J}\tau_3\otimes C_0^{-1}\sigma(\hat{n}))\widehat{\Psi}(\theta_P v).$$

For an antilinear map reversing the order of odd factors, the quadratic coefficient A is changed to $B^{\mathsf{T}}A^{\dagger}B$, where B is the displayed matrix. Inserting $A=\mathcal{J}M\otimes C_0^{-1}$ gives $\mathcal{J}(\tau_3M^{\dagger}\tau_3)\otimes C_0^{-1}$. The same computation on the generators gives $\Theta_P\alpha_S\Theta_P=\alpha_{(S^{\dagger})^{-1}}$, which equals α_S for $S\in G$.

Under a colour transformation, χ transforms by the same matrix as ψ, because $\epsilon u^{-\mathsf{T}}\epsilon^{-1}=u$. The form C_0^{-1} is invariant under this transformation, so every $\phi[M]$ is gauge invariant. Each is quadratic in the generators and hence even. □

Lemma 5.2 (A basis obtained from mesons)**.** The real space $\mathscr{S}_{\mathbb{R}}$ has a basis

$$\mathfrak{B}^{\mathrm{ext}}=\{S_b\widehat{\Gamma}_bS_b^{-1}:b=1,\dots,2N_f(4N_f-1)\},\qquad S_b\in G,\quad \Gamma_b\in\mathfrak{B}.$$

The matrices in this basis have operator norm at most one. There is a constant $c(N_f)$ such that the sum of the absolute values of the coefficients of any $M\in\mathscr{S}$ in this basis is at most $c(N_f)\|M\|$. ■

Proof. Let V be the real span of the displayed symplectic images of meson basis elements. Unitarity and $[S,\tau_3]=0$ imply $(SMS^{-1})^*=SM^*S^{-1}$, so $V\subset\mathscr{S}_{\mathbb{R}}$. We show directly that $V+iV=\mathscr{S}$. Write $r=(h,f)$ for a species–flavour index. Under $M\mapsto\mathcal{J}M$, the meson labels give all blocks with one ψ index and one χ index. Their matrix units are complex linear combinations of the meson basis. The remaining blocks are antisymmetric forms in two ψ indices or two χ indices, hence are spanned by pairs $r\neq s$. For such a pair let $S\in G$ be J on the plane spanned by ψ_r,χ_r and the identity elsewhere. Since $S^{-1}=-J$ on that plane, $\alpha_S\psi_r=\chi_r$ and $\alpha_S\chi_r=-\psi_r$. Thus

$$\alpha_S(\bar{\psi}_r\psi_s)=-\psi_r^{\mathsf{T}}C_0^{-1}\psi_s.$$

Applying partial charge conjugation to the other index instead gives the corresponding antidiquark. These images span every remaining block of $\mathcal{J}M$, proving the claim.

For the real-form conclusion, write any $M\in\mathscr{S}_{\mathbb{R}}$ as $v+iw$, with $v,w\in V$. Since $v^*=v$, $w^*=w$ and $M^*=M$, we have $w=0$. Thus $V=\mathscr{S}_{\mathbb{R}}$. Select a basis from the spanning set. Unitary conjugation preserves operator norm, and the coefficient estimate is equivalence of norms on this fixed finite-dimensional space. □

5.2. **Uniform estimates for the enlarged family.** Put $\mathcal{I}^{\mathrm{ext}}=\{0\}\cup\mathfrak{B}^{\mathrm{ext}}$, $\|f\|_0=\|f\|_\sharp$ and $\|f\|_M=\|f\|_\infty$ for $M\neq0$. An extended tuple is a list $\mathcal{X}=((\lambda_i,f_i))_{i=1}^n$ with these labels; its truncated function is the cumulant of the indicated fields. We extend it complex multilinearly in the labels and functions. Write

$$p=432C_{\mathscr{C}}/\log L,\qquad \kappa=5+p/2.$$

Here $C_{\mathscr{C}}$ is the source-energy counting constant in [QCDM1, Thm. F].

Proposition 5.3 (All scalar bilinears at separated supports)**.** For every $n \geq 2$, $R, d > 0$ and extended tuple whose real Lipschitz functions have supports in balls of radius R, at pairwise distances at least d,

$$|S_n^T(\mathcal{X})| \leq [B_{\mathrm{ext}}(R)(1+d^{-1})^{\kappa-1}]^n (n!)^\kappa \prod_i \|f_i\|_{\lambda_i}. \tag{73}$$

This holds on every torus of the construction and in every fixed-cutoff infinite-volume state obtained from these tori. The constant $B_{\mathrm{ext}}(R)$ is independent of cutoff and volume. For arbitrary labels in $\mathscr{S}$, the right side has the additional factor $\prod_{i:\lambda_i \neq 0} c(N_f)\|\lambda_i\|$. ∎

Proof. We first explain why the one-cell estimate of [QCDM1, Prop. 10.17] applies to a symplectic image of a meson and to cells of side less than one. Let a torus be divided into cubes of physical side $b = L^{-q}$, with an even number of cubes in each direction and walls through links. For a source s supported in one cube at distance $\delta > 0$ from its walls, the same fold used in that proposition has $\mathsf{N} = \ell^3/b^3$ copies. The upper and lower partition estimates cost $c_V \ell^3/\mathsf{N} = c_V b^3$ after taking the Nth root. The separated-energy estimate gives the same counting term $\mathscr{C}_K(\operatorname{supp} s; \delta)$; its proof is local at the support and uses the distance to the walls. Thus, writing $\lambda^{\mathfrak{b}}$ for the average of the reflected single-source energies,

$$\|e^{O(w)+\Phi_\Gamma(c)}\|_{\mathrm{ch}} \leq \exp\left(\operatorname{Re}\lambda^{\mathfrak{b}}(w, c\Gamma) + c_V b^3 + \mathscr{C}_K(\operatorname{supp}(w,c); \delta)\right). \tag{74}$$

The required source domains are unchanged, since reflection preserves the source norms and the source vanishes near the walls. To specify the seminorm, let $\operatorname{Fold} G$ be the product of the reflected copies of an even element G from one cell into all N cells. For the normalized torus functional ω_ℓ, put

$$\|G\|_{\mathrm{ch}} = \omega_\ell(\operatorname{Fold} G)^{1/\mathsf{N}}.$$

Reflection positivity makes the quantity under this root nonnegative. The multiple-cell chessboard inequality is

$$\left|\omega_\ell\left(\prod_i G_i\right)\right| \leq \prod_i \|G_i\|_{\mathrm{ch}},$$

when the G_i belong to distinct cells; the unused cells contain 1. This is the same inequality as in [QCDM1, Prop. 10.17 and its proof], with the finer board.

For $M = S\widehat{\Gamma}S^{-1}$ in the chosen basis,

$$e^{O(w)+\Phi[M](c)} = \alpha_S(e^{O(w)+\Phi_\Gamma(c)}).$$

The automorphism α_S commutes with every reflection in the fold and preserves the torus functional. Therefore the two exponentials have equal chessboard seminorms, and (74) bounds the extended source too. To apply the multiple-cell inequality, first bound each individual seminorm in this way and then multiply the bounds. Thus each cell may use its own matrix S: each individual fold is compared with its meson fold by one global symmetry.

Choose $q \geq 0$ least such that $\sqrt{3}L^{-q} < d$, and set $\eta = L^{-q}/(16n)$. Then

$$\eta^{-1} \leq 16\sqrt{3}Ln(1+d^{-1}).$$

First suppose $L^{K-q} \geq 16n$. Split each f_i by the product hat partition at mesh η. There are at most $(2R/\eta + 3)^3$ pieces for each entry. Each is supported in a cube of side 2η, its supremum norm does not increase, and its sharp norm increases by at most $1 + 80M/\eta$. Fix one piece from each entry. In each coordinate a piece forbids at most $4\eta L^K + 1$ positions for a wall of a board whose cells have $\Delta = L^{K-q}$ sites on a side. The n pieces forbid at most $\Delta/4 + n < \Delta$ positions. Choose walls at the remaining half-integral positions. They are at distance at least η from all pieces. The diameter of a cell is less than d, so no cell contains pieces from two different entries.

For the chosen pieces, let s_i be the corresponding meson source before its symplectic transformation, and let $\Lambda(z)=\sum_i \lambda^{\flat}(z_i s_i)$. The extended generating function is entire, and the chessboard estimate gives, on each coordinate face indexed by $A\subset\{1,\dots,n\}$,

$$\left|e^{-\Lambda(z)}\frac{Z^{\mathrm{ext}}(z)}{Z}\right|\le \exp\left(|A|c_V+\sum_{i\in A}\mathscr{C}_K(\operatorname{supp} s_i;\eta)\right),\qquad |z_i|<\frac{2\varrho_{\lambda_i}}{\|f_i\chi_i\|_{\lambda_i}}.$$

Here $\varrho_0=\varrho$, $\varrho_M=\varrho_J$ for $M\neq 0$, and $b^3\le 1$. Since every term of Λ depends on one variable, its mixed derivative in all $n\ge 2$ variables vanishes. The logarithmic derivative formula, as in the proof of [QCDM1, Thm. F], is

$$\partial_{[n]}\log Z^{\mathrm{ext}}(0)=\sum_{\pi}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{A\in\pi}\partial_A\left(e^{-\Lambda}Z^{\mathrm{ext}}/Z\right)(0).$$

The sum of the absolute combinatorial coefficients is at most $2^n n!$. Writing $G=e^{-\Lambda}Z^{\mathrm{ext}}/Z$, Cauchy's estimate on smaller coordinate circles, followed by the limit as their radii increase to the displayed radii, gives

$$|\partial_A G(0)|\le \exp\left(|A|c_V+\sum_{i\in A}\mathscr{C}_K(\operatorname{supp} s_i;\eta)\right)\prod_{i\in A}\frac{\|f_i\chi_i\|_{\lambda_i}}{2\varrho_{\lambda_i}}.$$

In the product over a partition, each index occurs exactly once. Hence both the source counts and the inverse Cauchy radii have the same total cost for every partition. The refined source count of that theorem gives

$$\mathscr{C}_K(\operatorname{supp} s_i;\eta)\le 216C_{\mathscr{C}}(1+\log_L(1/\eta))+C,$$

because the support has diameter bounded by a fixed multiple of η. Hence its exponential contributes at most $C^n\eta^{-np/2}$.

Summing over the pieces and restoring the original test functions therefore gives

$$|S_n^T(\mathcal{X})|\le n!\left[\frac{Ce^{c_V}}{\varrho_{\min}}(2R/\eta+3)^3(1+80M/\eta)\eta^{-p/2}\right]^n\prod_i\|f_i\|_{\lambda_i}.$$

Here $\varrho_{\min}=\min(\varrho,\varrho_J)$. The three powers of η^{-1} count the pieces, one comes from their sharp norms, and $p/2$ comes from the source-energy count. Since $\eta^{-1}\le Cn(1+d^{-1})$, the right-hand side is at most

$$\left[C'(1+R)^3(1+d^{-1})^{4+p/2}\right]^n n!\,n^{(4+p/2)n}\prod_i\|f_i\|_{\lambda_i}.$$

Applying $n^n\le e^n n!$ gives the factorial exponent $1+(4+p/2)=5+p/2$ in (73).

If $L^{K-q}<16n$, then $a^{-1}<16\sqrt{3}Ln(1+d^{-1})$. The Pfaffian identity above and $\|\widehat{D}_U^{-1}\|\le m_0^{-1}$ give

$$|Z_U[M]|\le \det D_U\,\exp\left(\frac{\|M\|_1}{2m_0}\right).$$

For the source of the extended tuple, $M(v)=-a\sum_i z_i f_i(x_v)\lambda_i$; the doubled fibre has $16N_f$ components and there are 48 chambers per site. Thus the fermion contribution is bounded by $384N_f m_-^{-1}\sum_x\|\sum_i z_i f_i(x)\lambda_i\|$. The action contribution is bounded by $2a^{-1}\sum_p|w(p)|$. A radius-R support has at most $27a^{-3}(1+R)^3$ sites and $81a^{-3}(1+R)^3$ plaquettes. Since $a\le 1$, both costs are bounded by

$$B_a(R)=C(1+N_f/m_-)(1+R)^3a^{-4}$$

times the appropriate test-function norm. Set $r_i=(B_a(R)\|f_i\|_{\lambda_i})^{-1}$, omitting zero test functions. On the face A, the normalized generating function has modulus at most $e^{|A|}$ for $|z_i|\le r_i$. Cauchy's estimate and the cumulant partition formula give

$$|S_n^T(\mathcal{X})|\le 2^n n!e^n B_a(R)^n\prod_i\|f_i\|_{\lambda_i}.$$

Use $a^{-1} < Cn(1+d^{-1})$ and $n^{4n} \le e^{4n}(n!)^4$ to obtain

$$|S_n^T(\mathcal{X})| \le [C(1+N_f/m_-)(1+R)^3(1+d^{-1})^4]^n (n!)^5 \prod_i \|f_i\|_{\lambda_i}.$$

This is bounded by the claimed estimate. Multilinearity and Lem. 5.2 give arbitrary labels.

Finally the expectation of any fixed local Grassmann polynomial converges along the tori defining a state, as proved in [QCDM1, Prop. 2.5]. A cumulant is a polynomial in finitely many such expectations. Passing to the limit proves the same estimate in the infinite-volume state. □

Proposition 5.4 (The enlarged limits)**.** Let S be a volume-first limit of [QCDM1], with a fixed defining sequence. There is a subsequence along which all extended truncated functions of orders $n \ge 2$ converge on pairwise disjoint C_c^1 supports. Their limits S^{ext} restrict to S on meson tuples and satisfy (73).

They are symmetric, translation invariant, invariant under simultaneous conjugation of the labels by $G_{\mathbb{C}}$. At fixed labels, these truncated functions determine distributions of locally finite order away from the diagonals. The centered functional σ^{ext}, obtained by the moment–cumulant formula with one-point functions zero, is reflection positive at every coordinate plane, with the involution $M \mapsto M^*$. It defines a separable reflection Hilbert space $\mathcal{H}^{\mathrm{ext}}$. Time translation gives a strongly continuous self-adjoint contraction semigroup e^{-tH}, $H \ge 0$; spatial translations give a strongly continuous unitary representation with generators P_1, P_2 commuting strongly with H. ■

Proof. Fix finitely many pairwise disjoint closed supports, each in a bounded ball. The uniform bound makes the truncated functions equicontinuous in the product of the indicated norms. Enumerate rational unions of closed cubes, a countable dense family of C_c^1 functions on these supports, and the finite label basis. A diagonal subsequence makes every corresponding truncated function converge. Equicontinuity extends the convergence to every tuple, and multilinearity extends the labels to $\mathscr{S}$. The original meson functions converge along the given defining sequence, so their limits remain S.

Symmetry and $G_{\mathbb{C}}$-invariance are exact at every cutoff. They pass to the limit. Lattice translations approximate every real translation; the estimate and translation continuity in C^1 give translation invariance. For a positive-side tuple, replace each field by its lattice field minus its expectation. The reflection sends this centered field to the centered field with label M^* and reflected conjugate test function. Every reflection form is therefore the finite sum of products of truncated functions specified by the centered moment–cumulant formula. Approximate a coordinate plane by bond planes. The supports stay a positive distance from the planes, and the reflected test functions converge in their insertion norms. Lattice reflection positivity and the bound pass the form inequality to the limit.

On a product of open sets with disjoint closures, the bound makes the limiting truncated functional continuous in the C^1 norms. The Schwartz kernel theorem gives a distribution there. Uniqueness makes these local distributions agree on intersections, so they define one distribution off the diagonals.

On finite linear combinations of positive-time tuples, define the positive semidefinite form by

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \sigma^{\mathrm{ext}}(\theta_3 \mathcal{X}^* \cdot \mathcal{Y}).$$

We first check that time translation descends to its quotient. Write $T_t\mathcal{X} = \tau_{te_3}\mathcal{X}$, $t \ge 0$. Translation invariance and $\theta_3 \tau_{te_3} = \tau_{-te_3}\theta_3$ give

$$\langle T_s v, T_t v \rangle = \langle v, T_{s+t} v \rangle, \qquad \|T_t v\|^2 = \langle v, T_{2t} v \rangle.$$

For a fixed v, these correlations are bounded uniformly for $s, t \ge 0$. Indeed, each is a finite sum of products of truncated correlations; the support radii stay fixed, the separations within each tuple stay fixed, and the reflected tuple stays a positive distance from the translated tuple.

Fix $t > 0$ and put $a_k = \|T_{kt} v\|^2$. Cauchy–Schwarz for the semidefinite form gives

$$a_k^2 = |\langle T_{(k-1)t} v, T_{(k+1)t} v \rangle|^2 \le a_{k-1} a_{k+1} \qquad (k \ge 1).$$

If $a_0 = 0$, this implies $a_1 = 0$. If $a_0 > 0$ and $a_1 > a_0$, all subsequent terms are positive and the ratios a_{k+1}/a_k are nondecreasing. They are then bounded below by $a_1/a_0 > 1$, contrary to boundedness. Thus $\|T_t v\| \le \|v\|$. In particular null vectors remain null, so T_t defines a contraction on the quotient and its completion $\mathcal{H}^{\mathrm{ext}}$. The empty tuple gives the unit vector Ω.

Translation is symmetric for the form, so each bounded operator T_t is self-adjoint. The semigroup law is the translation law on tuples. In particular

$$\langle v, T_t v\rangle = \|T_{t/2} v\|^2 \ge 0.$$

This is also the reflection form at the intervening plane. For a vector represented by finitely many tuples,

$$\|T_t v - v\|^2 = \langle v, T_{2t} v\rangle - 2\,\mathrm{Re}\langle v, T_t v\rangle + \|v\|^2 \longrightarrow 0 \quad (t \downarrow 0),$$

by continuity in the translated test functions. Contraction extends strong continuity to the completion. The spectral theorem for a strongly continuous self-adjoint contraction semigroup now gives $T_t = e^{-tH}$ with $H \ge 0$.

Spatial translations preserve the form and have inverses. They extend to strongly continuous unitary operators, and their commutation with the semigroup on tuples gives strong commutation of the generators. For separability, use the positive-time members of the countable family used for extraction, and take finite linear combinations with rational real and imaginary coefficients. Any positive-time tuple can be approximated by such tuples with supports in fixed closed sets that remain separated and stay away from the reflection plane. Its squared reflection-norm error is a finite combination of moments, each a polynomial in continuous truncated functions. The error therefore tends to zero, proving density of this countable set. □

Proposition 5.5 (The internal group on the enlarged space)**.** The map $[X] \mapsto [X]$ embeds the mesonic space $\mathcal{H}(S)$ isometrically into $\mathcal{H}^{\mathrm{ext}}$ and intertwines time and spatial translations. Moreover

$$V(S)[\mathcal{X}] = [S\mathcal{X}S^{-1}], \qquad S \in G,$$

is a strongly continuous unitary representation. It fixes Ω, commutes with H, P_1, P_2, and is trivial on $(-1,-1)$. Let Σ_3 be 1 on the ψ components and -1 on the χ components. The generator Q defined by

$$V(e^{i\varphi\Sigma_3}, e^{i\varphi\Sigma_3}) = e^{-i\varphi Q}$$

has spectrum in $2\mathbb{Z}$. The mesonic space lies in $\ker Q$, and both eigenspaces $Q = 2$ and $Q = -2$ are nonzero. Charge conjugation $V(J, J)$ exchanges them. ■

Proof. The two centered functionals agree on meson tuples, so the displayed embedding preserves inner products. Both translations act on the functions and are therefore intertwined. For $S \in G$, $(SMS^{-1})^* = SM^*S^{-1}$. Invariance of σ^{ext} proves that $V(S)$ preserves the form and its null space. Its inverse is $V(S^{-1})$. For a fixed tuple the orbit lies in a finite-dimensional span of label tuples and depends continuously on S; this proves strong continuity. The stated commutations hold on tuples and hence on the completion.

The central element $(-1,-1)$ is the identity on all quadratic labels. Setting $\varphi = \pi$ therefore gives $e^{-i\pi Q} = 1$, so the spectral theorem places the spectrum of Q in $2\mathbb{Z}$. Every meson is neutral, and the mesonic space lies in its kernel.

To see that the charge-two space is nonzero, choose the meson $N_f^{-1/2} \sum_f \bar\psi_{+,f}\psi_{-,f}$. Its matrix is $\tau_+ \otimes t_1$, $\tau_+ = (\tau_1 + i\tau_2)/2$, a linear combination of $\Gamma_{1,1}$ and $\Gamma_{2,1}$. The leading two-point formula of [QCDM1, Prop. 11.1] has a positive definite leading coefficient on this two-dimensional span. At a pair of reflected small positive test functions its relative error tends to zero as their scale tends to zero. Thus the associated one-field vector has positive norm. The element of G equal to $J \otimes 1$ on the plus half and to the identity on the minus half sends this meson to a nonzero multiple of $\sum_f \psi_{+,f}^{\mathsf{T}} C_0^{-1} \psi_{-,f}$. Its image under $V(S)$ has the same positive norm and charge two. Finally

$J\Sigma_3 J^{-1} = -\Sigma_3$, so $V(J,J)QV(J,J)^{-1} = -Q$. This gives the nonzero charge-minus-two space as well. □

6. Energy in the charged sectors

The estimate needed in the continuum has constant one. A direct propagator estimate initially gives a constant depending on the cutoff. Reflection positivity removes that constant before the cutoff is taken to zero.

6.1. **A sharp decay rate on the chamber lattice.** In this subsection lattice spacing is one. Let

$$L_\rho = \frac{1}{2}\sum_{s=1}^{3} t_s(1-\widehat{R}_s) \quad \text{on } \ell^2(W_3), \qquad E = \{g : ge_1 = \pm e_3\},$$

where $\widehat{R}_s$ is right multiplication by the within-site chamber reflection w_s. The kernel of L_ρ consists of the constants; let $\lambda_\rho > 0$ be its first positive eigenvalue. These are the within-site operator and its gap used in [QCDM1, Section 5]. There are 16 elements in E and 48 in W_3. Define

$$\begin{aligned} &h(k) = -\lambda_{\min}(L_\rho - k1_E), \qquad k(\alpha) = \tfrac{3}{2}(\cosh\alpha - 1) + 3\sinh\alpha, \\ &e(\alpha) = h(k(\alpha)), \quad 0 \le \alpha \le \operatorname{artanh}(1/2), \qquad \mu(m_0) = \sup\{\alpha : e(\alpha) < m_0\}. \end{aligned} \tag{75}$$

Lemma 6.1 (Directional decay)**.** If $\varphi(v) = \phi((x_v)_3)$ is real and bounded, with increments of modulus at most $\alpha \le \operatorname{artanh}(1/2)$, then

$$\operatorname{Re}(e^{\varphi} D_U e^{-\varphi}) \ge m_0 - e(\alpha).$$

If two sets of sites A, B are at separation at least d in their third coordinates, then, for $0 \le \alpha < \mu(m_0)$,

$$\|1_A D_U^{-1} 1_B\| \le (m_0 - e(\alpha))^{-1} e^{-\alpha d}.$$

The estimates are uniform in U. Moreover $\mu(x)/x \to 1$ as $x \downarrow 0$, and consequently $a^{-1}\mu(a\mathsf{m}) \to \mathsf{m}$ uniformly for m in compact subsets of $(0,\infty)$. ■

Proof. Write $R_s f(v) = U(v, v\cdot s) f(v\cdot s)$. This is a self-adjoint unitary involution. Conjugation changes the coefficient of the sth hop to $t_s(\frac{1}{2} - \gamma(u_s(v)))e^{-w_s(v)}$, where $w_s(v) = \varphi(v\cdot s) - \varphi(v)$ and $w_s(v\cdot s) = -w_s(v)$. Its Hermitian part is $B_s R_s$, with

$$B_s(v) = t_s\big(\tfrac{1}{2}\cosh w_s(v) + \gamma(u_s(v))\sinh w_s(v)\big).$$

The matrix B_s commutes with R_s. For $s = 1,2,3$, $w_s = 0$, since the hop stays at the same site. For $s = 0$, it vanishes outside E. Its two eigenvalues are $3(\frac{1}{2}\cosh w \pm \sinh w)$, nonnegative in the stated interval. Hence

$$\operatorname{Re}(e^{\varphi} D_U e^{-\varphi}) = m_0 + L_\rho + (\tfrac{3}{2} - B_0) + B_0(1 - R_0) \ge m_0 + L_\rho - k(\alpha)1_E.$$

This proves the first estimate.

Choose $\phi(s) = \alpha \min\{\operatorname{dist}(s, B_3), d\}$, where B_3 is the set of third coordinates of B. The weight is zero on B and αd on A. The first estimate makes the conjugated inverse bounded by $(m_0 - e(\alpha))^{-1}$, proving decay.

Let u be the constant unit vector in $\ell^2(W_3)$. There are 16 moving chambers among the 48 chambers, so $\langle u, 1_E u\rangle = 16/48 = 1/3$. Moreover

$$\|(1 - |u\rangle\langle u|)1_E u\|^2 = \langle u, 1_E u\rangle - |\langle u, 1_E u\rangle|^2 = \tfrac{1}{3} - \tfrac{1}{9} = \tfrac{2}{9}.$$

Writing a vector as $cu + v$, $v \perp u$, and using $L_\rho \ge \lambda_\rho$ on $u^\perp$, gives, for $0 \le k < 3\lambda_\rho/2$,

$$\frac{k}{3} \le h(k) \le \frac{k}{3} + \frac{2k^2}{9\lambda_\rho - 6k}.$$

Indeed the lower bound uses the trial vector u. For the upper bound the quadratic form is bounded below by $-(k/3+\delta)\|(c,v)\|^2$ when

$$\begin{pmatrix} \delta & -\sqrt{2}k/3 \\ -\sqrt{2}k/3 & \lambda_\rho - 2k/3 + \delta \end{pmatrix} \geq 0; \qquad \delta = \frac{2k^2}{9\lambda_\rho - 6k}$$

satisfies this condition. Since $k(\alpha) = 3\alpha + O(\alpha^2)$, we have $e(\alpha) = \alpha + O(\alpha^2)$. Inversion at zero gives $\mu(x) = x + O(x^2)$ and the claimed uniform limit. □

Lemma 6.2 (Reflection positivity fixes the prefactor)**.** Let T be symmetric on a vector space with a positive semidefinite Hermitian form. If $\langle v, T^{2k}v\rangle \leq C\lambda^{2k}$ for all $k \geq 0$, where $0 < \lambda \leq 1$, then

$$\langle v, T^{2k}v\rangle \leq \lambda^{2k}\langle v, v\rangle, \qquad |\langle v, T^k v\rangle| \leq \lambda^k \langle v, v\rangle.$$

■

Proof. Put $a_k = \langle T^k v, T^k v\rangle$. Cauchy–Schwarz gives $a_k^2 \leq a_{k-1}a_{k+1}$. If one a_k with $k \geq 1$ vanishes, this inequality forces all such a_k to vanish. Otherwise the ratios a_{k+1}/a_k are nondecreasing. A ratio larger than λ^2 would contradict $a_k \leq C\lambda^{2k}$ at large k. Thus $a_k \leq \lambda^{2k}a_0$. A final Cauchy–Schwarz inequality gives the second assertion. □

Proposition 6.3 (The lattice charged estimate)**.** Fix $m_0 > 0$ and an infinite-volume torus-limit state ω_ν. Let Θ be reflection in $x_3 = 1/2$ and let τ be translation by e_3. For an even local element F supported in $x_3 \geq 1$, form its reflection class $[F]$. Translation induces a positive self-adjoint contraction T on the resulting Hilbert space.

Let $\mathcal{F} = \{+,-\} \times \{1, \ldots, N_f\}$. The weight of a monomial is $n_f = \#\psi_f - \#\bar{\psi}_f$ and $|n|_1 = \sum_f |n_f|$. If F has weight n, then

$$0 \leq \omega_\nu(\tau^k F\,\Theta F) \leq e^{-\mu(m_0)|n|_1 k}\,\omega_\nu(F\,\Theta F), \qquad k \geq 0. \tag{76}$$

If the conditional Gaussian expectation $h_F(U) = \langle F\rangle_U$ vanishes at every U, the same estimate holds with 2 in place of $|n|_1$. ■

Proof. The fixed-spacing states of [QCDM1] are translation invariant and reflection positive at coordinate planes through sites and through links. Thus the form $\langle [G], [F]\rangle = \omega_\nu(F\Theta G)$ is positive, and $\Theta\tau = \tau^{-1}\Theta$ makes translation symmetric. Wick's determinant formula and $\|D_U^{-1}\| \leq m_0^{-1}$ bound $|\omega_\nu(\tau^k F\Theta F)|$ uniformly in k, with a constant depending on F, m_0. Apply Lem. 6.2 with $\lambda = 1$ to obtain a contraction on the quotient. Positivity follows from reflection through $x_3 = 1$, since $\Theta F = \Theta_1(\tau F)$ and $\omega_\nu(\tau F \Theta_1 \tau F) \geq 0$.

For clarity, consider a pair of monomials of total degree $2p$. Wick's formula is a sum of at most $p!$ products of p propagators. A propagator pairs only a quark with an antiquark of the same species. If the monomial on the positive side has weight n, let c_f^+ and c_f^- count its quarks and antiquarks of species f that are paired across the separation. Each pairing confined to this side removes one quark and one antiquark of the same species. Therefore

$$c_f^+ - c_f^- = n_f, \qquad c_f^+ + c_f^- \geq |n_f|.$$

Summing over species shows that at least $|n|_1$ propagators cross. Between the supports of ΘF and $\tau^{2k}F$, their separation in the third coordinate is at least $2k+1$. By Lem. 6.1, for every $\alpha < \mu(m_0)$,

$$\omega_\nu(\tau^{2k}F\Theta F) \leq C(F,\alpha)e^{-\alpha|n|_1(2k+1)}.$$

The finite sum over the monomials only changes $C(F,\alpha)$. Applying Lem. 6.2 removes this constant. Let $\alpha \uparrow \mu(m_0)$ to obtain (76). This removal is performed at each fixed cutoff. The preliminary constant may depend on the cutoff; the final estimate depends on F only through its squared reflection norm, which can subsequently pass to the continuum limit.

For the last assertion subtract the product of the two conditional Gaussian expectations in Wick's formula. The permutations preserving both sets of factors cancel. Every remaining permutation

crosses the separation at least twice: if each monomial is neutral, one crossing in each direction is required; otherwise its even nonzero charge already requires at least two crossings. Since $h_F = 0$, the subtracted product vanishes. This gives $C(F,\alpha)e^{-2\alpha(2k+1)}$ and the same argument proves the asserted bound. □

6.2. **Passing the estimate to the enlarged limit.** Let $\mathcal{H}_n^{\mathrm{ext}}$ be the closed span of positive-time tuples of total weight n. A quadratic label has quark number 0 or ± 2, and every label is a sum of weight vectors. For an irreducible representation π of G, write $|\pi|$ for the largest ℓ^1 norm of one of its maximal-torus weights. This is the ℓ^1 norm of its highest weight: the weights lie in the convex hull of its signed-permutation Weyl orbit.

The three estimates below distinguish three notions of charge. The weight $n = (n_f)_{f\in\mathcal{F}}$ records, for each species, the number of ψ_f factors minus the number of $\bar\psi_f$ factors. On its weight space, the maximal torus acts by $V(z)v = z^{-n}v$. Taking every phase equal gives the quark number

$$q = \sum_{f\in\mathcal{F}} n_f, \qquad \{Q = q\} = \bigoplus_{\sum_f n_f = q} \mathcal{H}_n^{\mathrm{ext}}.$$

Thus quark number retains only the sum of the species charges. A meson $\bar\psi_{+,f}\psi_{-,f}$ has quark number zero but weight of ℓ^1 norm two. The weight estimate below therefore gives a positive lower bound on such weight spaces within the quark-number-zero sector.

The π-isotypic space is the closed span of all copies of the irreducible representation π. It contains the whole multiplet, and may include vectors of weight zero. Since H commutes with G, a spectral subspace of H in this space contains whole copies of π. Applying the weight estimate to a highest-weight vector therefore bounds the energy throughout the multiplet. This is stronger than applying the weight estimate separately to its zero-weight vectors.

Theorem E (Energy and internal charges)**.** Let S^{ext} be an extended volume-first limit and let m be a limit point of the rescaled mass parameters $\mathsf{m}_j = a_j^{-1}m_{0,j} \in [m_-, m_+]$ along its defining sequence. Then

$$H|_{\mathcal{H}_n^{\mathrm{ext}}} \geq |n|_1\mathsf{m}, \qquad H|_{\mathcal{H}_\pi^{\mathrm{ext}}} \geq |\pi|\mathsf{m}, \qquad H|_{\{Q=q\}} \geq |q|\mathsf{m}.$$

Only even $|n|_1$ and even quark numbers occur. Every nontrivial representation that occurs has $|\pi| \geq 2$. In particular,

$$H \geq 2\mathsf{m} \quad \text{on } ((\mathcal{H}^{\mathrm{ext}})^G)^\perp, \qquad \ker H \subset (\mathcal{H}^{\mathrm{ext}})^G.$$

The spectral projection onto $\ker H$ vanishes on every nonzero-weight space. The charge-two and charge-minus-two spaces are nonzero. ■□

The gap $2\mathsf{m}$ here is on the complement of the G-invariant vectors. In the invariant sector these estimates give only $H \geq 0$. Since the vacuum vector is invariant and has energy zero, the conclusion is

$$\mathbb{C}\Omega \subset \ker H \subset (\mathcal{H}^{\mathrm{ext}})^G.$$

Neither uniqueness of the vacuum nor a positive gap above it within the invariant sector follows from this theorem.

Proof. Pass to a subsequence with $\mathsf{m}_j \to \mathsf{m}$. For a finite combination of positive-time tuples $\mathcal{X} = \sum_i \zeta_i\mathcal{X}_i$, with $\mathcal{X}_i = ((\lambda_{ir}, f_{ir}))_r$, put

$$F_j = \sum_i \zeta_i \prod_r \left(\Phi_j[\lambda_{ir}](f_{ir}) - \omega_{\nu_j}(\Phi_j[\lambda_{ir}](f_{ir}))\right).$$

Here $\Phi_j[0](f)$ means the action-density insertion. The moment–cumulant formula expresses the correlations of these centered polynomials using only blocks of at least two insertions. Thus only the truncated functions already known to converge occur; convergence of the individual one-point

functions is unnecessary. If $\mathcal{X}$ has weight n, then F_j has weight n: the mean of a nonzero-weight field vanishes by the torus symmetry, and subtraction does not change a zero weight.

For large j the support lies above the bond plane $x_3 = a_j/2$. Fix t in $\bigcup_k L^{-k}\mathbb{N}$. Then t/a_j is an integer for all large j, and Prop. 6.3 gives

$$\omega_{\nu_j}(\tau^{t/a_j} F_j \Theta_{a_j/2} F_j) \le e^{-|n|_1 t\mu(a_j \mathsf{m}_j)/a_j}\, \omega_{\nu_j}(F_j \Theta_{a_j/2} F_j).$$

Reflection in $a_j/2$ differs from reflection in zero by a translation of size a_j. For C^1_c functions that translation tends to the identity in both $\|\cdot\|_\infty$ and $\|\cdot\|_\sharp$; in the latter case use uniform continuity of the gradient. The supports remain separated. Hence (73), multilinearity and the moment–cumulant formula let both sides pass to the limit. The limiting rate is m by Lem. 6.1, and

$$\langle [\mathcal{X}], e^{-tH}[\mathcal{X}]\rangle \le e^{-|n|_1 \mathsf{m} t}\, \|[\mathcal{X}]\|^2.$$

Strong continuity extends this from the dense set of times to all $t \ge 0$.

The maximal-torus projections $\int z^n V(z)\, dz$ have ranges $\mathcal{H}_n^{\mathrm{ext}}$; they are orthogonal, sum to the identity and commute with H. The Laplace transform inequality forces the spectral measure of H on each such space to be supported in $[|n|_1\mathsf{m}, \infty)$. Indeed positive mass on $[0, c]$ with $c < |n|_1\mathsf{m}$ would contradict the inequality as $t \to \infty$.

Every spectral projection of H commutes with G. If the isotypic subspace of π had a nonzero spectral part below $|\pi|\mathsf{m}$, that part would contain a copy of π, hence a vector of its highest weight. This contradicts the weight estimate. An even monomial has $|n|_1 \equiv \sum_f n_f \equiv 0 \pmod 2$. The highest weight of any occurring representation is an occurring weight, so its norm is even and is at least two unless the representation is trivial. Compact-group decomposition proves the estimate on the complement of the invariant vectors. The charge estimate follows by summing weight spaces and using $|n|_1 \ge |\sum_f n_f|$. The conclusions about the kernel follow from $\mathsf{m} \ge m_- > 0$, and the last assertion is Prop. 5.5. □

7. Decay of parity-odd meson correlations

The parity argument needs control at coinciding supports only for a conditional expectation over the fermions. The loop-variance estimate of Thm. B supplies that control. The part with a quark propagator joining the two supports is estimated at separated supports.

Write $h_F(U) = \langle F\rangle_U$ and $z(f) = a\sum_x f(x)\,\mathrm{tr}\, D^{-1}_{U,+}(x,x)$, where this trace is over spin, colour and the chambers over x, for one flavour. We use the consequence

$$\mathrm{Var}_\nu\, z(f) \le \frac{\|f\|_\infty^2}{4\varrho_J^2} \min\left\{ C(R)^{1/N_f}, \frac{C(R)-1}{N_f^2}\right\} \tag{77}$$

of Thm. B, for real bounded f supported in a radius-R ball. Its hypotheses and the corresponding restriction on the coupling are assumed throughout this section.

7.1. Decay of the part joining the supports.

Let $c_h = 3(1+r) = 9/2$ and $G^+_U = D^{-1}_{U,+}$. For nonnegative functions define

$$\Pi_U(f,h) = a^2 \sum_{x,y} f(x) h(y) \|G^+_U(x,y)\|^2_{\mathrm{HS}}.$$

Lemma 7.1 (A uniform estimate for one quark loop)**.** Fix $R, d > 0$. There is a number $C'(R,d)$, independent of the cutoff and the state, such that, when $a \le d/4$ and $a\mathsf{m} \le 2c_h$, the following holds. Suppose $f, h \in C^1_c$ have supports in $B(x_0, R), B(y_0, R)$ and $|x_0 - y_0| \ge 2R + 3d$. For any two meson labels Γ, Γ', the mean of their conditional connected function is bounded in modulus by

$$N_f C'(R,d) \|\Gamma\| \|\Gamma'\| \|f\|_\infty \|h\|_\infty e^{-m_-(|x_0-y_0|-2R-3d)/(2c_h)}. \tag{78}$$

■

Proof. First, for the inverse between arbitrary site sets A, B, conjugating by a bounded weight with nearest-site increments at most α changes the hopping operator by norm at most $c_h(e^{\alpha}-1)$. Choosing $\alpha = \log(1 + m_0/(2c_h))$ gives

$$\|1_A G_U^{+} 1_B\| \le \frac{2}{m_0} e^{-\alpha \operatorname{dist}_{\mathrm{graph}}(A,B)}.$$

For $m_0 \le 2c_h$, $\alpha \ge m_0/(4c_h)$.

Let $d(x) = \operatorname{dist}(x, B(x_0, R))$ and $\eta(x) = \min\{1, \max\{0, 2 - d(x)/d\}\}$. The commutator $[D_{U,+}, \eta]$ is supported in

$$A_d = \{x : d - a \le d(x) \le 2d + a\}, \qquad \|[D_{U,+}, \eta]\| \le c_h a/d.$$

For f supported where $d(x) \le d$ and a set $B \subset \{d(x) \ge 2d\}$ at Euclidean distance D_0 from A_d, the resolvent commutator identity gives

$$f^{1/2} G_U^{+} 1_B = f^{1/2} G_U^{+} [D_{U,+}, \eta] G_U^{+} 1_B.$$

Take a Hilbert–Schmidt norm in the first factor and an operator norm in the last factor, and square. The preceding decay estimate yields

$$\Pi_U(f, 1_B) \le \left(\frac{2c_h}{d\mathsf{m}}\right)^2 e^{-\mathsf{m} D_0/(2c_h)} \Pi_U(f, 1_{A_d}). \tag{79}$$

Choose fixed smooth nonnegative functions φ, g such that $\varphi = 1$ on $B(x_0, R)$ and vanishes where $d(x) \ge d/4$, while $g \ge 1$ where $3d/4 \le d(x) \le 9d/4$ and vanishes outside $d/2 < d(x) < 5d/2$. Their supports have distance at least $d/4$. For $\Gamma_* = i\tau_1 \otimes N_f^{-1/2} 1$, Wick's formula and $G_U^{-} = (G_U^{+})^{\dagger}$ give

$$S_2^T[\nu]((\Gamma_*, \varphi), (\Gamma_*, g)) = 2 \int \Pi_U(\varphi, g)\, d\nu.$$

The conditional one-point functions vanish for this label. The separated two-point bound of [QCDM1] therefore bounds the nonnegative integral on the right, uniformly in the cutoff and the state.

For two general labels the conditional connected function is

$$-a^2 \sum_{x,y} f(x) h(y) \operatorname{tr}[\Gamma G_U(x,y) \Gamma' G_U(y,x)].$$

Decomposing into the two τ_3 blocks and using Cauchy–Schwarz for their Hilbert–Schmidt norms bounds this by $2N_f \|\Gamma\| \|\Gamma'\| \Pi_U(|f|, |h|)$. The transposed charge-conjugation identity gives

$$\|G_U^{+}(x,y)\|_{\mathrm{HS}} = \|G_U^{+}(y,x)\|_{\mathrm{HS}}.$$

Apply (79); the separation from $\operatorname{supp} h$ to A_d is at least $|x_0 - y_0| - 2R - 3d$. The displayed separated meson bound then proves (78), with, for example,

$$C'(R,d) = \left(\frac{2c_h}{d m_-}\right)^2 C_2(d/4; R + 3d) \|g\|_{\infty},$$

where C_2 is the companion's separated two-point constant. □

7.2. **The parity tail.** Fix a coordinate direction e_μ. Let c and $w > 0$ belong respectively to $\bigcup_k L^{-k} \frac{1}{2}\mathbb{Z}$ and $\bigcup_k L^{-k}\mathbb{Z}$. Let θ_c be reflection in $x_\mu = c$, and θ_0 reflection in the middle plane $x_\mu = c + w/2$. For

$$\mathcal{F} = \sum_{k=1}^{p} c_k(\Gamma_k, f_k), \qquad \|\mathcal{F}\| = \sum_k |c_k| \|\Gamma_k\| \|f_k\|_{\infty},$$

assume $\Gamma_k \in \mathfrak{B}$ and that the real functions $f_k \in C^1_c$ are supported in the open slab $c < x_\mu < c + w$ and in a ball $B(x_*, R)$ centered on its middle plane. The parity signs π_Γ are those of [QCDM1]. Put

$$\mathcal{F}^- = \frac{1}{2}\sum_k c_k\big((\Gamma_k, f_k) - \pi_{\Gamma_k}(\Gamma_k, f_k \circ \theta_0)\big), \qquad s_{\mathcal{F}}(t) = \sigma_S(\theta_c(\mathcal{F}^-)^* \cdot \tau_{te_\mu}\mathcal{F}^-).$$

Proposition 7.2 (Decay of the parity-odd part)**.** Let S be a volume-first limit satisfying the hypotheses of Thm. B. Then $s_{\mathcal{F}}$ is continuous, nonnegative and nonincreasing on $(0, \infty)$. For each cutoff of its defining sequence let F_j^- be the polynomial associated to $\mathcal{F}^-$ and set

$$C_{\mathcal{F}} = \limsup_j \int |h_{F_j^-}|^2 \, d\nu_j.$$

This number is finite, with

$$C_{\mathcal{F}} \le \varrho_J^{-2}\|\mathcal{F}\|^2 \min\{N_f^2 C(R)^{1/N_f}, C(R) - 1\}.$$

For every $d, t_0 > 0$ with $t_0 + w \ge 2R + 3d$,

$$\int_{t_0}^{\infty} s_{\mathcal{F}}(t)\, dt \le (t_0 + w)C_{\mathcal{F}} + N_f C'(R, d)\|\mathcal{F}\|^2 \frac{2c_h}{m_-} e^{-m_-(t_0+w-2R-3d)/(2c_h)}. \tag{80}$$

Consequently $s_{\mathcal{F}}(t) \to 0$ as $t \to \infty$. ∎

Proof. At all sufficiently fine cutoffs the two planes are lattice planes and $m = w/a$ is an integer. Let V_{θ_0} be the linear parity transformation on the fermions, using the matrix $\sigma(e_\mu) \otimes \tau_2$. Then $V_{\theta_0}\Phi_\Gamma(f) = \pi_\Gamma \Phi_\Gamma(f \circ \theta_0)$. Thus $V_{\theta_0} F_j^- = -F_j^-$ and its expectation vanishes. Put $A_j = h_{F_j^-}$ and $\Xi_j = [F_j^-]$ in the reflection space at $x_\mu = c$.

For a meson label $\Gamma = 1_{\mathrm{s}} \otimes X \otimes 1_{\mathrm{c}}$, Wick's rule gives

$$h_{\Phi_\Gamma(f)} = -\alpha_\Gamma z(f) - \beta_\Gamma \overline{z(f)}, \quad \alpha_\Gamma = \operatorname{tr}(e_+ X), \quad \beta_\Gamma = \operatorname{tr}(e_- X), \quad |\alpha_\Gamma|, |\beta_\Gamma| \le N_f\|\Gamma\|.$$

Since A_j has mean zero, subtracting the means of these loops does not change their combination. All functions involved are supported in the same ball, which is preserved by θ_0. Minkowski's inequality and (77) give

$$\|A_j\|_2 \le \varrho_J^{-1}\|\mathcal{F}\| \min\{N_f C(R)^{1/(2N_f)}, (C(R) - 1)^{1/2}\}.$$

This proves the stated uniform estimate.

We next retain the sign coming from parity. Define the antilinear conjugation on generators by $\psi^\sharp = (\bar\psi\tau_2)^{\mathsf{T}}$, $\bar\psi^\sharp = (\tau_2\psi)^{\mathsf{T}}$, and reverse products. The identities on the doubled field give

$$\Theta_0 = \alpha_{(1,-1)} V_{\theta_0} \sharp, \qquad h_{F^\sharp} = \overline{h_F}, \qquad h_{\alpha_S F} = h_F.$$

As $\theta_c = \tau^{-m}\theta_0$ in lattice units, putting $B_j = \tau^{-m}\alpha_{(1,-1)}((F_j^-)^\sharp)$ gives $\Theta_c F_j^- = -B_j$ and $h_{B_j} = \tau^{-m}\overline{A_j}$. Hence

$$s_j(n) := \langle \Xi_j, T^n \Xi_j\rangle = -\langle A_j, \tau^{n+m} A_j\rangle_{L^2(\nu_j)} - R_j(n), \tag{81}$$

where $R_j(n)$ is the averaged conditional connected function of $\tau^n F_j^-$ and B_j. The two supports are in radius-R balls whose centers have distance $na + w$. Applying Lem. 7.1 to their finitely many meson terms gives, for $na + w \ge 2R + 3d$,

$$|R_j(n)| \le N_f C'(R, d)\|\mathcal{F}\|^2 e^{-m_-(na+w-2R-3d)/(2c_h)}.$$

Each of the four reflected/unreflected pairings has coefficient $1/4$, so no extra factor is needed.

The sequence $l \mapsto \langle A_j, \tau^l A_j\rangle$ is of positive type, since translation is unitary on $L^2(\nu_j)$. The relevant finite inequality is

$$0 \le \frac{1}{N}\left\|\sum_{l=0}^{N-1} \tau^l A_j\right\|_2^2 = \|A_j\|_2^2 + 2\sum_{l=1}^{N-1}(1 - l/N)\operatorname{Re}\langle A_j, \tau^l A_j\rangle.$$

Let n_0 be the least integer with $n_0 a \geq t_0$ and take $N > m + n_0$. For $l < m + n_0$ bound the real part by $\|A_j\|_2^2$. For $l \geq m + n_0$, the minus sign in (81) gives

$$\mathrm{Re}\langle A_j, \tau^l A_j\rangle \leq -s_j(l-m) + |R_j(l-m)|.$$

Consequently

$$\begin{aligned} 2\sum_{n=n_0}^{N-m-1}\left(1-\frac{n+m}{N}\right)s_j(n) &\leq \left[1+2\sum_{l=1}^{m+n_0-1}\left(1-\frac{l}{N}\right)\right]\|A_j\|_2^2 \\ &\quad + 2\sum_{n=n_0}^{N-m-1}\left(1-\frac{n+m}{N}\right)|R_j(n)| \\ &\leq [2(m+n_0)-1]\|A_j\|_2^2 + 2\sum_{n\geq n_0}|R_j(n)|. \end{aligned}$$

Because $0 \leq T \leq 1$, every $s_j(n)$ is nonnegative. The weights on the left increase to one as $N \to \infty$, whereas the remainder on the right is summable by its exponential bound. Monotone convergence therefore gives

$$\sum_{n\geq n_0} s_j(n) \leq (m + n_0 - \tfrac{1}{2})\|A_j\|_2^2 + \sum_{n\geq n_0}|R_j(n)|.$$

Multiply by a. The first coefficient is at most $w + t_0 + a/2$, and the geometric-series factor in the second term is $a/(1 - e^{-m_- a/(2c_h)}) \to 2c_h/m_-$.

At every time t in $\bigcup_k L^{-k}\mathbb{N}$, $s_j(t/a_j) \to s_{\mathcal{F}}(t)$ by the defining correlation limits. The reflected supports stay separated, so continuity of test-function translation and the companion's two-point bound make $s_{\mathcal{F}}$ continuous. Nonnegativity and monotonicity pass from s_j on the dense set of times to all times. Fix a finite partition $t_0 < t_1 < \cdots < t_p$, with $t_1, \ldots, t_p$ in that dense set. For all sufficiently fine cutoffs, monotonicity gives the explicit lower sum

$$a_j \sum_{n\geq n_0} s_j(n) \geq \sum_{i=1}^{p}(t_i - t_{i-1} - a_j)s_j(t_i/a_j).$$

Only finitely many continuum correlation limits are needed on the right. Pass along the defining subsequence, use the upper bound on the left, and then take the supremum over partitions of each fixed interval. Finally let the upper endpoint tend to infinity. Thus the continuum integral is obtained after the finite-cutoff tail bound; no uniform integrable majorant for the correlations has been assumed. The preceding estimates give (80). Finally $(t - t_0)s_{\mathcal{F}}(t) \leq \int_{t_0}^{t} s_{\mathcal{F}}(s)\,ds$, proving decay. □

For a real u symmetric about a coordinate plane, the field $\Phi_{\tau_3\otimes 1}(u)$ is one of these parity-odd combinations. Choose a slightly wider lattice-compatible slab about that plane. Its ordinary two-point correlation with its translate is $s_{\mathcal{F}}$ with a fixed shift in the translation parameter. It therefore tends to zero and has the corresponding integrable tail.

8. The short-distance law for two currents

The conserved lattice current is supported at the two ends of a link. The source estimates in [QCDM1] concern bilinears at one site. We first compare these two insertions at a fixed gauge configuration. The error contains a covariant difference of a fermion propagator away from its source. An energy inequality controls this difference by a positive pseudoscalar two-point function. The source estimates then compare the site insertion with its free value, whose leading term is computed by a Pauli matrix trace.

The construction and all data remain those of the chamber model with $r = 1/2$. We retain r in a few local identities to display the coefficients of the hopping operator. For a neighbouring pair of places $v, v' = v \cdot s$, write

$$h(v,v') = t_s(r - \gamma(u_s(v))), \qquad t^T(v,v') = \bar\psi(v)h(v,v')TU(v,v')\psi(v'),$$

where $T = e_+ \otimes t_+ + e_- \otimes t_-$ acts on species and flavour. Thus T commutes with the gamma matrices and with colour. Its reflection involution is $T^* = \tau_3 T^\dagger \tau_3 = T^\dagger$. The chamber graph, wall directions and transporters are those in section 2. In this section we label sites by $x \in a\mathbb{Z}^3$: the place (x,q) corresponds to $(a^{-1}x, q)$ in (1). Set

$$\begin{aligned} \mathfrak{j}^T(v,v') &= t^T(v',v) - t^T(v,v'), \\ j^T_\mu(x) &= \sum_{g:\, ge_1 = e_\mu} \mathfrak{j}^T((x,g),(x,g)\cdot 0), \\ J^T(f) &= ia\sum_{x,\mu} f_\mu(x + \tfrac12 a e_\mu) j^T_\mu(x). \end{aligned} \tag{82}$$

The site sum has eight terms. The factor i is part of the definition of J^T; it determines the sign of its two-point function and its reflection law.

Proposition 8.1 (Lattice conservation)**.** Let δ^T_v be the derivation which fixes the links and satisfies

$$\delta^T_v \psi(u) = \delta_{uv} T\psi(u), \qquad \delta^T_v \bar\psi(u) = -\delta_{uv}\bar\psi(u)T.$$

At every gauge configuration, with $(\operatorname{div}\mathfrak{j}^T)(v) = \sum_{v'\sim v}\mathfrak{j}^T(v,v')$, the conditional Gaussian functional satisfies

$$\langle (\operatorname{div}\mathfrak{j}^T)(v) F\rangle_U = -\langle \delta^T_v F\rangle_U \tag{83}$$

for every polynomial F in finitely many fermion generators. The identity also holds after gauge averaging. At a site,

$$\sum_{v: x_v = x} (\operatorname{div}\mathfrak{j}^T)(v) = \sum_\mu [j^T_\mu(x) - j^T_\mu(x - ae_\mu)].$$

Thus the divergence vanishes in correlations away from the fermion insertions; at an insertion it gives the contact term on the right of (83). ■

Proof. The diagonal mass term is invariant under δ^T_v. Since T commutes with the hopping matrices and the links,

$$\delta^T_v(\bar\psi D_U \psi) = -(\operatorname{div}\mathfrak{j}^T)(v).$$

The changes $\psi(v) \mapsto e^{tT}\psi(v)$ and $\bar\psi(v) \mapsto \bar\psi(v)e^{-tT}$ have cancelling Berezin Jacobians. Differentiating their change-of-variables identity proves (83) on a torus. On the infinite lattice the same identity follows from $D_U D_U^{-1} = 1$ and Wick's rule. Summing over the chambers at one site cancels the internal edges and leaves the displayed site divergence. Finally, $\delta^T_v F = 0$ when F has no generator at v. □

Fix $0 < \vartheta < 1$. In this section a test pair at scale d means two nonzero, nonnegative Lipschitz functions f, h such that each support lies in a ball of radius d,

$$\begin{gathered} d \le \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \le 6d, \\ \int f \ge \vartheta d^3 \|f\|_\infty, \qquad \int h \ge \vartheta d^3 \|h\|_\infty, \qquad \operatorname{Lip} f \le \frac{\|f\|_\infty}{\vartheta d}, \quad \operatorname{Lip} h \le \frac{\|h\|_\infty}{\vartheta d}. \end{gathered} \tag{84}$$

On a torus we also require $20d \le \ell$ and read the supports in one Euclidean ball of radius $5d$. In particular $d \le |x-y| \le 10d$ on the product of the supports. Put

$$k_{\mu\nu}(x) = \frac{\delta_{\mu\nu}|x|^2 - 2x_\mu x_\nu}{|x|^6}, \qquad \mathcal{K}_4(f,h) = \iint f(x)h(y)|x-y|^{-4}\,dx\,dy.$$

The test-pair assumptions give

$$10^{-4}\vartheta^2 d^2\|f\|_\infty\|h\|_\infty \le \mathcal{K}_4(f,h) \le (4\pi/3)^2 d^2\|f\|_\infty\|h\|_\infty. \tag{85}$$

Theorem F (Two currents)**.** There are a constant $C'_V(g)$ and a nondecreasing function $\epsilon_V(g,\vartheta;d)$ tending to zero as $d\downarrow 0$ with the following properties. Let $0<d\le d_*:=\min\{1/200,1/(20m_+)\}$ and $16a\le d$, and let (f,h) be a test pair as in (84). The state may be a torus functional or a functional ω_ν with $\nu\in\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$. For unit vectors e,e', $a_0,b_0\in\{0,3\}$ and Hermitian flavour matrices t,t', put $T=\tau_{a_0}\otimes t$, $T'=\tau_{b_0}\otimes t'$. Then

$$\kappa_2(J^T(fe),J^{T'}(he')) = \frac{\mathrm{tr}(tt')}{2\pi^2}\left[\delta_{a_0b_0}\iint f(x)h(y)\langle e,k(x-y)e'\rangle\,dx\,dy+\varepsilon\mathcal{K}_4(f,h)\right], \tag{86}$$

where

$$|\varepsilon|\le \epsilon_V(g,\vartheta;d)+C'_V(g)\left[\left(\frac{a}{\vartheta^4 d}\right)^{1/2}+\frac{a}{\vartheta^3 d}\right]. \tag{87}$$

If $\mathrm{tr}(tt')=0$, the cumulant is exactly zero.

At fixed test pair, these cumulants are bounded along every sequence with $K\to\infty$. Every limit point satisfies (86) with $|\varepsilon|\le\epsilon_V(g,\vartheta;d)$. This holds on a fixed torus and for infinite-lattice states; it also holds along tori on which the fixed pair is read with $20d\le\ell$. ■□

The assertion about limit points concerns the indicated two-point functions. It does not by itself construct all joint current correlations. For an orthonormal Hermitian flavour basis, $\mathrm{tr}(t_jt_k)=\delta_{jk}$, the leading coefficient is $\delta_{a_0b_0}\delta_{jk}/(2\pi^2)$.

8.1. **Conditional loops and their signs.** We use the resolvent in lattice units in this section:

$$G_U=(D_+[U]+m_0)^{-1}=a^{-1}\mathsf{G}_+.$$

Let $h_+(v,v')=t_s(r-\sigma(u_s(v)))$ be the upper-species hop. Define $Y(v,v')$ by its two nonzero blocks

$$Y(v,v')[v',v]=h_+(v',v)U(v',v),\qquad Y(v,v')[v,v']=-h_+(v,v')U(v,v'),$$

and sum these over the eight chamber edges above a link to obtain $Y_\mu(x)$. For a site vector function F set

$$Y(F)=\sum_{x,\mu}F_\mu(x)Y_\mu(x),\qquad N(F)(x,g)=\sigma(F(x)).$$

For a scalar function f and a unit vector e use $F^{f,e}_\mu(x)=f(x+ae_\mu/2)e_\mu\cdot e$, and abbreviate $Y(fe)=Y(F^{f,e})$, $N(fe)=N(F^{f,e})$. The two loop expectations to be compared are

$$\begin{aligned}E_\nu(f,e;h,e')&=a^2\mathsf{E}_\nu\,\mathrm{Tr}(G_UY(fe)G_UY(he')),\\ E^{\mathrm{s}}_\nu(f,e;h,e')&=a^2\mathsf{E}_\nu\,\mathrm{Tr}(G_UN(fe)G_UN(he')).\end{aligned} \tag{88}$$

The trace includes places, spin and colour, and no flavour variable. Subscript 1 denotes the value at $U=1$. We also use the positive form

$$P_\nu(F,H)=2a^2\sum_{u,u'}F(x_u)H(x_{u'})\mathsf{E}_\nu\|G_U(u,u')\|^2_{\mathrm{HS}} \tag{89}$$

for scalar $F,H\ge 0$. For a site matrix A acting trivially on colour, write

$$\Phi_A=a\sum_v\bar\psi(v)A(x_v)\psi(v).$$

This is the bilinear in (3), now allowing A to act on spin as well.

Lemma 8.2. The conditional mean of every current is zero. For real f, h,

$$\kappa_2(J^T(fe), J^{T'}(he')) = \operatorname{tr}(t_+ t'_+) E_\nu(f, e; h, e') + \operatorname{tr}(t_- t'_-) \overline{E_\nu(f, e; h, e')}. \tag{90}$$

For real site vector functions F, H, choose a Hermitian t with $\operatorname{tr} t^2 = 1$ and let $N_F = \sigma(F) \otimes e_+ \otimes t$, acting as the identity on colour. Then

$$\omega_\nu(\Phi_{N_F}) = 0, \qquad \kappa_2(\Phi_{N_F}, \Phi_{N_H}) = -a^2 \mathsf{E}_\nu \operatorname{Tr}(G_U N(F) G_U N(H)).$$

For $\Gamma = i\tau_1 \otimes t$ the corresponding scalar bilinears satisfy $\kappa_2(\Phi_\Gamma(F), \Phi_\Gamma(H)) = P_\nu(F, H)$. ■

Proof. Let $C = \sigma_2 \otimes \epsilon$ on spin and colour, where $\epsilon U^{\mathsf{T}} \epsilon^{-1} = U^{-1}$ for $U \in \mathrm{SU}(2)$. Charge conjugation gives

$$G_U(u, u')^{\mathsf{T}} = C^{-1} G_U(u', u) C, \qquad \sigma_2 \sigma_\mu^{\mathsf{T}} \sigma_2 = -\sigma_\mu.$$

Consequently the two orders of $G_U(u, u')$ have equal Hilbert–Schmidt norms and $\operatorname{tr}(\sigma_\mu G_U(v, v)) = 0$. Let $\mathcal{C}$ be C extended by the identity on species and flavour. The change of Grassmann variables $\psi \mapsto (\bar\psi \mathcal{C})^{\mathsf{T}}$, $\bar\psi \mapsto -(\mathcal{C}^{-1}\psi)^{\mathsf{T}}$ preserves the action and its Berezin measure and sends j^T to $-\mathsf{j}^{T^{\mathsf{T}}}$. On the infinite lattice the same invariance follows directly from the covariance identity and Wick's rule.

At fixed U the current mean is a flavour-invariant linear form in T, hence a linear combination of $\operatorname{tr} t_+$ and $\operatorname{tr} t_-$. It is unchanged by transposition and changed in sign by charge conjugation, so it is zero. Thus the covariance of conditional means makes no contribution. Wick's rule gives $-\operatorname{Tr}(GYGY')$ for two bilinears, while the two factors ia in the currents give $-a^2$. In the lower species $G_- = G_+^*$ and $Y_- = -Y_+^*$; cyclicity of the trace gives the complex conjugate loop. This proves (90).

The vanishing Pauli trace proves the assertion about the site mean; its second cumulant is the stated Wick trace. Finally $i\tau_1$ exchanges the two species. Its two orientations in the loop give twice the squared Hilbert–Schmidt norm, since $G_- = G_+^*$ and the two orders have the same norm. This proves the last identity. □

8.2. **Replacing the two endpoints by a site.** For a chamber function φ, let $\nabla_s \varphi(v) = \varphi(v) - U(v, v \cdot s)\varphi(v \cdot s)$. For a set of sites S write

$$\|\varphi\|_S^2 = \sum_{x_v \in S} |\varphi(v)|^2, \qquad \mathfrak{D}_S(\varphi) = \sum_{x_v \in S} \sum_{s=0}^{3} |\nabla_s \varphi(v)|^2.$$

Lemma 8.3. For a site vector function F supported in S, $Y(F) = N(F) + R(F)$, where

$$|\langle \chi, R(F)\varphi \rangle| \le c_Y \|F\|_\infty (\mathfrak{D}_S(\chi)^{1/2} \|\varphi\|_S + \|\chi\|_S \mathfrak{D}_S(\varphi)^{1/2}), \qquad c_Y = \sqrt{3}(51 + 3r). \tag{91}$$

■

Proof. For $v = (x, g)$ with $g e_1 = e_\mu$, the wall-zero hops are $3(r - \sigma_\mu)$ and $3(r + \sigma_\mu)$. Expanding the two endpoint values about v gives

$$\begin{aligned}\langle \chi, Y(v, v \cdot 0)\varphi \rangle = {} & 6\chi(v)^* \sigma_\mu \varphi(v) - 3(\nabla_0 \chi(v))^* (r + \sigma_\mu)\varphi(v) \\ & + 3\chi(v)^* (r - \sigma_\mu) \nabla_0 \varphi(v).\end{aligned}$$

The last two terms sum to at most $3(1 + r)$ times the right-hand expression in (91) at x. To replace the first term by the sum over all 48 chambers, choose a map from these chambers onto the eight with $g e_1 = e_\mu$, with six preimages each. Put $b(g) = \chi(x, g)^* \sigma_\mu \varphi(x, g)$ and denote the map by π_μ. Then

$$6 \sum_{g:\, g e_1 = e_\mu} b(g) = \sum_g b(g) + \sum_g (b(\pi_\mu(g)) - b(g)).$$

The first sum is the site insertion. Join each chamber to its image by a shortest gallery using walls $1, 2, 3$. These walls connect the chambers of one site, since their positive quadratic form has precisely

the chamber-independent fields as its kernel. Their transporters are the identity. For one step $w \mapsto w \cdot s$ of a gallery,

$$b(w \cdot s) - b(w) = -\chi(w)^* \sigma_\mu \nabla_s \varphi(w) - (\nabla_s \chi(w))^* \sigma_\mu \varphi(w \cdot s).$$

A shortest gallery visits a chamber at most once. There are 48 galleries, so each chamber and each chamber-wall pair occurs at most 48 times. Cauchy–Schwarz therefore gives

$$\left| \sum_g (b(\pi_\mu(g)) - b(g)) \right| \le 48 (\|\chi\|_x \mathfrak{D}_x(\varphi)^{1/2} + \mathfrak{D}_x(\chi)^{1/2} \|\varphi\|_x).$$

Thus the coefficient is $51 + 3r$ for one direction. Sum over directions using $\sum_\mu |F_\mu(x)| \le \sqrt{3} |F(x)|$ and then apply Cauchy–Schwarz over sites. □

Lemma 8.4 (Energy away from the source)**.** Let $0 \le \eta \le 1$ be a site cutoff with finite support, let δ be the largest difference of η at neighbouring sites, and let A_η contain the places at both ends of every hop where η changes. Suppose $\eta = 1$ on S and its neighbours. If either $(D_+[U] + m_0)\varphi = 0$ or $(D_+[U]^* + m_0)\varphi = 0$ wherever $\eta > 0$, then

$$\mathfrak{D}_S(\varphi) \le C_E \delta \|\varphi\|^2_{A_\eta}, \qquad C_E = 2(1 + r)/r. \tag{92}$$

■

Proof. Write D for either of these two operators without its mass. Its Hermitian part is $\Delta = \sum_s r t_s (1 - R_s)$, where R_s is a self-adjoint unitary and $t_s \ge 3$. For every place v,

$$((D + m_0)(\eta\varphi))(v) = \eta(x_v)((D + m_0)\varphi)(v) + \sum_s D(v, v \cdot s)(\eta(x_{v \cdot s}) - \eta(x_v))\varphi(v \cdot s).$$

The first term is zero where $\eta > 0$ by the equation, and elsewhere by its factor η. Thus $(D+m_0)(\eta\varphi) = [D, \eta]\varphi$. Only wall zero changes the site. Its hop has norm $3(1 + r)$, and $v \mapsto v \cdot 0$ is a bijection, so $\|[D, \eta]\varphi\| \le 3(1 + r)\delta \|\varphi\|_{A_\eta}$. Let T_η be the set of places at which the commutator can be nonzero. It is contained in A_η. Since $m_0 > 0$ and $0 \le \eta \le 1$,

$$\begin{aligned} \langle \eta\varphi, \Delta\eta\varphi \rangle &\le \mathrm{Re} \langle \eta\varphi, (D + m_0)\eta\varphi \rangle \\ &= \mathrm{Re} \langle \eta\varphi, [D, \eta]\varphi \rangle \le \|\varphi\|_{T_\eta} \|[D, \eta]\varphi\| \le 3(1 + r)\delta \|\varphi\|^2_{A_\eta}. \end{aligned}$$

On the other hand this energy is $\frac{1}{2} \sum_{v,s} r t_s |\nabla_s(\eta\varphi)(v)|^2$, which is at least $(3r/2)\mathfrak{D}_S(\varphi)$. □

Lemma 8.5 (Point splitting)**.** Let $16a \le d$ and $\mathrm{dist}(\mathrm{supp}\, f, \mathrm{supp}\, h) \ge d$. Let B_f, B'_f be the sets of sites at distance at most a and $d/4 + 5a$, respectively, from $\mathrm{supp}\, f$, and similarly for h. Then

$$|E_\nu(f, e; h, e') - E^{\mathrm{s}}_\nu(f, e; h, e')| \le C_{\mathrm{split}} (a/d)^{1/2} \|f\|_\infty \|h\|_\infty (P_\nu(1_{B'_f}, 1_{B_h}) + P_\nu(1_{B_f}, 1_{B'_h})), \tag{93}$$

where $C_{\mathrm{split}} = 12(1 + r) c_Y C_E^{1/2}$. The same estimate holds at $U = 1$. ■

Proof. Write $Y(fe) = N(fe) + R_f$ and similarly for h. The difference of traces is

$$\mathrm{Tr}(G_U R_f G_U Y(he')) + \mathrm{Tr}(G_U N(fe) G_U R_h).$$

Let S_f be the site support of $F^{f,e}$; it lies within $a/2$ of $\mathrm{supp}\, f$. Choose a cutoff equal to one within $2a$ of S_f, decreasing linearly to zero over width $d/4$. Its neighbouring-site difference is at most $4a/d$, its transition region lies in B'_f, and it vanishes on B_h.

Expand $Y(he')$ in its nonzero blocks Y'_{pq}. Their norms are at most $3(1 + r)\|h\|_\infty$, and each place is a row and column of at most one such block. The matrix elements of $(G_U R_f G_U)(q, p)$ are $\langle G^*_U(\cdot, q)e_\alpha, R_f G_U(\cdot, p)e_\beta \rangle$. Both columns satisfy the homogeneous equation on the support of the cutoff. For $X_{q,\alpha} = G^*_U(\cdot, q)e_\alpha$ and $\Phi_{p,\beta} = G_U(\cdot, p)e_\beta$, Lems. 8.3 and 8.4 give

$$|\langle X_{q,\alpha}, R_f \Phi_{p,\beta} \rangle| \le 2 c_Y \|f\|_\infty (C_E \delta)^{1/2} \|X_{q,\alpha}\|_{B'_f} \|\Phi_{p,\beta}\|_{B'_f}.$$

Set

$$\mathcal{X}_q = \sum_{u\in B'_f} \|G_U(q,u)\|_{\mathrm{HS}}^2, \qquad \mathcal{Y}_p = \sum_{u\in B'_f} \|G_U(u,p)\|_{\mathrm{HS}}^2,$$

where a sum over a site set includes all its places. Summing the squares over the four spin-colour basis vectors gives

$$\|(G_U R_f G_U)(q,p)\|_{\mathrm{HS}} \le 2c_Y \|f\|_\infty (C_E\delta)^{1/2} (\mathcal{X}_q\mathcal{Y}_p)^{1/2}.$$

On this four-dimensional fibre the trace norm is at most twice the Hilbert–Schmidt norm. With $c_h = 3(1+r)$,

$$\begin{aligned} |\operatorname{tr}((G_U R_f G_U)(q,p) Y'_{pq})| &\le 4c_h c_Y \|f\|_\infty \|h\|_\infty (C_E\delta)^{1/2} (\mathcal{X}_q\mathcal{Y}_p)^{1/2} \\ &\le 2c_h c_Y \|f\|_\infty \|h\|_\infty (C_E\delta)^{1/2} (\mathcal{X}_q + \mathcal{Y}_p). \end{aligned}$$

Each p and q occurs at most once, and $(C_E\delta)^{1/2} \le 2C_E^{1/2}(a/d)^{1/2}$. The first trace is therefore at most

$$C_{\mathrm{split}}(a/d)^{1/2}\|f\|_\infty\|h\|_\infty \sum_{u\in B'_f,\, u'\in B_h} (\|G_U(u,u')\|_{\mathrm{HS}}^2 + \|G_U(u',u)\|_{\mathrm{HS}}^2).$$

For the second trace interchange f, h; the blocks of $N(fe)$ have norm at most $\|f\|_\infty$ and one block per place. Its constant is no larger. Multiply by a^2, integrate, and use the equality of the two Hilbert–Schmidt norms from Lem. 8.2. □

8.3. **Source estimates with a spin matrix.** The site current insertion contains σ_μ. We therefore need the mass source to range in

$$\mathcal{M} = \mathrm{End}(\mathbb{C}^2_{\mathrm{spin}} \otimes \mathbb{C}^2_\tau \otimes \mathbb{C}^{N_f}) \otimes 1_{\mathrm{colour}}.$$

A source $J(x) \in \mathcal{M}$ acts identically at the 48 chambers of a site. Write $\hat{J}$ for this multiplication operator, $\|J\| = \sup_x \|J(x)\|$, and $\mathsf{n} = -aJ$. The fine operator is $D[U] + m_0 + \hat{\mathsf{n}}$. There are three source domains in this argument:

$$\begin{aligned} \mathscr{O}_k^{\mathcal{M}} &= \{J : \lambda_k \|J\| < r_M/L\}, \\ \mathcal{P}^{\mathcal{M}} &= \{J : \|J\| \le 2\varrho_J\}, \\ \mathcal{O}_j^{\mathrm{lev},\mathcal{M}} &= \{J : \lambda_j \|J\| < R_J\}, \qquad R_J = \min\{r_M, \varrho_S, \varrho'_S/L\}. \end{aligned}$$

The radii are those of [QCDM1, Sections 10 and 11]. Here r_M is chosen from the inverse and walk estimates, and ϱ_S, ϱ'_S are the free-operator radii. The physical radius satisfies $2\varrho_J \le c_J m_-$, where $c_J \le \min\{1/2, \varrho_*, r_M/L\}$; hence $\mathcal{P}^{\mathcal{M}}$ lies strictly inside every $\mathscr{O}_k^{\mathcal{M}}$. The source chain through level k is formed on $\mathscr{O}_k^{\mathcal{M}}$. To estimate the energy made at level j we use the larger level domain $\mathcal{O}_j^{\mathrm{lev},\mathcal{M}} \subset \mathscr{O}_{j-1}^{\mathcal{M}}$. Indeed $\lambda_j = L\lambda_{j-1}$ and $R_J \le r_M$. Only the supremum norm of J enters these domains; no derivative of J is required.

For this proof let $\mathfrak{G}(d)$ consist of bounded site matrix pairs (F,H) whose supports lie in balls of radius $2d$ with centres at distance at most $8d$, and whose supports have separation at least $d/4$. The same notation will be used for scalar or vector pairs after placing them in fixed matrices of norm at most one.

Proposition 8.6. There is a nondecreasing function $A_g(d)$, finite on $(0, 1/200]$ and tending to zero as $d \downarrow 0$, such that for $(F,H) \in \mathfrak{G}(d)$, $16a \le d$, and $m_0 \le 1/2$,

$$\left|\kappa_2(\Phi_F, \Phi_H) + a^2 \operatorname{Tr}(\hat{F} G_0 \hat{H} G_0)\right| \le A_g(d) d^2 \|F\|\|H\|. \tag{94}$$

Here $G_0 = (D[1] + m_0)^{-1}$ includes the full fibre. The estimate holds on admitted tori and in their infinite-lattice limit states. There is also a constant $C_{\mathrm{fr}}(g)$ such that, for nonnegative scalar pairs in $\mathfrak{G}(d)$,

$$P_1(F,H) \le C_{\mathrm{fr}}(g) d^2 \|F\|\|H\|, \qquad P_\nu(F,H) \le (C_{\mathrm{fr}}(g) + A_g(d)) d^2 \|F\|\|H\|. \tag{95}$$

■

Proof. We first check the operator estimates and local extraction on $\mathcal{M}$. We then carry them through the source chain and the sum over histories, which supplies the cell bound used in the geometric reduction.

The inverse and background maps. On a history, let Ω_1 be the region of the fine fermion variables being integrated. The layered operator on this region has the form

$$\mathcal{K}_i(\mathsf{n}) = (D[U] + m_0)_{\Omega_1\Omega_1} + \mathsf{W}_i + \hat{\mathsf{n}} = \mathcal{K}_i + \hat{\mathsf{n}}.$$

Here W_i is the sum of block-average penalties and s_i is the smoothed local scale of [QCDM1, Section 6]. Let ι embed a site spinor isometrically as a field constant over its 48 chambers; then $\Pi_0 = \iota\iota^*$ is the chamber-constant projection. Multiplication by n commutes with s_i, with scalar site functions, and with the chamber projection Π_0. Moreover, $\iota^*\hat{\mathsf{n}}\iota = \mathsf{n}$ on the chamber-independent subspace. These identities hold because the same matrix acts at each chamber of a site.

The Neumann estimate can be written without a scalar-mass assumption. Put

$$B_i = \mathsf{s}_i^{-1/2}\mathcal{K}_i^{-1}\mathsf{s}_i^{-1/2}, \qquad N_i = \mathsf{s}_i^{1/2}\hat{\mathsf{n}}\mathsf{s}_i^{1/2}.$$

In the weighted row and column kernel norm, $\|B_i\| \le C_{\mathcal{K}}$ and $\|N_i\| \le \sup_x \mathsf{s}_i(x)\|\mathsf{n}(x)\|$. For $i \le j$, this supremum is at most $2r_M$. The choice $r_M \le (4C_{\mathcal{K}})^{-1}$ gives $\|B_iN_i\| \le 2C_{\mathcal{K}}r_M \le 1/2$. These are the source perturbation bounds of [QCDM1, Lem. 6.5 and Section 10]. Thus

$$\mathsf{s}_i^{-1/2}\mathcal{K}_i(\mathsf{n})^{-1}\mathsf{s}_i^{-1/2} = \sum_{q\ge 0}(-B_iN_i)^qB_i, \qquad \|\mathsf{s}_i^{-1/2}\mathcal{K}_i(\mathsf{n})^{-1}\mathsf{s}_i^{-1/2}\| \le 2\|B_i\|. \tag{96}$$

The same calculation after an exponential conjugation gives the weighted inverse bound. The local inverses in the walk construction use the pulled-back site matrix, so their support and decay bounds are unchanged. The heavy block has real part at least $\lambda_\rho + m_0 - \|\mathsf{n}\|$; its bound uses the norm of the perturbation and the commutation with Π_0. Positivity of the complex mass matrix itself is not needed.

For a background column with prescribed coarse data, write $\psi_P(\mathsf{n}) = \mathcal{K}_i(\mathsf{n})^{-1}g_P$. The resolvent identity gives

$$\psi_P(\mathsf{n}) - \psi_P(0) = -\mathcal{K}_i(\mathsf{n})^{-1}\hat{\mathsf{n}}\psi_P(0). \tag{97}$$

The row twin obeys the corresponding identity, with the order of the factors reversed. For the fluctuation covariance, let A_i be the integrated places at level i and let q_i be the one-step fermion average. Put $T_A = 1_{A_i}q_i\cdots q_1$. The block-kernel parameters are

$$\theta = b_F > 0, \qquad \theta_i = \frac{b_F(1-L^{-1})}{1-L^{-i}} \quad (i \ge 1),$$

as in [QCDM1, Lem. 5.4]. The Schur formula at the start of [QCDM1, Section 6] then reads, for $i \ge 1$,

$$\Gamma_i(\mathsf{n})^{-1} = c_i^{-1} + c_i^{-1}\theta_iT_A\mathcal{K}_{i+1}(\mathsf{n})^{-1}T_A^+\theta_ic_i^{-1}, \qquad c_i = \theta_i + \theta q_{i+1}^+q_{i+1}.$$

At $i = 0$ the identity is $\Gamma_0(\mathsf{n})^{-1} = \mathcal{K}_1(\mathsf{n})^{-1}$. Here $+$ denotes the holomorphic row twin, agreeing with the adjoint at real links. The averages do not depend on n. Substitution of (96) therefore gives the same covariance, background, and Gram estimates as [QCDM1, Prop. 6.7], with its fixed enlarged constants. All these objects are holomorphic in n on the stated balls.

The source coordinates and extraction. At zero gauge source the independent coordinates are

$$\mathfrak{c} = (0, \mathsf{n}, (\sigma_i)_{i\le k}), \qquad \mathscr{D}_k^{\mathcal{M}} = \{L^{k+1}\|\mathsf{n}\| < r_M,\ \|\sigma_i\| < B_\sigma\phi_i\ (i \le k)\}.$$

The doubled domain doubles each radius. Since $L^{k+1}\|\mathsf{n}\| < 2r_M$ there, the preceding operator estimates apply through $\mathcal{K}_{k+1}$ and Γ_k. The local matrix coordinates σ_i, the increments, and the extracted zeroth moments already take values in $\mathcal{M}$ in the construction of [QCDM1, Prop. 10.1].

More explicitly, let $k_X(\mathfrak{c})$ be the quadratic kernel at the trivial configuration before extraction, and set

$$\mathrm{ord}_0[k](y) = \tfrac{1}{2}\sum_{y'}[k(y,y') + k(y',y)].$$

On a torus these sums use the lifted paths in the definition of the local moment. The increment is

$$\nu_{k+1}(y;\mathfrak{c}) = \mu_{k+1} - \mathfrak{s}_{k+1}^{-1}\sum_X(\mathrm{ord}_0[k_X(\mathfrak{c})](y) - \mathrm{ord}_0[k_X(0)](y)),$$

with the same restriction to polymers of diameter less than one quarter of the torus side. Let $A_{k+1}(y,z)$ be the moment map of a local mass form, evaluated at the trivial gauge configuration and $\mathsf{n} = 0$. It is a linear map on $\mathcal{M}$. On the infinite lattice, constant background fields give

$$\sum_z A_{k+1}(y,z)N = \mathfrak{s}_{k+1}N \qquad (N \in \mathcal{M}). \tag{98}$$

On a torus, both this row sum and the corresponding column sum differ from $\mathfrak{s}_{k+1}N$ by at most $C_A e^{-4c_t\ell/\lambda_{k+1}}\|N\|$. Indeed, the local pieces which do not wrap around agree with their infinite-lattice counterparts; the remaining pieces are bounded by the exponential walk and polymer tails. For nonzero n, write A_{k+1}^{n} for the moment map. Its change from A_{k+1} has the weighted column-sum bound of [QCDM1, Prop. 10.1, proof], with factor $C_{\mathrm{bg}}^{\mathrm{w}}L^{k+1}\|\mathsf{n}\|/r_M$.

The extraction acts on the entire matrix coefficient. It need not leave a kernel with zero zeroth moment: a varying coefficient ν_{k+1} leaves differences of the form

$$\sum_z A_{k+1}(y,z)[\nu_{k+1}(z) - \nu_{k+1}(y)].$$

The cell estimates retain these differences, the source variation, and the torus error. The extraction forms have coefficients independent of the gauge configuration and commuting with colour, which preserves their joint gauge invariance. The weighted row sums and cell-unit estimates use operator norms of these coefficients, so apply to the larger source space with the same constants.

The source chain is

$$\sigma_0 = 0, \qquad \sigma_{k+1} = L\sigma_k + \nu_{k+1}(\mathfrak{c}_k(J)) - \mu_{k+1}, \qquad \mathfrak{c}_k(J) = (0, -aJ, (\sigma_i(J))_{i\le k}).$$

The coordinate Cauchy estimate in the doubled domain has the same distance as before, with the gauge-source entry omitted:

$$\|\mathfrak{c} - \mathfrak{c}'\|_{k,y} = \sup_x e^{-\kappa_\sigma|x-y|/L^{k+1}}\max\left\{\frac{L^{k+1}|\mathsf{n}-\mathsf{n}'|(x)}{r_M}, \max_{i\le k}\frac{|\sigma_i - \sigma_i'|(x)}{B_\sigma\phi_i}\right\}.$$

Here x, y are in fine lattice coordinates. In particular

$$|\nu_{k+1}(\mathfrak{c})(y) - \nu_{k+1}(\mathfrak{c}')(y)| \le B_\nu\phi_{k+1}\|\mathfrak{c} - \mathfrak{c}'\|_{k,y}.$$

The induction of [QCDM1, Lem. 10.4] therefore keeps the chain in $\mathscr{D}_k^{\mathcal{M}}$ and gives

$$\|\sigma_k(J)\| \le B_\nu S_\uparrow\phi_k\lambda_k\|J\|/r_M.$$

If J and J' agree outside S, the same induction gives the tail $C_t\phi_k e^{-\kappa_\sigma\,\mathrm{dist}(x,S)/\lambda_k}$, now in physical coordinates. Neither estimate uses a commutation of J with the spin matrices.

Symmetries and the sum over histories. The four source operations are

$$J^\star = \gamma_4 J^\dagger\gamma_4, \quad J^{\mathsf{c}} = C_s J^{\mathsf{T}} C_s^{-1}, \quad J \mapsto VJV^{-1}, \quad J \mapsto S_{\mathsf{t}}(J\circ\mathsf{t}^{-1})S_{\mathsf{t}}^{-1}, \qquad C_s = \sigma_2\otimes 1_\tau\otimes 1_{N_f}.$$

Here $V = e_+\otimes V_1 + e_-\otimes V_2$, with $V_1, V_2 \in \mathrm{U}(N_f)$, acts trivially on spin. The map t is a lattice translation followed by a signed coordinate permutation preserving the block partition; S_{t} is its spinor lift from [QCDM1, proof of Prop. 2.5]. The sign ambiguity in the lift disappears in conjugation. They preserve $\mathcal{M}$ and its norm. At real links, for example, write $D_J = D[U] + m_0 - a\hat{J}$. Then

$$D_J^* = \gamma_4 D_{J^\star}\gamma_4, \qquad D_J^{\mathsf{T}} = \mathcal{C}^{-1}D_{J^{\mathsf{c}}}\mathcal{C}.$$

The flavour and isometry identities are obtained by conjugating D_J with their respective lifts. The finite-dimensional integral gives the exact identity $Z[J^\star] = \overline{Z[J]}$. The diameter restriction in local extraction requires the torus qualifications of [QCDM1, Prop. 10.1]; it does not give an exact covariance identity for every local increment.

The exact conjugation identity for the sum of the source energies follows separately, as in [QCDM1, Prop. 10.6, proof]. Sum the new activities and localized normalizers before evaluating their scalar parts. On the background pairs used in the density identity, their sum is the logarithm of the whole step integral. The trivial pair belongs to this domain, and conjugation carries its scalar part to the complex conjugate at the conjugated source. Subtracting the real zero-source value proves the identity for the level sum. The assignment to individual polymers need not be preserved.

The history comparison uses the same fine matrix n in the chain and on the history; only the coordinates σ_i differ. Its distance estimate therefore applies with the same constants. In the history integrals, the fermion covariance and stored background columns are those of (96) and (97). Schur's test, the row bounds, and the Berezin Gram bound depend only on their norms. Products such as $(1 - \mathfrak{s}_j^{-1} \sum_y A_j^{\mathsf{n}}(y,z))a_{X,p}(z)$ are estimated in that order, so no commutation of their matrix factors is assumed.

Write $\mathcal{H}$ and $\bar{\mathcal{H}}$ for the composed column and row background maps from the variables before step j to the original fermion variables. They are the maps obtained by successive square completion in [QCDM1, Section 5.3]. In the following formula, the subscript AA restricts both coarse arguments to the variables integrated at that step. The determinant variation used in the local energy is

$$\log\det\Gamma_{j-1}(\mathsf{n}) - \log\det\Gamma_{j-1}(0) = \int_0^1 \mathrm{Tr}\left[\Gamma_{j-1}(t\mathsf{n})^{-1}(\bar{\mathcal{H}}\hat{\mathsf{n}}\mathcal{H})_{AA}(t\mathsf{n})\right] dt.$$

Expand the inverse and the two background maps into walks and assign each trace to their joint support. The fibre trace is bounded by $8N_f$ times the product of the operator norms. This proves the same local energy estimate as [QCDM1, Prop. 10.6]. A change of source outside a polymer can be seen only by a walk reaching that change; reserving part of its decay gives the source-distance tail.

At zero gauge source, every variation of the gauge weight and every associated Gaussian or linear term vanishes. The remaining closed fermion loops are gauge invariant. Their first derivative in curvature at the trivial configuration is zero, since a linear form invariant under the adjoint action of SU(2) vanishes. Consequently the history majorants and their sums in [QCDM1, Prop. 10.14] apply with the same constants. The bare density is entire in J, and its coefficients are continuous on the compact link space, as required when that sum is identified with the original partition function. This proves the first-order source and energy bounds needed here. No second-order improvement in ϕ for the enlarged source space is being used.

Reflection and the cell bound. The cell bound requires the correct reflection of a spin matrix. Directly on generators,

$$\Theta_P(\bar{\psi}(v)J(v)\psi(v)) = \bar{\psi}(\theta v)\gamma(\hat{n})J(v)^\dagger\gamma(\hat{n})\psi(\theta v).$$

For the reflection lift $S_P = \gamma_4\gamma(\hat{n})$ this is the linear reflected source followed by $\star$. The norm is preserved. The operation $\star$ commutes with conjugation by the reflection lift, since $\gamma_4 S_P = -S_P\gamma_4$ and the sign drops out of the conjugation. For a word r in the board reflections, define

$$\mathsf{r}_{\mathrm{lin}}J = S_{\mathsf{r}}(J \circ \mathsf{r}^{-1})S_{\mathsf{r}}^{-1}, \qquad \mathsf{r}_\theta J = \begin{cases} \mathsf{r}_{\mathrm{lin}}J, & |\mathsf{r}| \text{ even}, \\ (\mathsf{r}_{\mathrm{lin}}J)^\star, & |\mathsf{r}| \text{ odd}. \end{cases}$$

If J is supported strictly inside one cell, the reflected exponentials are even and have disjoint supports. Hence their product is

$$\mathrm{Fold}\, e^{\Phi_J} = e^{\Phi_{J^{\mathrm{ch}}}}, \qquad J^{\mathrm{ch}} = \sum_{\mathsf{r}} \mathsf{r}_\theta J, \qquad \|J^{\mathrm{ch}}\| = \|J\|.$$

The chessboard estimate applies to this product. The partition-function bound just obtained and the separated energy estimate in [QCDM1, Prop. 10.16] then give the cell bound of [QCDM1, Prop. 10.17]. Odd reflection words cause no change in its real energy bound, because $E_K[J^\star] = \overline{E_K[J]}$. For sources in the central ball of radius $1/4$ this bound takes the form

$$\Psi(J) := e^{-\Lambda(J)}\frac{Z[J]}{Z[0]}, \quad \Psi(0) = 1, \quad |\Psi(J)| \le B'_Z, \quad B'_Z = e^{c_V + \mathscr{C}_*}, \tag{99}$$

Here $\mathscr{C}_*$ bounds the source count uniformly for supports in the central ball of radius $1/4$, as in [QCDM1, Prop. 10.17], and $\Lambda(J) = \mathsf{N}^{-1}\sum_{\mathsf{r}}(E_K[\mathsf{r}_{\mathrm{lin}}J] - E_K[0])$ and $\mathsf{N} = \ell^3$.

Packing and averaging. A pair in $\mathfrak{G}(d)$ lies in a ball of radius $6d$. Choose a grid of mesh $14d$ inside the ball of radius $1/4 - 7d \ge 0.215$, and move each grid point to the nearest permitted lattice translate of the centre of this pair. The displacement is at most $\sqrt{3}a/2 < d/8$. The translated balls stay inside the cell ball, and their separation is at least $2d - \sqrt{3}a \ge d$. An inscribed cube supplies at least

$$N \ge \left\lfloor \frac{0.43}{14\sqrt{3}d} \right\rfloor^3 \ge (112d)^{-3}$$

points: $0.43/(14\sqrt{3}) > 0.0177$, and $(0.0177 - 1/112)d^{-1} \ge 1$ for $d \le 1/200$.

For separated matrix sources put $\|F\|_\circ = \|F\|/\varrho_J$ and similarly for H. The function $\Psi(z_1F + z_2H)$ is bounded by B'_Z on $|z_1| < 2/\|F\|_\circ$, $|z_2| < 2/\|H\|_\circ$. On the bidisc smaller by $1/(4B'_Z)$, Schwarz's lemma on complex lines gives $|\Psi - 1| \le 1/2$. Its logarithm exists there, and Cauchy's estimate gives

$$|\partial_1\partial_2 \log \Psi(0)| \le 4(B'_Z)^2\|F\|_\circ\|H\|_\circ.$$

Apply this to $\sum_t \varepsilon_t F^t$ and $\sum_t \varepsilon_t H^t$, with independent signs $\varepsilon_t \in \{-1, 1\}$. The supports are disjoint, so these sums have the norms of one pair. Since $\mathsf{E}_\varepsilon \varepsilon_t\varepsilon_{t'} = \delta_{tt'}$, averaging the bilinear derivative retains only matching translates. Translation invariance then yields

$$\kappa_2(\Phi_F, \Phi_H) = \frac{1}{N\mathsf{N}}\sum_{t,\mathsf{r}} \partial_1\partial_2 E_K(F^{t,\mathsf{r}}, H^{t,\mathsf{r}}) + R_{\mathrm{cell}}, \quad |R_{\mathrm{cell}}| \le C_{\mathrm{cell}}(g) d^3 \varrho_J^{-2}\|F\|\|H\|,$$

where $\mathsf{N} = \ell^3$ is the number of unit cells. One may take $C_{\mathrm{cell}}(g) = 4 \cdot 112^3 \exp(2c_V + 2\mathscr{C}_*)$ in the notation of that lemma.

The energy decomposition. At zero gauge source the energy decomposition [QCDM1, Lem. 11.4] has just its fermion and interaction parts. The relative gauge weight is zero, as are the linear gauge source and its Gaussian variations. In particular the term of fourth order in the gauge source is zero. Completing the fermion square at each step is an algebraic identity for every $J \in \mathcal{M}$. The coarse determinant from one step is the input determinant of the next. Write $\mathbb{D}_K(1; J)$ for the final coarse Dirac matrix after these K square completions at the trivial gauge configuration with source J. Their product gives

$$\sum_{j,X} E_j^{\mathrm{F}}(X; J) = \log \frac{\det(D[1] + m_0 - a\hat{J})}{\det(D[1] + m_0)} - \log \frac{\det \mathbb{D}_K(1; J)}{\det \mathbb{D}_K(1; 0)}.$$

This is an identity of logarithmic germs at zero. On the larger level-source domains its continuation is defined by the product of the invertible fluctuation determinants. Separate global logarithms of the two displayed determinants are not required.

For completeness, the small factor in the interaction energy comes from the part of the step potential left after the explicit normalizers and Gaussian terms. On the level-j domain the induction in the proof of that lemma applies to the independent matrix coordinates $(0, -aJ, (\sigma_k)_{k<j})$, with $\|\sigma_k\| < 2B_\sigma\phi(x_k)$, and gives

$$\|\mathcal{V}_i^{\mathrm{rest}}\| \le C_{\mathrm{rest}}\phi(x_{i-1}), \qquad i \le j, \qquad \|\sigma_i\| < 2B_\sigma\phi_i.$$

At zero gauge source its gauge vertices are independent of J, its fermion vertices have the source-step bound proved above, and its closed-loop variations have the weighted trace bound. The localized variation of an older activity has the bound from the preceding induction level. Thus the scalar output at the trivial configuration, after removing the normalizers, is at most $C_w^{\mathrm{s}}\phi(x_{j-1})$. Subtracting its value at zero source gives

$$|E_j^{\mathrm{I}}(X;J)| \le 2C_w^{\mathrm{s}}\phi(x_{j-1})e^{-\kappa d_j(X)}. \tag{100}$$

This induction also retains holomorphy and locality in the independent coordinates. The matrix-source vertices and covariances are holomorphic and read J on their assigned walks. Scalar localization preserves the parameter dependence of the old arrays; a member assigned to X reads only their restrictions to X. The cluster expansion, its pointed families and the Cauchy integrals defining the remainder therefore have the same properties. These are exactly the coordinate clauses used in the interaction-tail proof of [QCDM1, Lem. 11.4]; the matrix extension above verifies their inputs.

Let $S = \operatorname{supp}(J - J')$ and $D = \operatorname{dist}(X, S) > 0$. Multiply the remainder potential by z. Its scalar output at the trivial pair, before the normalizers are removed, is a holomorphic function $\mathcal{E}(z;\mathfrak{c})$ on the full independent coordinate domain, with

$$|\mathcal{E}(z;\mathfrak{c})| \le K_M(c_{\mathrm{cl}}+1)e^{-\kappa d_j(X)}, \qquad |z| \le R = \frac{c_{\mathrm{cl}}}{4C_{\mathrm{rest}}\phi(x_{j-1})} \ge 2.$$

The two source chains have the same J on X, and their induced coordinates differ there by at most $C_{\mathrm{t}}\phi(x_k)e^{-\kappa_\sigma D/\lambda_j}$. Keep J' fixed and replace all $\sigma_k(J')$ simultaneously by $\sigma_k(J')+\zeta(\sigma_k(J)-\sigma_k(J'))1_X$. Since the chain lies in the inner window $\|\sigma_k(J')\| \le B_\sigma\phi(x_k)$, this curve stays in the coordinate domain for $|\zeta| < 1/\theta$, where $\theta = (C_{\mathrm{t}}/B_\sigma)e^{-\kappa_\sigma D/\lambda_j}$. Locality and Schwarz's lemma bound the difference of the two outputs by

$$C_{\mathrm{tail}}K_M(c_{\mathrm{cl}}+1)e^{-\kappa d_j(X)}e^{-\kappa_\sigma D/\lambda_j}, \qquad C_{\mathrm{tail}} = 2\max\{1, C_{\mathrm{t}}/B_\sigma\}.$$

For $\theta \ge 1$ the two separate bounds suffice; for $\theta = 0$ the outputs agree. There is no sum over k in this interpolation.

The output difference vanishes at $z = 0$, where its scalar potential and quadratic origin read only the zero gauge source. At $z = 1$ it equals the difference of E_j^{I}: the gauge normalizer and Gaussian part agree, and the free fermion normalizer cancels with E_j^{F}. Schwarz's lemma in z now supplies R^{-1}. Thus the constant is at most $4C_{\mathrm{rest}}C_{\mathrm{tail}}K_M(1+c_{\mathrm{cl}}^{-1})$, already included in the companion's C_I, and

$$|E_j^{\mathrm{I}}(X;J) - E_j^{\mathrm{I}}(X;J')| \le C_I\phi(x_j)e^{-\kappa d_j(X)}e^{-\kappa_\sigma \operatorname{dist}(X,S)/\lambda_j}. \tag{101}$$

If X meets S, use the two bounds (100); the fixed comparison of $\phi(x_{j-1})$ and $\phi(x_j)$ is included in C_I. This explains why the interaction tail, as well as its undifferentiated size, retains a factor ϕ.

Summing the mixed derivatives. For the averaged interaction sum, the level-j source radii are $R_J/(\lambda_j\|F\|)$ and $R_J/(\lambda_j\|H\|)$. At levels with $\lambda_j < d$ the separation $d/4$ gives the tail $e^{-\kappa_\sigma d/(16\lambda_j)}$ and at most $C(d/\lambda_j)^3$ cubes near one support. At levels $\lambda_j \ge d$, use the independent signs for the packed sources before applying Cauchy's estimate. The packed pairs and their board images meet at most $CN\mathsf{N}(d/\lambda_j)^3$ cubes: their union occupies a fixed-volume region in each unit cell and $N \ge cd^{-3}$. The signs have the same source norm as a single member. Dividing by $N\mathsf{N}$ therefore leaves the factor $C(d/\lambda_j)^3$, even at these coarser levels.

After summing the polymers through each cube, Cauchy's estimate and (101) bound the mean by $CR_J^{-2}\|F\|\|H\|$ times

$$\sum_j \phi(x_j)\lambda_j^2(d/\lambda_j)^3 \begin{cases} e^{-\kappa_\sigma d/(16\lambda_j)}, & \lambda_j < d, \\ 1, & \lambda_j \ge d. \end{cases} \tag{102}$$

The two powers of λ_j are the inverse Cauchy radii; the other factor is the number of cubes after averaging. To see its summability, put $t = \mathsf{b}/d$. The scale comparison $x_j \asymp \mathsf{b}/\lambda_j$ and $\phi(u) = u^{-1/2}(1+\log u)^{p_0}$ give, with fixed comparison constants,

$$\frac{\phi(\mathsf{b}/\lambda)}{\phi(t)} \le \begin{cases} (\lambda/d)^{1/2}(1+\log(d/\lambda))^{p_0}, & \lambda \le d, \\ (\lambda/d)^{1/2}, & \lambda \ge d. \end{cases}$$

All arguments are in the admitted decreasing range of ϕ. Consequently, after division by $d^2\phi(t)$, the terms with $\lambda_j \ge d$ form a geometric sum with ratio $L^{-1/2}$. For $\lambda_j < d$, the sum is bounded by a fixed multiple of

$$\sum_{u\ge 0} L^{u/2}(1+(u+1)\log L)^{p_0} e^{-(\kappa_\sigma/16)L^{u-1}} < \infty.$$

This is the summation in [QCDM1, Lem. 11.6], with the support separation $d/4$. It gives

$$|E^{\mathrm{I}}_{\mathrm{averaged}}| \le C^{\natural}_M R_J^{-2}\phi(\mathsf{b}/d)d^2\|F\|\|H\|.$$

The last determinant. Put $A(z_1,z_2) = \mathbb{D}_K(1; z_1F + z_2H)$ and $A = A(0,0)$. The real-part bound gives $\|A^{-1}\| \le (c_b\mathsf{m})^{-1}$. Differentiating the inverse formula inserts $-a\hat{F}$ between two background maps. Their squared row sums are $O(a^2)$, so the first derivative satisfies

$$\|A_1\|_1 \le Ca^3\sum_x \|F(x)\| \le Cd^3\|F\|, \qquad \|A_2\|_1 \le Cd^3\|H\|.$$

Here $\|\cdot\|_1$ is the trace norm. The supports lie in balls of radius $2d$, so their site counts are bounded by a constant times $(d/a)^3$. In the mixed derivative the inserted matrices are separated by one free propagator, with bound $Ca^2(16d^{-2} + \mathsf{m}^{-1})$ at separation $d/4$. The same row estimate therefore gives

$$\|A_{12}\|_1 \le C(d^4 + d^6\mathsf{m}^{-1})\|F\|\|H\|.$$

The logarithmic derivative is

$$\partial_1\partial_2 \log\det A(z)\big|_{z=0} = \mathrm{Tr}(A^{-1}A_{12} - A^{-1}A_2A^{-1}A_1).$$

Using the trace norm for the localized derivatives, and the operator norm for A^{-1}, bounds it by

$$C^{\natural}_{\mathrm{last}}(d^6\mathsf{m}^{-2} + d^4\mathsf{m}^{-1})\|F\|\|H\|,$$

as in the last paragraph of [QCDM1, Lem. 11.5]. No factor proportional to the total volume enters these two traces. The free loop is unchanged by the translates and reflected matrix sources. Thus (94) holds with

$$\alpha_g(d) = C_{\mathrm{cell}}(g)\varrho_J^{-2}d + C^{\natural}_M R_J^{-2}\sup_{\mathsf{b}\in I_g}\phi(\mathsf{b}/d) + C^{\natural}_{\mathrm{last}}(d^4 m_-^{-2} + d^2 m_-^{-1}), \qquad A_g(d) = \sup_{0<s\le d}\alpha_g(s).$$

This function is finite and tends to zero. The mass labels lie in a compact interval, and the same is true of $\mathsf{b} \in I_g$ at fixed g. On the infinite lattice, pass from the tori by convergence of local Gaussian functionals and by the vanishing of the massive free image terms.

For the positive free form, the propagator estimate in the proof of [QCDM1, Lem. 11.5] gives $\|G_1(u,u')\| \le C'_G(g)|n|^{-2}$, with $n = (x_u - x_{u'})/a$. Its leading term has this bound because $(1+t)e^{-t} \le 1$ and $te^{-t} \le e^{-1}$. The remainder is $O(|n|^{-3})$, and the torus images are bounded by $C'_{\mathrm{im}}(\ell/a)^{-2}$. These constants are included in $C'_G(g)$. The support balls contain at most

$$\frac{4\pi}{3}(2d/a + \sqrt{3}/2)^3 \le 36\pi(d/a)^3$$

sites each, with 48 places per site, and $|n| \geq d/(4a)$. The four-dimensional spin-colour Hilbert–Schmidt norm is at most twice the operator norm. Thus

$$\begin{aligned} P_1(F,H) &\leq a^2\, 8C'_G(g)^2\, 48^2 (36\pi)^2 (d/a)^6 (4a/d)^4 \|F\|\|H\| \\ &= 2048(36\pi)^2 48^2 C'_G(g)^2 d^2 \|F\|\|H\|. \end{aligned}$$

This gives the stated choice of $C_{\mathrm{fr}}(g)$. Apply (94) with $\Gamma = i\tau_1 \otimes t$ and use Lem. 8.2 to obtain the second bound in (95). □

Theorem G (Comparison with the free current loop)**.** Suppose $16a \leq d \leq 1/200$ and $m_0 \leq 1/2$. Let f, h be real bounded functions, each supported in a ball of radius d, with support separation between d and $6d$. Then

$$|E_\nu(f,e;h,e') - E_1(f,e;h,e')| \leq \left(A_g(d) + C_{\mathrm{pt}}(g)(a/d)^{1/2}\right) d^2 \|f\|_\infty \|h\|_\infty, \tag{103}$$

where $C_{\mathrm{pt}}(g) = 2C_{\mathrm{split}}(2C_{\mathrm{fr}}(g) + A_g(1/200))$. ■□

Proof. The pairs $(1_{B'_f}, 1_{B_h})$ and $(1_{B_f}, 1_{B'_h})$ in Lem. 8.5 belong to $\mathfrak{G}(d)$. Indeed the support enlargement is at most $d/4 + 5a \leq 9d/16$, so each enlarged support fits in a ball of radius $2d$, and the remaining separation is at least $d - 9d/16 - a \geq d/4$. The shifted site pair $(F^{f,e}, F^{h,e'})$ also belongs to this class. Apply (95) to the two point-splitting errors, one under ν and one at $U = 1$, and apply (94) to N_F, N_H. Adding the three differences gives (103). □

8.4. **The free tensor and the current classes.**

Lemma 8.7. For a test pair, $16a \leq d$ and $10\mathsf{m}d \leq 1$,

$$\left| E_1^{\mathrm{s}}(f,e;h,e') - \frac{1}{4\pi^2} \iint f(x)h(y)\langle e, k(x-y)e'\rangle\, dx\, dy \right| \leq C_{\mathrm{cr}} \left(\frac{a}{\vartheta d} + \mathsf{m}d + (d/\ell)^2 \right) d^2 \|f\|_\infty \|h\|_\infty. \tag{104}$$

The last term is absent on the infinite lattice. The constant depends only on the fixed model and $m_-\ell_G$, where ℓ_G is the least admitted side. ■

Proof. The free propagator expansion in the proof of [QCDM1, Lem. 11.5] gives, in its light upper block,

$$G_1(n) = v(n)\sigma(n) + s(n) + O(|n|^{-3}), \qquad v(n) = \frac{(1 + m_0|n|)e^{-m_0|n|}}{4\pi|n|^3}, \quad s(n) = \frac{m_0 e^{-m_0|n|}}{4\pi|n|}.$$

The light factor between individual chambers is 48^{-1}; the sum over the two chamber indices cancels its square. The mixed and heavy parts have bounds $C|n|^{-3}$ and $C|n|^{-4}$, hence cost $O(a/d)$ in this loop. Use

$$\mathrm{tr}(\sigma(n)\sigma(F)\sigma(n)\sigma(H)) = 2(2(n\cdot F)(n\cdot H) - |n|^2 F\cdot H).$$

The sign of the second propagator and the colour factor two give for the v^2 term

$$\frac{(1+t)^2 e^{-2t}}{4\pi^2} \langle F, k(n) H\rangle, \qquad t = m_0|n|.$$

The two vs terms involve three Pauli matrices. Their ratio to the size $4v^2|n|^2|F||H|$ of the leading term is at most $2t$, since

$$\frac{s(n)}{v(n)|n|} = \frac{t}{1+t} \leq t.$$

The s^2 term has ratio at most t^2, and $(1+t)^2 e^{-2t} = 1 + O(t^2)$. On the shifted supports,

$$15d/16 \leq d - a \leq |x-y| \leq 10d + a \leq 11d, \qquad t \leq 11\mathsf{m}d \leq 2.$$

The two mixed terms therefore give an $O(\mathsf{m}d)$ error; they cannot be replaced by an $O((\mathsf{m}d)^2)$ error in this estimate.

Homogeneity gives $a^2k(n) = a^6k(x-y)$. The discrete error just estimated is bounded by

$$C(\mathsf{m}d + a/d)a^6 \sum_{x,y} |F(x)||H(y)||x-y|^{-4} \le C(\mathsf{m}d + a/d)d^2\|f\|_\infty\|h\|_\infty.$$

Replacing midpoint values by site values uses

$$|F^{f,e}(x) - f(x)e| \le \tfrac{\sqrt{3}}{2} a \operatorname{Lip} f \le \frac{a}{\vartheta d}\|f\|_\infty,$$

and the same bound for H. For the remaining Riemann sum, $|k(x-y)| \le Cd^{-4}$ and $|\nabla k(x-y)| \le Cd^{-5}$ on the separated supports. The change across a lattice cube is therefore bounded by a constant times

$$a(d^{-4}\operatorname{Lip} f\|h\|_\infty + d^{-4}\|f\|_\infty \operatorname{Lip} h + d^{-5}\|f\|_\infty\|h\|_\infty).$$

Its support in the product space has volume $O(d^6)$. Summing the cube errors gives $Ca/(\vartheta d)$ times $d^2\|f\|_\infty\|h\|_\infty$. On the torus the antiperiodic image sum is bounded entrywise by $C(m_-\ell_G)(\ell/a)^{-2}$; its contribution to the loop is $C(d/\ell)^2$ at the same normalization. These bounds prove the claim. □

Proof of Thm. F. For $T = \tau_{a_0} \otimes t$ use $t_+ = t$, $t_- = \epsilon_{a_0} t$, with $\epsilon_0 = 1$ and $\epsilon_3 = -1$. Equation (90) is therefore $\operatorname{tr}(tt')$ times $E_\nu + \epsilon_{a_0}\epsilon_{b_0}\overline{E_\nu}$. The leading tensor in Lem. 8.7 is real; adding its conjugate multiplies it by $1 + \epsilon_{a_0}\epsilon_{b_0} = 2\delta_{a_0 b_0}$. The remaining error is at most twice the sum of the interacting-to-free difference, the free point-splitting error, and (104). Divide by the lower bound in (85). For example, take

$$\epsilon_V(g,\vartheta;d) = \frac{2\cdot 10^4}{c_B\vartheta^2}\left[A_g(d) + C_{\mathrm{cr}}(m_+ d + (d/\ell_G)^2)\right], \qquad c_B = \frac{1}{2\pi^2},$$

and

$$C_V'(g) = \frac{2\cdot 10^4}{c_B}(C_{\mathrm{pt}}(g) + 2C_{\mathrm{split}}C_{\mathrm{fr}}(g) + C_{\mathrm{cr}}).$$

This gives (87); when the flavour trace vanishes the exact loop formula gives zero. At fixed d, f, h, both mesh terms tend to zero with K. The displayed formulas bound the cumulants and every limit point as claimed. If torus sides vary, the assumption $20d \le \ell$ makes the Euclidean pair and its two kernel integrals the same at every cutoff. □

Proposition 8.8 (A nonzero current class)**.** For every sufficiently small fixed $d > 0$ and all sufficiently large K, every nonzero Hermitian flavour matrix t and $T = \tau_{a_0} \otimes t$, $a_0 \in \{0, 3\}$, admit a current supported on the positive side of $P_{1/2} = \{x_3 = a/2\}$ whose class in the lattice reflection-positive Hilbert space is nonzero. More precisely, let $\vartheta_c = 1/500$, $z_d = (5d/8 + a/2)e_3$, and $f(x) = (d/8 - |x - z_d|)_+$. For $\mu = 1, 2, 3$,

$$\omega_\nu(J^T(fe_\mu)\Theta_{P_{1/2}}J^T(fe_\mu)) \ge \tfrac{1}{4}c_B \operatorname{tr}(t^2)\mathcal{K}_4(f, f\circ\theta) > 0. \tag{105}$$

The bound is uniform in admitted labels and torus-limit states. It also holds on admitted tori with $20d \le \ell$. ■

Proof. The two tents form a test pair at scale d and ϑ_c. Their centres are $5d/4$ apart and their support radii are $d/8$. Also $\operatorname{Lip} f = 1$, $\|f\|_\infty = d/8$, and $\int f = (\pi/1536)d^3\|f\|_\infty \ge \vartheta_c d^3\|f\|_\infty$. For large K, every endpoint of a bond with midpoint in its support is strictly above $P_{1/2}$.

On generators, reflection reverses the ordered edge and sends $\mathrm{j}^T(v, v')$ to $-\mathrm{j}^{T^*}(\theta v, \theta v')$. Antilinearity also sends i to $-i$. After summing the oriented bonds this gives

$$\Theta_P J^T(f) = J^{T^*}(\varrho\bar{f}\circ\theta), \tag{106}$$

where ϱ is the linear reflection. The positive form of [QCDM1, Thm. B and Thm. A] therefore evaluates the left side of (105) by the two-current law at $e = e_\mu$ and $e' = \varrho e_\mu$; the current means are zero.

On these supports $n = x - y$ has $n_3 \geq d$ and $|n_\perp| \leq d/4$, so

$$\langle e_\mu, k(n)\varrho e_\mu\rangle \geq \frac{15}{17}|n|^{-4}.$$

Choose d so that $\epsilon_V(g, \vartheta_c; d) < 1/4$, and then K so that the two mesh errors in (87) total at most $1/4$. The leading term less the error is at least $(15/17 - 1/2)c_B \operatorname{tr}(t^2)\mathcal{K}_4$, which proves (105). The same supports lie in the positive slab of the torus when $20d \leq \ell$. □

9. Analytic dependence at fixed volume

A constant source changes the bare coupling or the bare mass. This observation connects the source estimates to analytic dependence on the physical parameters. The partition function itself need not be bounded as the cutoff is removed. The useful quantity is the partition function after the same vacuum energy has been removed from its upper and lower bounds.

Throughout this section the physical torus T_ℓ is fixed. Fix $0 < g \leq \gamma$ and an allowed side $\ell \in \mathfrak{L}(g)$, and write

$$I_g = [g^{-2} + \bar{B},\, L(g^{-2} + B_\sharp) - \bar{B}], \qquad W = I_g \times [m_-, m_+].$$

The constants and the cofinal set $\mathfrak{L}(g)$ are those of [QCDM1, Prop. 10.20]. In particular the estimates in this section hold for every $K \geq 1$ and every $(\mathsf{b}, \mathsf{m}) \in W$. We use the chamber operator with $r = 1/2$.

For a source $s = (w, J)$ put

$$Z_K[s] = \int e^{-(\mathsf{b}/a)A(U) + O_K(w)} \det\bigl(D[U] + a\mathsf{m} - a\widehat{J}\bigr)\, dU, \qquad a = L^{-K}.$$

The determinant includes both internal blocks and all flavours. Thus $Z_K[0] > 0$. The source w is a function on plaquette centres and J takes values in the complex span of the physical meson basis $\mathfrak{B}$. The definitions of O_K and the meson insertions are those in section 2. The action-source norm is $\|w\|_\sharp = \max\{\|w\|_\infty, 40M \operatorname{Lip} w\}$. Let

$$\mathcal{P} = \{(w, J) : \|w\|_\sharp \leq 2\varrho,\ \|J\|_\infty \leq 2\varrho_J\}.$$

The radii are independent of K and ℓ, and $2\varrho_J \leq c_J m_-$, $c_J \leq 1/2$.

We first recall precisely the source input. It is the combination of the local-energy, history and counting estimates of [QCDM1, Props. 10.6 and 10.14, and the counting estimate preceding Prop. 10.16]. There is a holomorphic energy

$$E_K(s) = E_K(0) + \sum_{j=1}^{K} \sum_{Y \in \mathcal{D}_j} e_j(Y; s), \qquad Z_K[s] = e^{E_K(s)} W_K[s], \tag{107}$$

defined on a neighbourhood of $\mathcal{P}$. Here Y is a connected union of blocks of physical side $M\lambda_j$, $\lambda_j = L^{j-K}$, and $d_j(Y)$ is the block-tree length used in the construction. For fixed positive constants,

$$|e_j(Y; s)| \leq C_E e^{-\kappa d_j(Y)}, \tag{108}$$

$$|e_j(Y; s) - e_j(Y; s')| \leq C_E e^{-\kappa d_j(Y)} e^{-\kappa_\sigma \operatorname{dist}(Y, \operatorname{supp}(s - s'))/\lambda_j}. \tag{109}$$

Distances in this section are physical. The geometry gives $\operatorname{diam} Y \leq c_D M\lambda_j(1 + d_j(Y))$. The construction chooses κ so that summing $e^{-\kappa d_j(Y)/2}$ over polymers containing one specified block costs at most C_{cnt}. Finally,

$$|W_K[s]| \leq e^{(C+2)\ell^3}, \qquad W_K[0] \geq e^{-C_P(1 + \log X_g)\ell^3}, \qquad X_g = L(g^{-2} + B_\sharp). \tag{110}$$

The conjugation of the source makes E_K real for constant real coupling and mass shifts. No positivity at complex sources is used.

9.1. **A zero-free neighbourhood.** Set

$$V_\ell = \exp\{[C + 2 + C_P(1 + \log X_g)]\ell^3\}, \qquad \rho_b = \frac{\varrho}{1 + V_\ell}, \qquad \rho_m = \frac{\varrho_J}{1 + V_\ell}. \tag{111}$$

Let $W_{\mathbb{C}}$ be the union, over $(b, m) \in W$, of the open polydiscs $|b' - b| < \rho_b$, $|m' - m| < \rho_m$.

Proposition 9.1 (Complex parameters)**.** For every K, the partition function $Z_{K,\ell}(b', m') = Z_K[0]$ formed with bare parameters b'/a and am' has no zero on $W_{\mathbb{C}}$. For a real base point $(b, m) \in W$, write $c = (-(b' - b)1, -(m' - m)1)$ and $R_K(b', m') = W_K[c]/W_K[0]$, with the energies and partition functions based at (b, m). Then

$$|R_K - 1| \le (1 + V_\ell) \max\left\{\frac{|b' - b|}{2\varrho}, \frac{|m' - m|}{2\varrho_J}\right\} < \frac{1}{2}. \tag{112}$$

On this polydisc the pressure has the decomposition

$$\ell^{-3} \log Z_{K,\ell}(b', m') = \ell^{-3} E_K(c) + \ell^{-3} \log W_K[c].$$

The logarithm is chosen real at the real base point, and

$$|\log R_K| \le \log 2, \qquad \ell^{-3} |\log W_K[c]| \le C + 2 + C_P(1 + \log X_g) + \ell^{-3} \log 2.$$

Consequently every sequence of cutoffs has a subsequence on which $\ell^{-3} \log R_K$ converges locally uniformly on this polydisc, with all parameter derivatives. ■

Proof. The constant source identity $Z_{K,\ell}(b', m') = Z_K[c]$ holds before any integration: the action coefficient becomes b'/a, and the matrix in the determinant becomes $D + am'$. The ratio R_K is holomorphic on the larger polydisc with radii $2\varrho, 2\varrho_J$, $R_K(b, m) = 1$, and $|R_K| \le V_\ell$ by (110). For a nonzero displacement let r be the maximum in (112). Restrict $R_K - 1$ to the complex line through that displacement, rescaled so that its unit disc lies in the larger polydisc. Dividing by $1 + V_\ell$ gives a function of modulus at most one, vanishing at zero. The given displacement corresponds to radius r on this disc. Schwarz's lemma proves (112). Thus R_K has a logarithm with $|\log R_K| \le \sum_{n \ge 1} 2^{-n}/n = \log 2$. The remaining bounds follow from the positive real lower bound and the upper bound in (110). Local boundedness gives a normally convergent subsequence; Cauchy's formula gives convergence of every derivative. □

For example, if $p + q \ge 1$, differentiation at a real base point gives

$$\left| S^T_{p+q,K}((0,1)^p, (1,1)^q) - (-\partial_{b'})^p (-\partial_{m'})^q E_K(c(b', m'))\big|_{(b',m')=(b,m)} \right| \le p!q! \log 2\, \rho_b^{-p} \rho_m^{-q}. \tag{113}$$

The RG base point (b, m) is held fixed in this derivative. Here the second label 1 in $(1, 1)$ is the constant test function and the first is the identity meson matrix, extended linearly from the chosen basis. The sign comes from the constant-source identity. The vacuum term may diverge with K; this estimate concerns its difference from the cumulant. The radii in (111) depend on the physical volume. No infinite-volume analyticity follows from this argument.

9.2. **Supports with empty total intersection.** For closed nonempty sets $S_1, \dots, S_n \subset T_\ell$ define

$$d_\cap(S_1, \dots, S_n) = \min_{x \in T_\ell} \max_i \operatorname{dist}(x, S_i).$$

Compactness shows that $d_\cap > 0$ precisely when $\bigcap_i S_i = \emptyset$. Pairwise intersections are allowed. A tuple $X = ((\alpha_i, f_i))_{i=1}^n$ is admissible in this section when $n \ge 2$, $\alpha_i \in \{0\} \cup \mathfrak{B}$, f_i is smooth, and its supports have empty total intersection. We write $d_\cap(X)$ for this distance. As usual $\|f\|_0 = \|f\|_\sharp$ and $\|f\|_\Gamma = \|f\|_\infty$ for $\Gamma \in \mathfrak{B}$; these basis matrices have operator norm at most one.

Lemma 9.2 (Mixed derivatives of the vacuum energy)**.** Let $s_* + \sum_i z_i s_i \in \mathcal{P}$ for $|z_i| \le r_i$, and suppose the supports of s_i have $d_\cap = d > 0$. There is a constant C_{ov}, independent of K, ℓ and the sources, such that

$$\left|\partial_1 \cdots \partial_n E_K(s_* + \textstyle\sum_i z_i s_i)\big|_{z=0}\right| \le 2^{n-1} C_{\mathrm{ov}} \ell^3 (1 + d^{-3}) \prod_i r_i^{-1}. \tag{114}$$

■

Proof. Let P_i set $z_i = 0$. A mixed derivative kills every term independent of one variable. It can therefore be estimated by Cauchy's formula applied to $\prod_i (1 - P_i) e_j(Y; s_* + s(z))$. Explicitly, this finite difference is

$$\prod_i (1 - P_i) e_j(Y; s_* + s(z)) = \sum_{A \subset \{1,\dots,n\}} (-1)^{n-|A|} e_j\left(Y; s_* + \sum_{i \in A} z_i s_i\right).$$

The operators $1 - P_i$ commute. We may therefore choose one index i separately for each polymer, apply the source-tail bound to $(1 - P_i)e_j$, and bound the other $n - 1$ differences by their 2^{n-1} terms. This is why empty total intersection suffices: the source that is distant from a small polymer need not be the same for every polymer.

Choose $y \in Y$. Some support S_i has $\operatorname{dist}(y, S_i) \ge d$. If $\operatorname{diam} Y \ge d/2$, half the polymer weight pays for this diameter:

$$e^{-\kappa d_j(Y)/2} \le e^{\kappa/2} e^{-\kappa d/(4 c_D M \lambda_j)}.$$

Otherwise $\operatorname{dist}(Y, S_i) \ge d/2$, and the source tail supplies $e^{-\kappa_\sigma d/(2\lambda_j)}$. Set

$$\eta = \min\{\kappa/(4c_D M), \kappa_\sigma/2\}.$$

The derivative of one energy is at most

$$2^{n-1} C_E e^{\kappa/2} \prod_i r_i^{-1} e^{-\eta d/\lambda_j} e^{-\kappa d_j(Y)/2}.$$

Assigning each polymer to one of its blocks and using the polymer count bounds the sum over Y by this expression times $C_{\mathrm{cnt}}(\ell/(M\lambda_j))^3$. Finally

$$\sum_{j=1}^{K} \lambda_j^{-3} e^{-\eta d/\lambda_j} \le C_\eta (1 + d^{-3}).$$

Indeed, split at $\lambda_j = d$. The coarse part is a geometric sum bounded by $C d^{-3}$, and the fine part is bounded by $C d^{-3} \sum_{u \ge 0} L^{3u} e^{-cL^u}$; when $d \ge 1$ only the latter estimate is needed. Summing the energy decomposition proves the assertion. □

Let $k_n = \sum_\pi (|\pi| - 1)!$, with the sum over partitions of $\{1, \dots, n\}$, and let $\varrho_{\min} = \min(\varrho, \varrho_J)$.

Proposition 9.3 (Uniform cumulants at complex parameters)**.** For every admissible tuple X the lattice cumulant $S^T_{n,K}(b, m; X)$ is holomorphic on $W_{\mathbb{C}}$, and

$$|S^T_{n,K}(b, m; X)| \le \left(\frac{n}{\varrho_{\min}}\right)^n \left[k_n (2V_\ell)^n + 2^{n-1} C_{\mathrm{ov}} \ell^3 (1 + d_\cap(X)^{-3})\right] \prod_i \|f_i\|_{\alpha_i}. \tag{115}$$

For all $p, q \ge 0$,

$$(-\partial_b)^p (-\partial_m)^q S^T_{n,K}(X) = S^T_{n+p+q,K}(X, (0,1)^p, (1,1)^q). \tag{116}$$

The tuple on the right has the same $d_\cap$. ■

Proof. Work about a real base point and represent the new parameters by the constant source c of Prop. 9.1. For a zero test function the assertion follows from multilinearity. Otherwise choose $r_i = \varrho_{\alpha_i}/(n\|f_i\|_{\alpha_i})$. The sum of the action-source norms is at most ϱ, and the sum of the meson-source norms is at most ϱ_J. Since $V_\ell \geq 1$, the constant displacement has norms less than $\rho_b \leq \varrho/2$ and $\rho_m \leq \varrho_J/2$. In particular

$$\Big\| c_w + \sum_i z_i w_i \Big\|_\sharp < \tfrac{3}{2}\varrho < 2\varrho, \qquad \Big\| c_J + \sum_i z_i J_i \Big\|_\infty < \tfrac{3}{2}\varrho_J < 2\varrho_J.$$

Thus the whole Cauchy polydisc stays inside $\mathcal{P}$. The triangle inequality gives this source budget even when the supports overlap.

Set $G(z) = W_K[c + s(z)]/W_K[c]$. The zero-free estimate gives $G(0) = 1$ and $|G(z)| \leq 2V_\ell$. As germs at zero,

$$\log Z_K[c + s(z)] - \log Z_K[c] = E_K(c + s(z)) - E_K(c) + \log G(z).$$

The mixed derivative of the last term is

$$\sum_\pi (-1)^{|\pi|-1}(|\pi| - 1)! \prod_{B\in\pi} \partial^B G(0).$$

For each block B of a partition, Cauchy's estimate gives

$$|\partial^B G(0)| \leq 2V_\ell \prod_{i\in B} r_i^{-1}.$$

The product over the blocks contains each inverse radius once and costs $(2V_\ell)^{|\pi|} \leq (2V_\ell)^n$. Summing the partition coefficients therefore bounds the mixed logarithmic derivative by $k_n(2V_\ell)^n \prod_i r_i^{-1}$. This calculation uses the logarithmic germ at zero; the generating function may have zeros elsewhere in its source polydisc. Apply Lem. 9.2 to the energy and use $\prod_i r_i^{-1} \leq (n/\varrho_{\min})^n \prod_i \|f_i\|_{\alpha_i}$. This proves (115).

The finite-dimensional integral is entire in the parameters and the sources. Its partition function is nonzero at zero source throughout $W_\mathbb{C}$; local logarithms therefore give holomorphic cumulants there. Differentiating the constant-source identity proves (116). Adding an entry supported on the whole torus adds zero to the maximum defining $d_\cap$, so it leaves that distance unchanged. □

Theorem H (Holomorphic families of continuum correlations)**.** Every sequence $K_j \to \infty$ has a subsequence for which $S^T_{n,K_j}(b, m; X)$ converges for every admissible tuple X, locally uniformly in $(b, m) \in W_\mathbb{C}$ and with all parameter derivatives. Denote the limit by $S_n(b, m; X)$. It obeys (115) and (116) with the cutoff removed. At real parameters its restriction to separated tuples is a continuum limit of the original lattice model on T_ℓ. For each fixed choice of indices it determines a unique distribution on

$$T_\ell^n \setminus \{x_1 = \cdots = x_n\}.$$

This distribution is symmetric and translation invariant, has the flavour and cubic covariance of the lattice correlations, and depends holomorphically on the parameters when tested on admissible tensors. ■□

Proof. Choose a countable collection of products whose factors are finite unions of closed dyadic boxes and have empty total intersection. On each product choose countable dense sets of smooth test functions, in the C^1 norm for an action-density entry and the uniform norm for a meson entry. There are finitely many basis indices at each position. The bound (115) and Montel's theorem, followed by a diagonal selection, give locally uniform convergence on every tuple in these sets. Use one enumeration containing all orders n, all label strings, all these support tuples, and their dense test families. A countable compact exhaustion of $W_\mathbb{C}$ makes the local uniform convergence part of this same selection. Thus one subsequence works for all these data.

For an arbitrary admissible tuple the supports have $d = d_\cap(X) > 0$. Enlarge them by less than $d/4$ and cover the enlarged sets by sufficiently fine closed dyadic cubes. The resulting sets still have $d_\cap \geq d/2$. Approximation within these sets extends the convergence to X uniformly on every compact parameter set. To see the uniformity, write $T_K(f_1, \dots, f_n) = S^T_{n,K}(b, m; X)$ with fixed labels and these enlarged supports. The bound gives a common constant B there. Multilinearity gives

$$|T_K(f_1, \dots, f_n) - T_K(h_1, \dots, h_n)| \leq B \sum_{i=1}^{n} \|f_i - h_i\|_{\alpha_i} \prod_{r<i} \|h_r\|_{\alpha_r} \prod_{r>i} \|f_r\|_{\alpha_r}.$$

The approximating functions may be chosen with bounded norms, so the right-hand side tends to zero uniformly in K and the parameters. Applying the same estimate to two cutoffs proves convergence for the arbitrary tuple.

For a compact parameter set, choose slightly larger compact polydiscs still contained in $W_\mathbb{C}$. Cauchy's formula on their boundaries gives convergence of every parameter derivative on the smaller sets. The constant insertions in (116) also belong to the common extraction: adding a whole-torus support preserves $d_\cap > 0$. Hence both sides of the insertion identity converge to the stated identity.

A point outside the total diagonal has two distinct coordinates. It therefore has a product neighbourhood with two disjoint closed factors. On this neighbourhood the estimate defines a continuous multilinear functional on smooth functions, hence a distribution. These local distributions agree on overlaps: products of test functions are dense, and the same limiting functional gives their values. A partition of unity glues them uniquely.

Permutation, flavour and cubic identities hold at each cutoff and pass to the limits. Approximate any torus translation by lattice translations. Translation is continuous in the test-function norms, so the uniform bound gives the full translation identity. At real fixed parameters the same subsequence defines the stated continuum limit. If parameters also converge inside W, local uniform convergence shows that the limit is the value of this same holomorphic family at the limiting parameters. □

Corollary 9.4 (Mass differentiation)**.** For every admissible tuple the function $m \mapsto S_n(b, m; X)$ is real analytic on the allowed real interval and

$$\partial_m S_n(b, m; X) = -S_{n+1}(b, m; X, (1, 1)).$$

The inserted cumulants are bounded uniformly in the cutoff at fixed g, ℓ, n and supports. The same holds for every higher mass derivative and for mixed coupling and mass derivatives. ■□

Proof. Apply the preceding theorem and the insertion identity. The uniform bound for the inserted tuple uses its unchanged $d_\cap > 0$. □

Remark 9.5. The theorem fixes correlations on partial diagonals. It does not fix an extension to the total diagonal. If two translation-invariant distributional extensions exist, their difference is supported there. In relative coordinates it is a finite sum of derivatives of the delta distribution with constant coefficients. Thus the remaining ambiguity is local; it cannot be prescribed independently on a proper partial diagonal.

10. Continuation in the mass

At fixed real coupling and fixed volume, one subsequence gives meromorphic continuations of all separated correlations in the mass, with a common discrete exceptional set. We prove that this set misses a neighbourhood of a longer positive real mass interval. Its width is at least a constant times $[\ell^3(1 + \log \ell)]^{-1}$; see Thm. I and Prop. 10.5. The proof first makes the vacuum normalization independent of the real mass used to define it. Reflection into small cells then gives a differential inequality on the real axis, and a harmonic majorant extends its control to complex masses. The final two propositions retain the short-distance kernels and the volume-independent

real-mass cumulant bounds. The continuum tests in this section are C^1 at the action index and continuous at the scalar-bilinear indices; the finite-cutoff estimates also apply to Lipschitz tests.

10.1. **A common normalization as the mass varies.** We now fix the real coupling $b \in I_g$ and continue in the mass alone. Write $I = [m_-, m_+]$ and

$$U_r = \{z \in \mathbb{C} : \operatorname{dist}(z, I) < r\}, \qquad r_J = r_M/L.$$

The source radii satisfy

$$2\varrho_J \le \tfrac{1}{2} r_J, \qquad 2\varrho_J \le \tfrac{1}{2} m_-.$$

The torus, g, and b remain fixed in every complex continuation below. All cutoff limits are subsequential.

For a real centre $x \in I$, denote its source energy by $E_K^{(x)}(s)$. Put

$$q_z^{(x)} = (0, -(z-x)1), \qquad Z_K(z;t) = Z_K^{(x)}\big[q_z^{(x)} + \textstyle\sum_i t_i s_i\big].$$

Here s_i is the source of the entry (α_i, f_i). The right side is independent of the centre: in its determinant the mass matrix is

$$ax - a\big(J - (z-x)1\big) = az - aJ.$$

Thus $Z_K(z;t)$ is the partition function at bare mass az, with the indicated insertions. It is entire in z, t. Write

$$W_K^{(x)}[s] = e^{-E_K^{(x)}(s)} Z_K^{(x)}[s].$$

This is the normalized partition function denoted by W_K above. When a centre is fixed, we suppress its superscript. The energies at two centres need a separate comparison.

Choose $n_\pi \ge 1$ with $(1 + 1/n_\pi)^{p_0} \le 4/3$, and put

$$\pi_j = (1 + (K-j)/n_\pi)^{p_0}, \qquad \|w\|_{\sharp,j} = \max\{\|w\|_\infty, 40M\lambda_j\pi_j \operatorname{Lip} w\}.$$

Here p_0 is the exponent already fixed in section 4; no new RG weight is chosen. We use the source construction of [QCDM1] on

$$\mathscr{O}_j = \{(w, J) : \|w\|_{\sharp,j} < 4\varrho,\ \lambda_j\|J\|_\infty < r_J\}, \qquad \lambda_j = L^{j-K}.$$

In particular its local energies are holomorphic on $\mathscr{O}_{j-1}$, with the size and tail bounds recalled above. The history bound applies on

$$\mathcal{P}_j^{\#} = \{(w, J) : \|w\|_\sharp < 2\varrho,\ \lambda_j\|J\|_\infty \le r_J/2\}.$$

We will also use the almost-additivity estimate at a base source and its bilinear improvement from [QCDM1, Props. 10.15 and 10.18]. For bilinear sources $s_0 = (0, J_0)$, $s_i = (0, J_i)$ with

$$\|J_0\|_\infty \le r_J/2, \qquad \|J_i\|_\infty \le \epsilon \le r_J/8,$$

whose supports have mutual distance at least δ and lie in balls of radii R_i, that improvement is

$$\left| E_K\left(s_0 + \sum_i s_i\right) - \sum_i E_K(s_0 + s_i) + (N-1)E_K(s_0) \right| \le C_\flat \sum_i \left(1 + (R_i/\delta)^3\right). \tag{117}$$

The constant $C_\flat$ is independent of the cutoff and side. At the fine scales this bound uses the separation exponential. At the coarse scales Schwarz's lemma in the bilinear source adds a factor $\lambda_{j-1}/4$. This makes the sum independent of the number of coarse scales. That improvement is needed when the cells below become small.

Proposition 10.1 (Compatibility of the vacuum energies)**.** Let $x, x' \in I$, $\zeta = x' - x$, and $s_\zeta = (w, J - \zeta 1)$ for $s = (w, J)$. For $1 \le k \le K$, if $s, s_\zeta \in \mathscr{O}_k$, then

$$E_k^{(x)}(s_\zeta) + \mathfrak{f}_k^{(x)} = E_k^{(x')}(s) + \mathfrak{f}_k^{(x')}.$$

Here $\mathfrak{f}_k^{(x)}$ is the real finite-size constant in the empty-history scalar normalization of [QCDM1, Lem. 9.15]; $|\mathfrak{f}_k^{(x)}| \le C_{\mathrm{fs}}$. ∎

Proof. The two bare densities agree by the mass-source identity. Their block integration also agrees before it is expressed in terms of extracted coefficients. Indeed the fermion block kernel reads only b_F and the block averages, while the gauge chart integral and the density test read the calibrations $\mathfrak{t}_j = L^{K-j} b$. None of these depends on the mass centre or on a coefficient extracted by the flow.

More explicitly, if d_j is the density of the empty-history term at level j, the next density is

$$d_{j+1}(W) = \chi_{j+1}(W)\, \mathsf{I}_j[\mathrm{Ber}_j(\mathrm{Ker}_j\, d_j)]\,(W).$$

Here χ_{j+1} is the density-test indicator, I_j integrates the gauge chart, Ber_j integrates the old Grassmann variables, and Ker_j is the fermion block kernel. The density is zero when W fails the test. This is the exact tested chart and block-integration identity of [QCDM1, Lem. 7.1 and Prop. 10.1]. The two inputs have $d_0 = d_0'$; linearity and the identical kernel, test and chart give $d_j = d_j'$ at every subsequent level.

We compare the scalar coefficients at the trivial block field. In the extracted form of the empty-history density all fermion quadratic terms and activities with legs have positive Grassmann degree. Thus its degree-zero coefficient, at a tested block field V, is

$$g_0(V) = \exp\left[E_k^{(x)}(s_\zeta) - A_k(U; s_\zeta) + \sum_{j \le k} \sum_X E_0^{(j)}(X; \mathbf{p}_X^{(j)}(U); s_\zeta) \right], \tag{118}$$

where U is any background in the minimal orbit over V. Gauge invariance makes this expression independent of that choice. The analogous formula defines g_0'.

The tested block fields form an open set containing 1. On a neighbourhood of 1, the canonical critical-background theorem of [YMM1, Thm. E] provides a continuous representative $V \mapsto U^\circ(V)$ of the minimal orbit. To apply it, write $V = e^{iB_V}$ bond by bond using the principal logarithm. Write r_{B} for the positive logarithmic row radius in that theorem. For $\|V - 1\| < \min\{1, 2r_{\mathrm{B}}/\pi\}$, $\|B_V\| \le (\pi/2)\|V - 1\| < r_{\mathrm{B}}$. The logarithm is continuous and the representative is holomorphic in B_V on this ball. Their composition has the required continuity. The geometric hypotheses are those already verified in the tested-block-field lemma of [QCDM1]. Its restricted pairs depend continuously on V; the scalar activities are holomorphic on an open pair domain containing them. The action and the currents entering those pairs are continuous functions of the link matrices. There are only finitely many polymers at fixed cutoff and volume. Therefore (118) is continuous near 1, and so is g_0'.

The exact density identity gives equal integrals of g_0 and g_0' over every Borel subset of the block-field space. The reference measure there is Haar measure and charges every nonempty open set. Two continuous densities agreeing almost everywhere must consequently agree throughout this neighbourhood. At $V = 1$ we may take $U = 1$. The action vanishes, and the scalar normalization in the source step gives

$$g_0(1) = \exp(E_k^{(x)}(s_\zeta) + \mathfrak{f}_k^{(x)}), \qquad g_0'(1) = \exp(E_k^{(x')}(s) + \mathfrak{f}_k^{(x')}).$$

Their equality determines the difference of the exponents modulo $2\pi i\mathbb{Z}$.

The common domain $\{s : s, s_\zeta \in \mathscr{O}_k\}$ is convex. The difference of the two exponents is continuous there and takes values in $2\pi i\mathbb{Z}$, hence is constant. If the domain is nonempty, its mass inequalities imply $|\zeta| < 2r_J/\lambda_k$; it therefore contains $s_* = (0, \zeta 1/2)$. The two sources at this point are $(0, \zeta 1/2)$

and $(0,-\zeta 1/2)$. They are fixed by source conjugation, so both energies and both finite-size constants are real. Their difference is therefore zero. This fixes the branch and proves the identity. □

Proposition 10.2 (A common normalizer)**.** For every $K \geq 1$ there is a holomorphic function

$$\mathcal{E}_K(z) = E_K^{(x)}(q_z^{(x)}) + \mathfrak{f}_K^{(x)}, \qquad |z-x| < r_J, \quad x \in I,$$

on U_{r_J}. It satisfies $\mathcal{E}_K(\bar{z}) = \overline{\mathcal{E}_K(z)}$. Set

$$M_0 = \exp\Big(\max\{C+2, C_P(1+\log X_g)\}\ell^3\Big), \qquad h_K(z) = Z_K(z)e^{-\mathcal{E}_K(z)}.$$

Then

$$|h_K(z)| \leq M_0 e^{C_{\mathrm{fs}}} \quad (\mathrm{dist}(z,I) \leq 2\varrho_J), \qquad h_K(x) \geq (M_0 e^{C_{\mathrm{fs}}})^{-1} \quad (x \in I). \tag{119}$$

For $0 < r < 2\varrho_J$, put

$$\rho_{n,r} = \min\{\varrho/n, 2\varrho_J - r\}, \qquad M_{n,d} = M_0 \exp\Big[nC_{\mathrm{count}}(1+d^{-3})\ell^3\Big].$$

Here C_{count} is the torus source-count constant. If $n \geq 2$, the supports of $X = ((\alpha_i, f_i))_{i=1}^n$ are separated by at least d, and $\|f_i\|_{\alpha_i} \leq \rho_{n,r}$, then there are holomorphic functions $\mathcal{E}_{K,i}(z,t_i)$ on $U_r \times \{|t_i| < 1\}$, zero at $t_i = 0$, such that

$$\Big|Z_K(z;t)e^{-\mathcal{E}_K(z)-\sum_i \mathcal{E}_{K,i}(z,t_i)}\Big| \leq M_{n,d}e^{C_{\mathrm{fs}}}. \tag{120}$$

■

Proof. If $|z-x|, |z-x'| < r_J$, apply Prop. 10.1 to $s = q_z^{(x')}$. Its shifted source is $q_z^{(x)}$. Thus the displayed definitions of $\mathcal{E}_K$ agree. Each is holomorphic on its disc, and the discs cover U_{r_J}. Source conjugation gives the stated reality identity. The upper bound follows from (110) at $q_z^{(x)}$, and the lower bound at $x \in I$ follows from the zero-source lower bound at that centre. The factor $e^{C_{\mathrm{fs}}}$ accounts for $\mathfrak{f}_K^{(x)}$.

Define locally

$$\mathcal{E}_{K,i}(z,t_i) = E_K^{(x)}(q_z^{(x)} + t_i s_i) - E_K^{(x)}(q_z^{(x)}).$$

For $|z-x| < r$, $|t_i| < 1$, every partial sum of these sources lies in the interior of $\mathcal{P}$: the action norm is less than $n\rho_{n,r} \leq \varrho$, and disjointness of the supports gives the matrix-source norm

$$|z-x| + \max_i |t_i|\,\|f_i\|_\infty < 2\varrho_J.$$

Apply Prop. 10.1 with and without $t_i s_i$, and subtract. The finite-size constants cancel, proving that $\mathcal{E}_{K,i}$ is independent of the chosen centre.

Almost additivity at the base $q_z^{(x)}$ gives

$$\left|E_K^{(x)}\left(q_z^{(x)} + \sum_i t_i s_i\right) - E_K^{(x)}(q_z^{(x)}) - \sum_i \mathcal{E}_{K,i}(z,t_i)\right| \leq \sum_i \mathscr{C}_K(\mathrm{supp}\, s_i; d) \leq nC_{\mathrm{count}}(1+d^{-3})\ell^3.$$

Combining this with the partition upper bound proves (120). □

10.2. **A lower bound on a longer real mass interval.** The common normalizer controls possible zeros uniformly on compact subsets of $U_{2\varrho_J}$. To obtain a useful zero-free neighbourhood we first bound the normalized partition function on a longer real interval.

Fix $0 < \epsilon \leq \varrho_J/2$, and set

$$R_\epsilon = 2\varrho_J - 2\epsilon, \qquad c_g = C_P(1+\log X_g),$$
$$c_\epsilon = 512\epsilon^{-1}L^3 e^{1+19C_\flat}, \qquad A_\epsilon = (C+3+c_g)e^{c_\epsilon R_\epsilon},$$
$$K_\epsilon = 1 + \left\lceil \tfrac{1}{3}\log_L A_\epsilon \right\rceil.$$

These constants are independent of the side and cutoff.

Proposition 10.3 (The normalized partition function at real masses)**.** Let $x_0 \in I$, $K \geq K_\epsilon$, and

$$h_{K,x_0}(x) = Z_K(x) e^{-E_K^{(x_0)}(q_x^{(x_0)})}.$$

For real $|x - x_0| \leq R_\epsilon$,

$$h_{K,x_0}(x) \geq e^{-A_\epsilon \ell^3}, \qquad |(\log h_{K,x_0})'(x)| \leq c_\epsilon A_\epsilon \ell^3. \tag{121}$$

■

Proof. Write $E = E_K^{(x_0)}$, $q_x = q_x^{(x_0)}$, $h = h_{K,x_0}$, and $u(x) = -\ell^{-3} \log h(x)$. All these real masses exceed $m_-/2$. The positive real-mass density gives $Z_K(x) > 0$, and $E(q_x)$ is real, so $h > 0$ and u is real analytic.

For $1 \leq i \leq K-1$, divide the torus into cells of physical side $\lambda = L^{-i}$. Their lattice side $\lambda_{\mathrm{l}} = L^{K-i}$ is odd and at least L; the number of cells in each direction is even. Let $N_{\mathrm{c}} = (\ell/\lambda)^3$. In a cell centred at y^c, in lattice coordinates, put

$$\eta_c(y) = \prod_{\nu=1}^{3} \tau(y_\nu - y_\nu^c), \qquad \tau(v) = \min\left\{1, \max\left\{0, \frac{3\lambda_{\mathrm{l}}/8 - |v|}{\lambda_{\mathrm{l}}/4}\right\}\right\}.$$

These functions are carried to one another by wall reflections. Their supports have mutual distance at least $\lambda/4$, and each lies in a ball of radius $3\sqrt{3}\lambda/8$. Consequently $1 + (R/\delta)^3 \leq 19$.

For $|t| \leq \epsilon$, apply the chessboard inequality to $e^{\Phi(q_x + t\eta_c 1)}$ in one cell and to $e^{\Phi(q_x)}$ in the others, with each constant source restricted to its own cell. Since q_x is real and scalar, its fold is the same constant source. Thus

$$|Z[q_x + t\eta_c 1]| \leq Z[q_x + (t\eta)^{\mathrm{ch}}]^{1/N_{\mathrm{c}}} Z[q_x]^{1-1/N_{\mathrm{c}}},$$

where the coefficient in a reflected cell is t or $\bar{t}$. The folded source has norm at most $R_\epsilon + \epsilon < 2\varrho_J$. Use (117) and the partition upper bound. With

$$\Lambda(t) = N_{\mathrm{c}}^{-1} \sum_{c'} [E(q_x + t\eta_{c'} 1) - E(q_x)],$$

source conjugation identifies the real parts of the odd and even reflected terms. We obtain

$$\psi_c(t) = e^{-\Lambda(t)} \frac{Z[q_x + t\eta_c 1]}{Z[q_x]}, \qquad \psi_c(0) = 1,$$

$$|\psi_c(t)| \leq \exp\left[19C_\flat + (C + 2 + u(x))\lambda^3\right] =: B_1(x, \lambda).$$

These functions are holomorphic past $|t| = \epsilon$, so $|\psi_c'(0)| \leq B_1/\epsilon$.

For a site function v, define the linear functional

$$\mathcal{V}(v) = \frac{d}{dt} \log W_K[q_x + tv1]\Big|_{t=0}.$$

For one offset o of the board, $\sum_c \psi_c'(0) = \mathcal{V}(\eta^{(o)})$, where $\eta^{(o)} = \sum_c \eta_c$. Hence $|\mathcal{V}(\eta^{(o)})| \leq N_{\mathrm{c}} B_1/\epsilon$. Summing over the λ_{l}^3 offsets gives

$$\sum_o \eta^{(o)} = \sigma^3, \qquad \sigma = \sum_{v \in \mathbb{Z}} \tau(v) \geq \lambda_{\mathrm{l}}/8.$$

Since $q_x + (0, t1) = q_{x-t}$, this yields

$$|u'(x)| \leq 512\epsilon^{-1}\lambda^{-3} \exp\left[19C_\flat + (C + 2 + u(x))\lambda^3\right].$$

Set $Y = C + 2 + \max\{u, 1\}$. At a point where $Y \leq A_\epsilon$, choose $i = \lceil \frac{1}{3} \log_L Y \rceil$. Then

$$1 \leq i \leq K - 1, \qquad \lambda^3 Y \leq 1 < L^3 \lambda^3 Y.$$

The preceding bound gives $|u'| \le c_\epsilon Y$ and $|Y'| \le c_\epsilon Y$ almost everywhere. At x_0, the zero-source floor gives $u(x_0) \le c_g$, so $Y(x_0) \le C + 3 + c_g$. Integrate the differential inequality on the maximal interval where $Y \le A_\epsilon$. If its right endpoint were less than $x_0 + R_\epsilon$, then

$$Y(x) \le (C + 3 + c_g) e^{c_\epsilon (x - x_0)} < A_\epsilon$$

at that endpoint, contradicting maximality and continuity. The same argument to the left gives $Y \le A_\epsilon$ throughout $|x - x_0| \le R_\epsilon$. The two assertions now follow from $u \le A_\epsilon$ and $|(\log h)'| = \ell^3 |u'|$. □

Lemma 10.4 (A complex neighbourhood from a real derivative bound)**.** Let h be holomorphic near $\{|z - x| \le r\}$, where x is real, and suppose

$$h(\bar z) = \overline{h(z)}, \qquad h > 0 \text{ on } [x - r, x + r],$$
$$\log |h(z)/h(x)| \le W, \qquad \log(h(x')/h(x)) \le \Lambda |x' - x| \quad (x' \in [x - r, x + r]).$$

For $z = x + \xi + i\eta$, $|z - x| \le r/2$, $\eta \ne 0$,

$$\log |h(z)/h(x)| \le \frac{16W}{3\pi} \frac{|\eta|}{r} + \Lambda |\xi| + \frac{2}{\pi} \Lambda |\eta| \log \frac{r}{|\eta|}. \tag{122}$$

Consequently, if $0 < \rho \le r/e$ and

$$\rho \left[\frac{16W}{3\pi r} + \Lambda + \frac{2}{\pi} \Lambda \log(r/\rho) \right] \le 1,$$

then $|h(z)| \ge h(x)/2$ for $|z - x| \le \rho/(2(e + 1))$. ■

Proof. By conjugation it suffices to consider $\eta > 0$. On the upper unit half-disc put

$$\zeta = (z - x)/r, \qquad u(\zeta) = \log |h(x + r\zeta)/h(x)|.$$

This function is subharmonic and bounded above, including when h has zeros. The harmonic measure of the semicircle is

$$\omega(\zeta) = \frac{2}{\pi} \arctan \frac{2 \operatorname{Im} \zeta}{1 - |\zeta|^2}.$$

The fractional linear map $(1 + \zeta)/(1 - \zeta)$ sends the half-disc to the first quadrant and its semicircle to the positive imaginary axis, giving this formula. For $|\zeta| \le 1/2$, $\omega(\zeta) \le 16 \operatorname{Im} \zeta/(3\pi)$. The second harmonic function

$$q(\zeta) = \operatorname{Re} \left(\zeta + \frac{2i}{\pi} \zeta \log \zeta \right) = \operatorname{Re} \zeta \left(1 - \frac{2}{\pi} \arg \zeta \right) + \frac{2}{\pi} \operatorname{Im} \zeta \log \frac{1}{|\zeta|}$$

uses the principal logarithm. It extends continuously to the diameter with value $|s|$, and on the semicircle its value is $\cos t (1 - 2t/\pi) \ge 0$. The boundary assumptions therefore give $u \le W\omega + \Lambda r q$ on the half-disc by the maximum principle. The two endpoints cause no difficulty because u is bounded above. Bound $|1 - 2 \arg \zeta/\pi| \le 1$ and $\log(1/|\zeta|) \le \log(1/\operatorname{Im} \zeta)$. This proves (122).

Since $y \log(r/y)$ increases for $0 < y \le r/e$, the hypothesis on ρ gives $|h| \le e\, h(x)$ on the disc of radius ρ. Schwarz's lemma applied to $h - h(x)$, bounded there by $(e + 1)h(x)$, gives

$$|h(z) - h(x)| \le (e + 1) h(x) |z - x|/\rho.$$

The asserted lower bound follows. □

Set

$$I_\epsilon = [m_- - 2\varrho_J + 3\epsilon,\ m_+ + 2\varrho_J - 3\epsilon], \quad W_\ell = (C + 2 + A_\epsilon)\ell^3, \quad \Lambda_\ell = c_\epsilon A_\epsilon \ell^3.$$

Let ρ_ℓ be the largest number in $(0, \epsilon/e]$ such that

$$\rho_\ell \left[\frac{16 W_\ell}{3\pi\epsilon} + \Lambda_\ell + \frac{2}{\pi} \Lambda_\ell \log(\epsilon/\rho_\ell) \right] \le 1, \qquad w_\ell = \frac{\rho_\ell}{2(e + 1)}. \tag{123}$$

The left side is increasing on this interval and tends to zero at zero. Thus ρ_ℓ exists. Define

$$\Omega_\epsilon = \{z : \operatorname{dist}(z, I_\epsilon) < w_\ell\}, \qquad \rho_n^\epsilon = \min\{\varrho/n, 2\epsilon\}.$$

All real points of Ω_ϵ exceed $m_-/2$.

Proposition 10.5 (A zero-free neighbourhood of the longer interval)**.** For $K \geq K_\epsilon$ and $z \in \Omega_\epsilon$,

$$|Z_K(z)| \geq \tfrac{1}{2} e^{-A_\epsilon \ell^3 - C_{\mathrm{fs}}} |e^{\mathcal{E}_K(z)}| > 0. \tag{124}$$

If X has $n \geq 2$ entries with pairwise separation at least $d > 0$, then

$$|S^T_{n,K}(z;X)| \leq \left[\frac{2(2e^{A_\epsilon \ell^3} M_{n,d} + 1)}{\rho^\epsilon_n}\right]^n \prod_i \|f_i\|_{\alpha_i}. \tag{125}$$

There is $c_w > 0$, depending on the fixed constants, g and ϵ, but not on the side, such that

$$w_\ell \geq \frac{c_w}{\ell^3(1 + \log \ell)}.$$

■

Proof. For $z \in \Omega_\epsilon$, choose $x \in I_\epsilon$ with $|z - x| < w_\ell$, and let $x_0 \in I$ be nearest to x. Then

$$|x - x_0| \leq R_\epsilon - \epsilon, \qquad \{|\zeta - x| \leq \epsilon\} \subset \{|\zeta - x_0| \leq R_\epsilon\}.$$

On this disc h_{K,x_0} is holomorphic, $|h_{K,x_0}| \leq e^{(C+2)\ell^3}$, and $h_{K,x_0}(x) \geq e^{-A_\epsilon \ell^3}$. On its real diameter Prop. 10.3 gives the logarithmic derivative bound Λ_ℓ. Apply Lem. 10.4 with $r = \epsilon$, $W = W_\ell$, and $\Lambda = \Lambda_\ell$. It gives

$$|h_{K,x_0}(z)| \geq \tfrac{1}{2} e^{-A_\epsilon \ell^3}.$$

Since $\mathcal{E}_K(z) = E^{(x_0)}_K(q^{(x_0)}_z) + \mathfrak{f}^{(x_0)}_K$, this is (124).

Also $|z - x_0| < R_\epsilon$. For $\|f_i\|_{\alpha_i} \leq \rho^\epsilon_n$, the local insertion bound used in Prop. 10.2 applies at this centre with radius $R_\epsilon = 2\varrho_J - 2\epsilon$. The holomorphic function

$$\Psi(t) = \frac{Z_K(z;t) e^{-E^{(x_0)}_K(q^{(x_0)}_z) - \sum_i \mathcal{E}_{K,i}(z,t_i)}}{h_{K,x_0}(z)}$$

has $\Psi(0) = 1$ and $|\Psi| \leq 2e^{A_\epsilon \ell^3} M_{n,d}$ on $\max_i |t_i| < 1$. On each line through zero, Schwarz's lemma bounds $|\Psi(t) - 1|$ by $(2e^{A_\epsilon \ell^3} M_{n,d} + 1) \max_i |t_i|$. On the polydisc of radius $[2(2e^{A_\epsilon \ell^3} M_{n,d} + 1)]^{-1}$, its logarithm has modulus at most $\log 2$. For $n \geq 2$, its full mixed derivative equals that of $\log Z_K(z;t)$: each subtracted insertion energy depends on only one t_i. Cauchy's inequality and multilinearity give (125).

For the width, put

$$\alpha = \frac{16(C + 2 + A_\epsilon)}{3\pi\epsilon} + c_\epsilon A_\epsilon, \quad \beta = \frac{2c_\epsilon A_\epsilon}{\pi}, \quad L_\ell = \max\{1, \log(2\epsilon\ell^3(\alpha + \beta))\}.$$

The defining expression in (123) is $\rho\ell^3[\alpha + \beta \log(\epsilon/\rho)]$. For $\rho_* = [2\ell^3(\alpha + 2\beta L_\ell)]^{-1}$,

$$\log(\epsilon/\rho_*) \leq L_\ell + \log(2L_\ell) \leq 2L_\ell.$$

Thus its value at ρ_* is at most one when $\rho_* \leq \epsilon/e$; if $\rho_* > \epsilon/e$, its value at ϵ/e is less than $1/2$. It follows that

$$\rho_\ell \geq \min\{\epsilon/e, \rho_*\}.$$

For $\ell \geq 1$,

$$L_\ell \leq a_w + 3 \log \ell \leq 3a_w(1 + \log \ell), \qquad a_w = 1 + |\log(2\epsilon(\alpha + \beta))|.$$

One may therefore take

$$c_w = \frac{1}{2(e+1)} \min\{\epsilon/e, [2(\alpha + 6\beta a_w)]^{-1}\}.$$

□

10.3. **A common meromorphic family.**

Theorem I (Continuation in the mass)**.** Fix $b \in I_g$ and $0 < \epsilon \le \varrho_J/2$. Every sequence $K_j \to \infty$ has a subsequence and a holomorphic function h on $U_{2\varrho_J}$, not identically zero, with the following properties.

For every separated tuple X of order $n \ge 2$, $S^T_{n,K_j}(z;X)$ converges locally uniformly off the discrete set $\mathcal{Z} = \{h = 0\}$ to a meromorphic function $\mathcal{S}_n(z;X)$ on $U_{2\varrho_J}$, and $h^n\mathcal{S}_n(\cdot;X)$ is holomorphic there. The same h, subsequence and exceptional set work for all orders, labels and tuples. Moreover

$$h_{K_j} \longrightarrow h \quad \text{locally uniformly on } U_{2\varrho_J}.$$

The set $\mathcal{Z}$ is symmetric under conjugation and is the set of accumulation points, in this open domain, of zeros of Z_{K_j}.

The set $\mathcal{Z}$ is disjoint from Ω_ϵ. On Ω_ϵ the limits are holomorphic, obey (125), and converge with all mass derivatives. If $z_j \to z \in \Omega_\epsilon$, then

$$S^T_{n,K_j}(z_j;X) \longrightarrow \mathcal{S}_n(z;X).$$

At $m \in I$ these values are continuum limits with labels (b,m). Every continuum limit at fixed b, obtained with masses $m_j \to m \in I$, occurs this way after a subsequence. On all real masses in Ω_ϵ the centered limiting functional is reflection positive on the separated polynomial words used in its construction. It has the permutation, translation, cubic, flavour and charge-conjugation symmetries of the finite lattice theory.

Two such meromorphic families which agree on a subset of I having an accumulation point agree throughout $U_{2\varrho_J}$. This is a uniqueness assertion for analytic continuation from those values; it does not assert that distinct cutoff subsequences have the same limits. ■□

Proof. The functions h_K are uniformly bounded on $U_{2\varrho_J}$ by (119). We construct bounded holomorphic numerators for the cumulants on the same domain. Fix $0 < r < 2\varrho_J$, a separated tuple X, and first rescale its entries so that $\|f_i\|_{\alpha_i} \le \rho_{n,r}$. Put

$$\Psi_K(z,t) = Z_K(z;t)e^{-\mathcal{E}_K(z)-\sum_i \mathcal{E}_{K,i}(z,t_i)}.$$

It is bounded by $M_{n,d}e^{C_{\mathrm{fs}}}$ on $U_r \times \{\max_i |t_i| < 1\}$, and $\Psi_K(z,0) = h_K(z)$. Cauchy's formula on smaller circles, followed by their radius limit, gives the same bound for $\partial_B\Psi_K(z,0)$, $B \subset \{1,\dots,n\}$. The logarithmic derivative formula used in Prop. 9.3 gives, where $h_K \neq 0$,

$$S^T_{n,K}(z;X) = \frac{N_K(z;X)}{h_K(z)^n},$$

$$N_K(z;X) = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)!\, h_K(z)^{n-|\pi|} \prod_{B\in\pi} \partial_B\Psi_K(z,0).$$

The one-variable insertion energies have zero full mixed derivative because $n \ge 2$. Each numerator is holomorphic. The formulas obtained on different U_r's agree away from the discrete zeros of Z_K, hence everywhere by continuity. They therefore define one holomorphic $N_K(\cdot;X)$ on $U_{2\varrho_J}$. For the original entries its bound on U_r is

$$|N_K(z;X)| \le k_n \left(\frac{M_{n,d}e^{C_{\mathrm{fs}}}}{\rho_{n,r}}\right)^n \prod_i \|f_i\|_{\alpha_i}.$$

Apply Montel's theorem to h_K and these numerators on a countable compact exhaustion of $U_{2\varrho_J}$. Use one enumeration of all orders, basis labels and dense smooth test families with supports in fixed separated closed sets. The same diagonal selection gives convergence for every member of this enumeration. The displayed multilinear bound extends convergence to arbitrary separated tuples: enlarge their supports slightly while keeping positive separation, approximate in the action C^1 norm and meson uniform norm, and telescope the difference of the multilinear forms one entry at a time. The resulting error tends to zero uniformly on each compact mass set and uniformly in K.

Denote the limits by h, N. On I, (119) gives $h \geq (M_0 e^{C_{\rm fs}})^{-1}$, so $h \not\equiv 0$. Its zeros are discrete. On a compact set avoiding them, h_{K_j} is eventually bounded away from zero, and hence

$$S^T_{n,K_j} \longrightarrow \mathcal{S}_n = N/h^n$$

locally uniformly there. Source conjugation makes h_K real under conjugation, and the same is true of h. If $h(z) \neq 0$, uniform convergence excludes nearby zeros of h_{K_j} for all sufficiently large j. Conversely, on a small circle around a zero of h and containing no other zero, Rouché's theorem gives zeros of h_{K_j} inside for every sufficiently large j. Since h_K and Z_K have the same zeros, this proves the assertion about $\mathcal{Z}$.

By (124),

$$|h_K(z)| \geq \tfrac{1}{2} e^{-A_\epsilon \ell^3 - C_{\rm fs}} \quad (z \in \Omega_\epsilon,\ K \geq K_\epsilon).$$

Thus h has no zero there. The cumulant bound passes to the limit; Cauchy's formula on slightly larger compact sets gives convergence of all mass derivatives. Local uniform convergence and continuity give convergence at moving masses. In particular a continuum limit already obtained at $m_j \to m \in I$ is preserved when this further extraction is made.

For reflection positivity no convergence of one-point functions is required. At a real positive mass, center each lattice insertion by its own mean. Every centered moment is the sum, over partitions into blocks of size at least two, of products of cumulants. It therefore converges by the preceding result. On a positive-time polynomial the finite lattice reflection form is nonnegative, where each monomial has internally separated supports. Approximate a coordinate reflection plane by a bond plane at distance at most a; the reflected tests differ by a translation of at most $2a$. For smooth tests supported away from the plane, their norms converge and all separations in the reflection form stay bounded below. The uniform cumulant bound and multilinearity pass the inequality to the limit. This proves reflection positivity of the centered functional at every real mass in Ω_ϵ.

The lattice permutation and internal identities hold for the same finite integrals at these masses and pass to the limits. Lattice translations approximate all torus translations in the test norms, as in Thm. H; cubic covariance passes directly. Equivalently, the finite-cutoff identities are identities of rational functions of the mass and hold throughout their common domain of definition. Finally the identity theorem, applied after clearing denominators, proves the last assertion. □

10.4. **Correlation estimates at the continued masses.** The continuation preserves the two leading short-distance kernels. We first estimate the change of the source energy. The estimate uses Schwarz's lemma in the constant mass source; its scale factor is needed before the sum over levels.

Use the test pairs of (84), with $0 < d \leq 1/200$, and put

$$\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,dx\,dy, \qquad H_d = 1 + \log_L(1/d).$$

Thus

$$\mathcal{K}_q(f,h) \geq 10^{-q}\vartheta^2 d^{6-q}\|f\|_\infty \|h\|_\infty. \tag{126}$$

Only action-density and scalar-bilinear indices are used below.

Lemma 10.6 (Changing the mass in the mixed energy)**.** Fix a centre $x_0 \in I$ and $|\zeta| < 2\varrho_J$, and set $q[\zeta] = (0, -\zeta 1)$. Let $(F_\nu, H_\nu)_{\nu=1}^N$ be translates and coordinate mirror images of one test pair, including the induced action on its labels. Suppose the pairs have mutual distance at least d and $N \geq (\ell/(96d))^3$. Write

$$Q_K(\zeta) = \frac{1}{N}\sum_{\nu=1}^{N} \partial_{t_1}\partial_{t_2} E_K^{(x_0)}(q[\zeta] + t_1 F_\nu + t_2 H_\nu)\Big|_{t=0}.$$

There are constants A_B, A_0, independent of K, ℓ, b, x_0, such that

$$|Q_K(\zeta) - Q_K(0)| \leq A_B|\zeta| d^3 H_d \|f\|_\infty \|h\|_\infty$$

at two bilinear indices, and

$$|Q_K(\zeta) - Q_K(0)| \le A_0 \left(\frac{40M}{\vartheta\varrho}\right)^2 |\zeta| d H_d^{1+2p_0} \|f\|_\infty \|h\|_\infty$$

at two action-density indices. Here p_0 is the exponent in the RG weight $\phi(t) = t^{-1/2}(1+\log t)^{p_0}$. ■

Proof. Write $E = E_K^{(x_0)}$ and

$$e_j^\zeta(X;s) = e_j(X; q[\zeta] + s) - e_j(X;s).$$

The desired difference is the sum over j, X of the mean mixed derivatives of e_j^ζ. For a fixed level let $f_1 = f, f_2 = h$, and take source radii

$$r_i = \frac{r_J}{2\lambda_{j-1}\|f_i\|_\infty} \quad \text{at a bilinear index,} \qquad r_i = \frac{2\varrho}{\|f_i\|_{\sharp,j-1}} \quad \text{at the action index.}$$

For sources F, H with the disjoint supports and norms of the given pair, $s(t) = t_1 F + t_2 H$ lies in the level source domain when $|t_i| < r_i$. Indeed disjoint supports make the norm of the sum the maximum of the two norms. Adding $q[\zeta']$, with $|\zeta'| < r_J/(2\lambda_{j-1})$, still gives $\lambda_{j-1}\|J\|_\infty < r_J$ and $\|w\|_{\sharp,j-1} < 2\varrho$.

If s, s' are two such sources, the function

$$\zeta' \longmapsto e_j^{\zeta'}(X;s) - e_j^{\zeta'}(X;s')$$

vanishes at zero. The local source-tail estimate bounds it on this mass disc by

$$2C_E e^{-\kappa d_j(X)} \exp\left[-\kappa_\sigma \operatorname{dist}(X, \operatorname{supp}(s-s'))/L^j\right].$$

Schwarz's lemma therefore multiplies this bound by $2\lambda_{j-1}|\zeta|/r_J$. Apply the estimate with s' obtained by setting either t_1 or t_2 equal to zero. The finite-difference formula and Cauchy's estimate on the two source circles give

$$\begin{aligned} |\partial_{12} e_j^\zeta(X;F,H)| \le{}& \frac{8C_E\lambda_{j-1}|\zeta|}{r_J r_1 r_2} e^{-\kappa d_j(X)} \\ &\times \exp\left[-\frac{\kappa_\sigma}{L^j}\max\{\operatorname{dist}(X,\operatorname{supp}F), \operatorname{dist}(X,\operatorname{supp}H)\}\right]. \end{aligned} \tag{127}$$

The distances in this local display are in the original lattice units. Their physical scale is λ_j.

The energy-count estimate of [QCDM1, Prop. 10.16 and its proof] sums the last two factors over polymers. At $\lambda_j \le d$, each support lies in a radius-d ball and $\#_j^+(\operatorname{supp} F_\nu) \le 1728(d/\lambda_j)^3$. Separation retains $e^{-\kappa_E d/\lambda_j}$, where $\kappa_E > 0$ is the separation rate in the cited counting estimate; it is chosen at most $\min\{\kappa_\sigma/8, \kappa/(4c_{\mathcal{D}}M)\}$. Consequently the level contribution to the mean is at most

$$\frac{8\cdot 1728 C_{\mathscr{C}}\lambda_{j-1}|\zeta|}{r_J r_1 r_2}(d/\lambda_j)^3 e^{-\kappa_E d/\lambda_j}.$$

At $\lambda_j > d$, give the pairs independent signs ϵ_ν. The two sums $F_\epsilon = \sum_\nu \epsilon_\nu F_\nu$ and $H_\epsilon = \sum_\nu \epsilon_\nu H_\nu$ retain the norms and disjointness required in (127). The sign average of any bilinear form at these two sums is its sum over the diagonal pairs. The total number of level cubes is bounded by

$$(\ell/\lambda_j)^3 \le 96^3 N (d/\lambda_j)^3.$$

The mean contribution of this level is therefore at most

$$\frac{8\cdot 96^3 C_{\mathscr{C}}\lambda_{j-1}|\zeta|}{r_J r_1 r_2}(d/\lambda_j)^3.$$

This use of the total cube count avoids multiplying the coarse-scale bound by N.

For bilinears,

$$(r_1 r_2)^{-1} = 4\lambda_{j-1}^2 r_J^{-2}\|f\|_\infty\|h\|_\infty, \qquad \lambda_{j-1}^3(d/\lambda_j)^3 = d^3/L^3.$$

There are at most H_d coarse levels. The fine levels have distinct integers $p \ge 0$ with $d/\lambda_j \in [L^p, L^{p+1})$; their exponentials have a bounded sum. For example one may take

$$C'_\kappa = \sum_{p\ge 0} L^{3p} e^{-\kappa_E L^p}, \qquad A_B = \frac{2^{25} C_{\mathscr{C}}(1 + C'_\kappa)}{L^3 r_J^3}.$$

For action sources let $\pi_{j-1} = (1 + (K - j + 1)/n_\pi)^{p_0}$, the weight in the level source norm. The test-pair Lipschitz bound gives

$$\|f\|_{\sharp,j-1} \le \frac{40 M \pi_{j-1}}{\vartheta} \max\{1, \lambda_{j-1}/d\} \|f\|_\infty.$$

At the coarse levels,

$$\pi_{j-1} \le (3 + \log_L(1/d))^{p_0}, \qquad \lambda_{j-1}(d/\lambda_j)^3 \max\{1, \lambda_{j-1}/d\}^2 \le d/L.$$

At the fine level indexed by p, the corresponding bounds are $((3 + \log_L(1/d))(1 + p))^{p_0}$ and dL^{2p+1}. Thus the fine-scale sum is paid by

$$C''_\kappa = \sum_{p\ge 0} L^{2p} (1 + p)^{2p_0} e^{-\kappa_E L^p}.$$

One may take

$$A_0 = \frac{2^{21} 9^{p_0} C_{\mathscr{C}}(1 + L C''_\kappa)}{r_J}.$$

Substitution proves the two estimates. □

The next estimate keeps the partition function normalized at the shifted real mass. Its loss is proportional to the volume of an occupied cell, rather than to the volume of the torus.

Lemma 10.7 (A cell at a shifted real mass)**.** Let $x_0 \in I$, $|x - x_0| < 2\varrho_J$, and x be real. Put $q = q_x^{(x_0)}$, $E = E_K^{(x_0)}$, and

$$u(x) = -\ell^{-3} \log\bigl(Z_K(x) e^{-E(q)}\bigr).$$

Consider a reflection board with N cells of physical volume $|\square|$. Suppose $s = (w, J)$ is supported in one cell, at distance at least $\delta > 0$ from its walls, and

$$\|w\|_\sharp < 2\varrho, \qquad \|J\|_\infty \le 2\varrho_J - |x - x_0|.$$

If r_{lin} denotes the linear part of each board reflection on sources, set

$$\lambda_x(s) = \frac{1}{\mathsf{N}} \sum_r \{E(q + r_{\mathrm{lin}} s) - E(q)\}.$$

Then

$$\left| \frac{Z[q + s]}{Z[q]} \right| \le \exp\left[\operatorname{Re} \lambda_x(s) + (C + 2 + u(x))|\square| + \mathscr{C}_K(\operatorname{supp} s; \delta) \right].$$

For a sum $s = \sum_\square s_\square$ satisfying these conditions in its respective cells, the right side is the product of the displayed factors for the occupied cells. ■

Proof. The constant scalar source q is real and is fixed by reflection. Its exponential is the product of the even factors $e^{\Phi(q 1_\square)}$ in the cell algebras. Apply the chessboard inequality at the original positive mass x_0, using $e^{\Phi(q1_\square)} e^{O(w)+\Phi(J)}$ in the occupied cell and $e^{\Phi(q1_\square)}$ elsewhere. Folding an unoccupied cell gives $e^{\Phi(q)}$. Folding the occupied cell gives the source $q + s^{\mathrm{ch}}$, where the images of s have conjugated coefficients after an odd number of reflections. Hence

$$\left| \frac{Z[q + s]}{Z[q]} \right| \le \left(\frac{Z[q + s^{\mathrm{ch}}]}{Z[q]} \right)^{1/\mathsf{N}}.$$

The folded numerator is nonnegative. Its source lies in $\mathcal{P}$: folding preserves the norm of w, and the matrix norm is at most $|x-x_0|+\|J\|_\infty$. The history bound and base-source almost additivity give

$$Z[q+s^{\mathrm{ch}}] \le \exp\bigl[\operatorname{Re} E(q+s^{\mathrm{ch}}) + (C+2)\ell^3\bigr],$$

$$\Bigl|E(q+s^{\mathrm{ch}}) - E(q) - \sum_r \{E(q+r_\theta s) - E(q)\}\Bigr| \le \mathsf{N}\mathscr{C}_K(\operatorname{supp} s;\delta).$$

The reflected supports have mutual distance at least 2δ. The reflected cube count costs at most 27; this factor is included in $\mathscr{C}_K$, as in the one-cell estimate of [QCDM1, Prop. 10.17]. Since $q^\star = q$ and $E(v^\star) = \overline{E(v)}$, the real part of the sum is $\mathsf{N}\operatorname{Re}\lambda_x(s)$. Use $Z[q] = e^{E(q)-u(x)\ell^3}$ and $\ell^3/\mathsf{N} = |\square|$, then take the N-th root.

For several occupied cells use their respective factors in the same chessboard inequality. Each cell contributes $(Z[q]/Z[0])^{1/\mathsf{N}}$; their product is $Z[q]/Z[0]$. For an occupied cell its remaining factor is bounded by the single-cell computation. Dividing out the common product proves the multiple-cell bound. □

Proposition 10.8 (The two kernels at the continued masses)**.** Let $c_2 = 9/(16\pi^2)$, $c_B = 1/(2\pi^2)$ and fix ϵ, ϑ as above. Require the test pair (f,h) to be of class C^1. There are functions $\epsilon_0(d), \epsilon_B(d)$, tending to zero as $d \downarrow 0$, and constants D_0, D_B such that every family in Thm. I satisfies, on Ω_ϵ,

$$\mathcal{S}_2(z;(0,f),(0,h)) = c_2 b^{-2}\mathcal{K}_6(f,h)(1+\eta_0),$$
$$\mathcal{S}_2(z;(\Gamma,f),(\Gamma',h)) = c_B\mathcal{K}_4(f,h)(\delta_{\Gamma\Gamma'}+\eta_B),$$

where

$$|\eta_0| \le \epsilon_0(d) + D_0\{d + \operatorname{dist}(z,I)dH_d^{1+2p_0}\},$$
$$|\eta_B| \le \epsilon_B(d) + D_B\{d + \operatorname{dist}(z,I)dH_d\}.$$

The functions are uniform in the admitted side and mass centre; the displayed constants may depend on the side. At real $z \in \Omega_\epsilon$, the same conclusion holds with constants independent of the side. In particular the short-distance coefficient of the action two-point function determines b^{-2} throughout Ω_ϵ. ■

Proof. We prove uniform finite-cutoff bounds and then pass to the subsequence. Choose the centre $x_0 \in I$ nearest to z and put $\zeta = z - x_0$. On Ω_ϵ, $|\zeta| < R_\epsilon$. For a source associated to one entry define

$$\|F\|_\circ^\epsilon = \begin{cases} \|f\|_\sharp/\varrho, & \alpha = 0,\\ \|f\|_\infty/\epsilon, & \alpha \in \mathfrak{B}. \end{cases}$$

For the test pair, $\|f\|_\sharp \le 40M\|f\|_\infty/(\vartheta d)$. On $|t_i| < 2/\|F_i\|_\circ^\epsilon$, the source $q + t_1F + t_2H$ lies in $\mathcal{P}$: the extra bilinear norm is less than 2ϵ, while $|\zeta| + 2\epsilon < 2\varrho_J$.

Write $E = E_K^{(x_0)}$. At a nonzero partition function,

$$S_{2,K}^T(z;F,H) = \partial_{12}E(q;F,H) + \mathcal{B}_\zeta(F,H), \qquad \mathcal{B}_\zeta(F,H) = \partial_{12}\log W_K[q+t_1F+t_2H]\big|_0.$$

Both terms are bilinear in F, H. The zero-free estimate, the finite-size normalization and the history upper bound imply

$$\left|\frac{W_K[q+t_1F+t_2H]}{W_K[q]}\right| \le B_c, \qquad B_c = 2e^{(C+2+A_\epsilon)\ell^3+2C_{\mathrm{fs}}}.$$

For a holomorphic function ψ on the indicated bidisc with $\psi(0) = 1$ and $|\psi| \le B$, Schwarz's lemma on complex lines gives $|\psi - 1| \le 1/2$ on the bidisc smaller by $1/(4B)$. Its logarithm and Cauchy's formula therefore give

$$|\partial_{12}\log\psi(0)| \le 4B^2\|F\|_\circ^\epsilon\|H\|_\circ^\epsilon.$$

This applies to $\mathcal{B}_\zeta$ with $B = B_c$.

Take a torus grid with mesh at least $12d$, rounded to the lattice. For $8a \le d$, the radius-$5d$ balls containing its translated pairs are at distance at least d. There are $N \ge (\ell/(24d))^3$ such pairs. Translation invariance holds at complex mass: the finite integrals and their insertion derivatives are polynomials in z, so the real-mass identity extends as an identity of rational functions. The sign average used in the proof of Lem. 10.6 now yields

$$S^T_{2,K}(z;F,H) = Q_K(\zeta) + \mathcal{R}_c, \qquad |\mathcal{R}_c| \le 4B_c^2(24d/\ell)^3 \|F\|^\epsilon_\circ \|H\|^\epsilon_\circ. \tag{128}$$

For real x with $\operatorname{dist}(x,I) \le R_\epsilon$ one can replace the volume loss by a cell loss. The lower bound of Prop. 10.3 gives $u(x) \le A_\epsilon$. Translate the pair into the radius-$1/4$ ball of a unit cell. The cell packing of [QCDM1, Lem. 11.2 and its proof] has at least $(96d)^{-3}$ translates inside this ball. For their two sums with signs, Lem. 10.7 applies with margin $1/4$ and gives

$$\left| e^{-\lambda_x(t_1F+t_2H)} \frac{Z[q+t_1F+t_2H]}{Z[q]} \right| \le B_r, \qquad B_r = e^{C+2+A_\epsilon+\mathscr{C}_*},$$

where $\mathscr{C}_*$ is the fixed upper bound for $\mathscr{C}_K(S;1/4)$ when S lies in that ball. The logarithmic derivative subtracts the board average of $\partial_{12}E$. Averaging the signs retains its diagonal pairs, now including their images in all ℓ^3 cells. Thus

$$S^T_{2,K}(x;F,H) = Q_K(x-x_0) + \mathcal{R}_r, \qquad |\mathcal{R}_r| \le 4B_r^2(96d)^3 \|F\|^\epsilon_\circ \|H\|^\epsilon_\circ. \tag{129}$$

Here $N \ge (\ell/(96d))^3$, so Lem. 10.6 applies to both packings.

For clarity, the estimate of $Q_K(0)$ uses the free-template and interaction-error estimates in [QCDM1, proofs of Lems. 11.5 and 11.6], before adding the partition-function remainder in its two-point theorem. Their proofs apply to either packing just used: at fine levels they use the separated pair alone, and at coarse levels they use precisely $(\ell/\lambda_j)^3 \le 96^3 N (d/\lambda_j)^3$. The fourth-order source term has zero mixed derivative at these indices. The free terms give $c_2 b^{-2}\mathcal{K}_6$ and $c_B\delta_{\Gamma\Gamma'}\mathcal{K}_4$. After division by (126), the interaction errors are bounded by constants times $\phi(b/d)(1+\log(b/d))^{2p_0}$ and $\phi(b/d)$, respectively. The free remainders contain powers of d, d/ℓ, and the positive mass bounds, and hence tend to zero uniformly for admitted ℓ. Their mesh terms tend to zero for each fixed test pair as $K \to \infty$. Denote the resulting cutoff-independent remainders by $\epsilon_0(d), \epsilon_B(d)$.

Add the change $Q_K(\zeta) - Q_K(0)$ from Lem. 10.6 and the appropriate packing remainder. For the action term the denominator is at least

$$c_2\, 10^{-6}\vartheta^2 X_g^{-2} \|f\|_\infty \|h\|_\infty;$$

for the bilinear term it is at least $c_B 10^{-4}\vartheta^2 d^2 \|f\|_\infty\|h\|_\infty$. The mass-shift terms therefore have the displayed orders $|\zeta| d H_d^{1+2p_0}$ and $|\zeta| d H_d$. Each packing remainder has relative order d. Its constant depends on ℓ in (128), and is independent of ℓ in (129). Absorb these explicit factors into D_0, D_B.

Finally take the cutoff subsequence defining the holomorphic family. The tests are separated and of class C^1, so the cumulants converge; all mesh remainders vanish. The bounds survive and their errors tend to zero as $d \downarrow 0$. The coefficient conclusion follows after dividing the first formula by $c_2\mathcal{K}_6$. □

Proposition 10.9 (Uniform real-mass cumulant bounds)**.** Let $n \ge 2$, and let the real Lipschitz tests f_i have pairwise distance at least $d > 0$, each support lying in a ball of radius R. For

$$K \ge K_\epsilon, \qquad L^K \ge 16n, \qquad x \in \mathbb{R}, \quad \operatorname{dist}(x,I) \le R_\epsilon,$$

one has

$$|S^T_{n,K}(x;(\alpha_i,f_i)_i)| \le (1+d^{-1})^3 B_\epsilon(R)^n (n!)^{5+p/2} \prod_i \|f_i\|_{\alpha_i},$$

where

$$B_\epsilon(R) = \frac{C_B e^{C+2+A_\epsilon}(1+R)^3}{\min\{\varrho,\epsilon\}}, \qquad p = \frac{432 C_{\mathscr{C}}}{\log L}.$$

The constants $C_B, C_{\mathscr{C}}$ are those of the separated-cumulant bound in [QCDM1, Thm. F]. This bound is independent of the side, cutoff, coupling within I_g, and mass in the displayed interval. For tests in the continuum class specified at the start of this section, it passes to every continued family at its real masses. ∎

Proof. Choose $x_0 \in I$ nearest to x, and use $q = q_x^{(x_0)}$. Then $Z[q] > 0$ and $u(x) \le A_\epsilon$. Set $\eta = (16n)^{-1}$. The product hat partition of unity at mesh η splits each f_i into at most $(2R/\eta + 3)^3$ pieces $f_i\chi_a$. Each piece is supported in a cube of side 2η, with insertion norm at most $(1 + 80M/\eta)\|f_i\|_{\alpha_i}$.

Fix one piece from every insertion. The board selection in the proof of [QCDM1, Thm. F] leaves distance at least η between these supports and the walls of unit cells. To recall why the choice is available, in each coordinate a support forbids at most $4\eta L^K + 1$ half-integral wall positions. The n supports therefore forbid fewer than L^K positions because $L^K \ge 16n$. The resulting board has an even number of slabs on every admitted torus.

Write s_i for the source of the chosen piece and choose radii

$$\rho_i = \begin{cases} 2\varrho/\|f_i\chi_{a_i}\|_\sharp, & \alpha_i = 0, \\ 2\epsilon/\|f_i\chi_{a_i}\|_\infty, & \alpha_i \in \mathfrak{B}. \end{cases}$$

Group $s(z) = \sum_i z_i s_i$ by its cells, and put

$$\Lambda_x(z) = \sum_{\square} \lambda_x(s_\square(z)), \qquad G(z) = e^{-\Lambda_x(z)} Z[q + s(z)]/Z[q].$$

Disjoint supports ensure that every group has action norm less than 2ϱ and matrix norm less than $2\epsilon \le 2\varrho_J - |x - x_0|$. For a coordinate face indexed by B, at most $|B|$ cells are occupied. Lem. 10.7 and subadditivity of the support count give

$$|G(z_B)| \le \exp\left[|B|(C + 2 + A_\epsilon) + \sum_{i \in B} \mathscr{C}_K(\operatorname{supp} s_i; \eta)\right].$$

The refined cube count from [QCDM1, proof of Thm. F] gives

$$\mathscr{C}_K(\operatorname{supp} s_i; \eta) \le 216 C_{\mathscr{C}}(1 + \log_L(1/\eta)) + C.$$

Cauchy's formula on each face consequently bounds $\partial_B G(0)$ by this exponential times $\prod_{i \in B} \rho_i^{-1}$.

Differentiate $\log Z = \Lambda_x + \log G + \log Z[q]$:

$$S^T_{n,K} = \partial_{[n]}\Lambda_x + \sum_{\pi} (-1)^{|\pi|-1}(|\pi| - 1)! \prod_{B \in \pi} \partial_B G(0).$$

The sum of the absolute partition coefficients is at most $2^n n!$. Thus its contribution is bounded by

$$C^n e^{n(C+2+A_\epsilon)} (16n)^{np/2} 2^n n! \prod_i \rho_i^{-1}.$$

The full mixed derivative of Λ_x is zero unless all the pieces occupy one cell. In that case its reflected energy derivatives obey [QCDM1, Prop. 10.18(i)], since every partial source remains in $\mathcal{P}$. It bounds this term by

$$2^{n-1} \prod_i \rho_i^{-1} \sup_Q \mathscr{C}_K(Q; d) \le C^n (1 + d^{-1})^3 \prod_i \rho_i^{-1},$$

where Q ranges over cubes of side 2η. The reflected count, including its factor 27, is the one already used in Lem. 10.7.

Sum over the choices of pieces. Their number and norm enlargement cost at most

$$(32nR + 3)^{3n}(1 + 1280Mn)^n.$$

The inequality $n^n \le e^n n!$ turns these four powers of n into $(n!)^4$. The partition sum contributes one further factorial and the logarithmic count contributes $(n!)^{p/2}$. Absorbing fixed constants gives

the stated $B_\epsilon(R)$. No estimate used the torus side. For the stated continuum tests, passing along the cutoff subsequence proves the last assertion; no one-point limit is involved. □

Appendix A. Index of notation and conventions

The table lists the principal notation in the order in which it is used. A cited result includes the definitions immediately preceding it.

Notation	Meaning and location
$a = L^{-K}$, λ_j, β, m_0	The fine spacing, the level spacing $\lambda_j = L^{j-K}$, and the bare parameters $\beta = \mathsf{b}/a$, $m_0 = a\mathsf{m}$; section 2.
I_g, $\mathfrak{L}(g)$, T_ℓ	The admitted interval for b, the admitted torus sides, and $(\mathbb{R}/\ell\mathbb{Z})^3$; section 2.
$D[U]$, $D_\pm$, $e_\pm$, $\mathsf{G}_\pm$	The massless chamber operator, its two internal blocks, their projections $(1 \pm \tau_3)/2$, and the physical resolvents $a(D_\pm + m_0)^{-1}$; (1).
$\mathfrak{B}$, $O_K(w)$, $\Phi_K(J)$	The reflection-real meson basis, the action insertion, and the site bilinear insertion; (2)–(3). The label 0 denotes the action density.
$\|f\|_\alpha$, $\|f\|_\sharp$	$\|f\|_0 = \|f\|_\sharp = \max(\|f\|_\infty, 40M \operatorname{Lip} f)$ and $\|f\|_\Gamma = \|f\|_\infty$ for $\Gamma \in \mathfrak{B}$; section 2.
ν_K, $\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$	The finite-torus gauge law and the set of its infinite-lattice torus limits; section 3.
$\mathsf{L}(f)$, $\Xi^f_\nu(z)$	The loop $\operatorname{Tr}(\mathsf{G}_+ \hat{f})$ and its determinant moment; (4). The hat denotes multiplication by the test function.
D^{mix}, E^{mix}_K, h_K	Mixed extraction by inclusion–exclusion, the resulting energy in (13), and the holomorphic normalization in the circle bound (6).
F_s, $Z_K(s)$, $\nu_{K,s}$, $\mathfrak{l}_s$	The determinant at physical mass s, its partition function, its gauge law, and $\mathfrak{l}_s = \partial_s \log \lvert F_s\rvert^2$; (17)–(19).
R_b, R^{mix}, P_{col}	Deletion of species b, the mixed projection $(1-R_1)(1-R_2)$, and independent constant colour averaging in the two species at the trivial pair; Lem. 4.2.
t_j, x_j, ϕ_j, δ_j, $\mathfrak{l}_j$	The calibration coefficient $\lambda_j^{-1}\mathsf{b}$, the running coefficient, $\phi(x_j)$, and the gauge fluctuation radius; section 4. The additional scale weight is $\mathfrak{l}_j = \mathsf{t}_j^{-1/2}(\log \mathsf{t}_j)^{-p_0}$; Prop. 3.3.
$\mathfrak{c}$, $\mathscr{D}^\circ_k$, σ_i	The free coordinates $\mathfrak{c} = (0, \mathsf{n}, (\sigma_i)_{i\le k})$, their domain at zero gauge source, and the extracted mass-source corrections; (44).
H_k, $\bar{H}_k$, Γ_k^{-1}, W	The unbarred background map, its row twin, the fermion fluctuation covariance and the fermion fluctuation; section 4. The gauge fluctuation is B.
ϵ_F, $\|m^{\mathrm{tor}}_c\|_{\mathrm{nat}}$	The weighted norm of the chosen local decomposition of F, and the norm of its grouped cell zeroth moments; Lem. 4.5.
$\mathscr{S}$, M^*, $\mathscr{S}_\mathbb{R}$	The enlarged bilinear label space, the involution $M^* = \tau_3 M^\dagger \tau_3$, and its real form; (71). The insertion with label M is $\Phi[M](f)$.
$G_\mathbb{C}$, G, $V(S)$	$\mathrm{Sp}(2N_f, \mathbb{C})^2$, its compact form $\mathrm{USp}(2N_f)^2$, and the unitary action of $S \in G$; (72) and Prop. 5.5.
S^{ext}, σ^{ext}, $\mathcal{H}^{\mathrm{ext}}$, H, P_1, P_2	The extended truncated functions, their centered functional, the reflection Hilbert space, and its time and spatial translation generators; Prop. 5.4.
n, Q, $\mathcal{H}^{\mathrm{ext}}_n$, $\mathcal{H}^{\mathrm{ext}}_\pi$	A species weight, the quark-number operator Q with eigenvalue $q = \sum_f n_f$, the corresponding weight space, and the π-isotypic space. Here $\lvert\pi\rvert$ is the largest ℓ^1 norm of a weight of π; Prop. 5.5 and Thm. E.
j^T, j^T_μ, $J^T(f)$	The chamber-link current, its sum across a site bond, and its smeared physical insertion, including its factor i; (82).
$\mathcal{M}$, n, $\mathscr{O}^{\mathcal{M}}_k$, $\mathcal{P}^{\mathcal{M}}$, $\mathcal{O}^{\mathrm{lev},\mathcal{M}}_j$	The matrices acting on spin, internal and flavour indices and trivially on colour; the bare source $\mathsf{n} = -aJ$; and the chain, physical and level source domains; the definitions preceding Prop. 8.6.
$\mathcal{D}_j$, $d_j(Y)$, $E_K(s)$, $e_j(Y;s)$, $W_K[s]$	The polymers and their block-tree lengths, the extracted energy, its local increments, and the remaining factor in $Z_K[s] = e^{E_K(s)} W_K[s]$; (107).

Notation	Meaning and location (continued)
$\mathcal{P}$, ρ_b, ρ_m, $W_{\mathbb{C}}$	The physical source ball and the complex coupling–mass neighbourhood of the real parameter window; section 9 and (111).
$d_{\cap}(S_1, \dots, S_n)$	$\min_{x \in T_\ell} \max_i \operatorname{dist}(x, S_i)$. It is positive exactly when the closed supports have empty total intersection; the definition preceding Lem. 9.2.
I, U_r, $\mathcal{E}_K(z)$, $h_K(z)$	The mass interval, its complex r-neighbourhood, the common vacuum normalization, and $h_K(z) = Z_K(z)e^{-\mathcal{E}_K(z)}$; section 10 and Prop. 10.2.
I_ϵ, ρ_ℓ, w_ℓ, Ω_ϵ	The longer real mass interval, the radius used in its zero-free estimate, and its complex neighbourhood of width $w_\ell = \rho_\ell/[2(e+1)]$; (123) and Prop. 10.5.
h, $\mathcal{Z}$, $\mathcal{S}_n(z; X)$	A subsequential limit of the normalized partition functions, its zero set, and the common meromorphic continuation of the separated cumulants; Thm. I.

Conventions. The barred and unbarred variables are independent Grassmann generators. Haar measure has mass one, and the Berezin integral of $e^{-\bar{\psi}A\psi}$ is $\det A$. The traces defining L and $\mathfrak{l}_s$ include position, spin and colour but no flavour; when a source acts on flavour, the propagator is extended by the identity on that factor.

Operator formulas use lattice coordinates; supports and distances are physical unless stated otherwise. Links on T_ℓ are periodic and spinors are antiperiodic. Infinite-volume results take volume to infinity before removing the cutoff. The complex parameter results keep T_ℓ fixed.

The symbol M^* on $\mathscr{S}$ denotes the displayed reflection involution. Elsewhere $*$ on an operator denotes its Hilbert-space adjoint. The functions called h_K in the circle bound and in mass continuation have the separate definitions listed above. The former subtracts a one-species energy; the latter is a normalized partition function. Likewise $\mathfrak{l}_s$ in section 3.4 is a random loop at physical mass s, whereas $\mathfrak{l}_j$ in the scale estimates is the deterministic weight listed above. In section 8, $G_U = (D_+[U] + m_0)^{-1} = a^{-1}\mathsf{G}_+$ denotes the lattice propagator; G_+ includes the physical factor a.

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